MIL-A-20222
30 October 1951

MILITARY SPECIFICATION

ANNOUNCING-EQUIPMENT, SHIPBOARD

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Navy R17A11(SHIPS)
15 July 1948

This specification consists of this cover sheet and Bureau of Ships Ad Interim Specification R17A11(SHIPS), dated 15 July 1948 attached hereto, without modification.

Copies of this specification may be obtained upon application to the Bureau of Ships, Navy Department, Washington 25, D. C. When requesting, state title and number of the specification and the purpose for which required.

When a request for this specification is received by a supplying activity, it will be necessary to attach this cover sheet to the pertinent specification before issue.

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INDEX

Paragraph		Page
A.	Applicable Specifications, Other Publications and Drawings	1
1.	Specifications	1
	Navy Department Specifications	1
	Joint Army-Navy Specifications	2
	Federal Specifications	2
	Bureau of Ships Specifications	2
2.	Other Publications	2
3.	Drawings	3
		3
B.	Classes and Types	3
1.	Classes	3
2.	Nomenclature	4
3.	Microphones	4
4.	Amplifiers	4
5.	Loudspeakers	6
		6
C.	Material and Workmanship	6
1.	Material	6
2.	Aluminum Alloys	7
3.	Magnesium	7
4.	Iron and Steel	7
5.	Protection Against Corrosion	7
6.	Painting	7
7.	Plastic and Phenolic Materials	7
8.	Inflammable Materials	7
9.	Toxic Compounds	7
10.	Lubricants	7
11.	Workmanship	7
		7
D.	Requirements Common To All Classes and Types	7
1.	Scope	8
2.	Power Supply	8
3.	Batteries	8
4.	General Design Requirements (Mechanical)	8
a.	Bulkhead Mounting	8
b.	Deck Mounting	8
c.	Front Servicing	8
d.	Fabricated or Drawn Sheet Metal Enclosures	8
(1)	Preferred Design	8
(2)	Mounting Straps	8
(3)	Stiffening Grooves	8
(4)	Minimum Sheet Thickness	8
e.	Watertight Enclosures	8
f.	Gaskets	8
g.	Bolts, Nuts, Screws, Threads, Locking Devices, Lockwashers, Taper Pins, etc.	8

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Check the source to verify that this is the current version before use.

Paragraph		Page
D.4.	h. Interchangeability	9
	i. Standard Stock Parts	9
	j. Proprietary Parts	9
	k. Serviceability	9
	ℓ. Design for Production	9
	m. Rounded Edges and Corners	9
	n. Shock and Vibration Mountings	9
	o. Resistance to Shock	9
	p. Resistance to Vibration	9
	q. Resistance to Salt Spray	10
	r. Resistance to Moisture	10
5.	General Design Requirements (Electrical)	10
	a. Wiring	10
	(1) Hook-up Wire	10
	(2) Lacing	10
	(3) Sleeving or Tape	10
	(4) Cable Clamps	10
	(5) Grommets	10
	(6) Pre-formed Cables	10
	(7) Wiring to Terminals	10
	(8) Color Coding	10
	b. Mounting of Parts	10
	c. Shielding and R.f. Noise Interference	11
	(1) Objective	11
	(2) Bonding	11
	d. Segragation of Low Level Circuits	11
	e. Grounds	11
	(1) External	11
	(2) Internal	11
	f. Terminal Boards	11
	(1) General	11
	(2) Mounting	11
	(3) Marking	11
	g. Soldering	11
	(1) General	11
	(2) Strength of Joint	11a
	(3) Multiple Connections	11a
	(4) Soldering Flux	11a
	h. Temperature	11a
	(1) Operating Temperature	11a
	(2) Temperature Rise	11a
	i. Clearances	11a
	j. Dielectric and Insulation Characteristics	11a
	(1) Dielectric Strength	11a
	(2) Insulation Resistance	11a
	(3) Loudspeaker Voice Coil Circuits	11a

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Check the source to verify that this is the current version before use.

Paragraph | Page

D. 5. k. Interchangeability . . . 12
ℓ. Serviceability . . . 12
6. Electrical Parts . . . 12
a. General . . . 12
b. Electronic Tubes . . . 12
c. Electron-Tube Sockets . . . 12
d. Capacitors . . . 12
e. Resistors . . . 12
f. Potentiometers . . . 12
g. Transformers . . . 13
h. Relays . . . 13
i. Terminal Boards and Terminals . . . 13
j. Fuse Holders . . . 13
k. Fuses . . . 14
ℓ. Switches . . . 14
m. Indicator Lamps and Lenses . . . 14
n. Volume Indicator Meters . . . 14
7. Marking of Parts . . . 15
8. Engineering Services . . . 15
a. Services Required . . . 15
b. Report of Services . . . 15
c. Multiple Manufacture . . . 15
9. Definitions . . . 15
a. Definitions of Terminology . . . 15
(1) Part . . . 15
(2) Assembly . . . 16
(3) Component . . . 16
(4) Equipment . . . 16
(5) System . . . 16
(6) Accessory . . . 16
(7) Attachment . . . 16
b. Definitions of Degree of Enclosure . . . 16
(1) Pressure-proof . . . 16
(2) Watertight . . . 16
(3) Waterproof . . . 16
(4) Dripproof . . . 16
10. Nameplates and Instruction Plates . . . 17
a. Equipment Nameplates . . . 17
b. Spare Parts Box Nameplates . . . 17
c. Designation Plates . . . 17
d. Diagrams and Instruction Plates . . . 17
11. Plans . . . 17
a. Preliminary Plans . . . 17
b. Working Plans . . . 13
(1) Overall Schematic . . . 18
(2) Schematic Diagrams of Assemblies . . . 18
(3) Wiring Diagrams . . . 19
(4) Mechanical Assembly Plans . . . 19
(5) List of Spare Parts . . . 19
(6) Clarity and Legibility . . . 19
(7) Design Changes After Plan Approval . . . 19

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Check the source to verify that this is the current version before use.

Paragraph Page

D.12. Instruction Books 20
a. Title Page 20
b. General Data 20
c. Table of Contents 21
d. Section 1 - Introduction 21
e. Section 2 - System Description 21
f. Section 3 - Installation 21
g. Section 4 - Adjustments and Tests 21
h. Section 5 - Operation 21
i. Section 6 - Maintenance 22
j. Section 7 - List of Repair Parts 22
k. Illustrations 23
l. Blank Pages 23
m. Numbering 23
n. Approval 23
o. Instruction Books for Components 23
p. Additional Instruction Books 24
q. Abridged Instruction Sheets 24
13. Spare Parts and Tools 24
a. Spare Parts 24
b. Test Headset 24
c. Boxes 25
d. Marking 25
e. Protection and Identification 25
f. Multiple Manufacture 25
g. Bulk Spares 25
14. Qualification (Type or Brand Approval) 25

E. Detail Requirements 25
1. Microphones 25
a. General Description of Microphones Types 25
(1) Basic Microphone Unit, Type IC/MBU 25
(2) Microphone Control Station, Types IC/MSA, IC/MSB 25
(3) Microphone Control Box, Types IC/MBA, IC/MBB, IC/MBC, IC/MBD . . 26
(4) Microphone Control Box, Type IC/MBT 26
(5) Microphone Jack Box, Types IC/MJB, IC/MJT 26
(6) Microphone Jack Box, Type IC/MJP 26
(7) Portable Microphone, Type IC/MPM 26
(8) Handset, Type IC/MPH 26
(9) Microphone With Flexible Mounting, Type IC/MFM 26
b. Mechanical Requirements 26
(1) Type IC/MBU 26
(a) Maximum Size 26
(b) Serviceability 27
(1) Mike Element 27
(2) Ready Replacement 27
(3) Interchangeability 27
(4) Electrical Connections 27
(5) Polarization 27
(c) Air Gap Clearance 27
(d) Press-to-Talk Switch 27
(1) Contact Arrangement 27
(2) Switch Construction 27
(3) Switch Action 27
(4) Switch Endurance 27

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Check the source to verify that this is the current version before use.

Paragraph		Page
E.1.	b. (1)(e) Resistance to Gun Blast	27
	(f) Microphone Endurance Requirements	27
	(g) Mounting Facility	28
	(2) Microphone Station and Boxes	28
	(a) Parts	28
	(1) Tabulation	28
	(2) Description of Parts	28
	(a) Microphone	28
	(b) Volume Indicator Meter	28
	(c) Busy Lights	28
	(d) Speaker Selection Switches	29
	(e) "Power on" Light	29
	(f) Talkback Disconnect Switches	29
	(g) "IMC Tie-In" Switch	29
	(h) Fuses	29
	(i) Talk Switch	29
	(j) Jack	29
	(k) Designation Plates	29
	(ℓ) Instruction Plate	30
	(b) Provision for Cable Entrance	30
	(c) Maximum Weight and Dimensions	31
	(d) Resistance to Pressure	31
	(e) Resistance to Depth Charge	31
	(3) Portable Microphone , Type IC/MPM, and Handset, type IC/MPH	31
	(a) Parts	31
	(1) Tabulation	31
	(2) Description	32
	(a) Unit	32
	(b) Microphone Element	32
	(c) Receiver Element	32
	(d) Press-to-Talk Switch	32
	(e) Chest Plate	32
	(f) Neck Strap	32
	(g) Microphone Cable	32
	(h) Plug	32
	(i) Handset Shell	33
	(b) Maximum Weight	33
	(c) Strength of Cable Assembly	33
	(d) Resistance to Tumble	33
	(e) Microphone Mounting Attachment	33
	(4) Microphone, Type IC/MFM (Flexible Mounting)	33
	(a) Description	33
	(b) Cable Entrance	33
	(c) Maximum Weight and Dimensions	33
	(d) Microphone Clip	33
	c. Electrical Requirements	33
	(1) Basic Microphone Unit, Type IC/MBU	33
	(a) Impedance	33
	(b) Sensitivity	34
	(c) Frequency Response Characteristic	34
	(d) Induced Voltage	34
	(2) Electrical Circuits	34
	(3) Noise Pickup	34
	(4) Control Circuits, Type IC/MBT	34
	(a) Local Talk Switch	34
	(b) Remote Talk Switch	35
	(c) "IMC" Alarm Circuit	35

Source: https://assist.dla.mil -- Downloaded: 20261O-19T07:26Z
Check the source to verify that this is the current version before use.

Paragraph	Page
E.2. Amplifiers and Controls	35
a. Mechanical Requirements	35
(1) Amplifier Assemblies and Signal Generators	35
(a) Definition	35
(b) Construction	35
(1) Accessibility	35
(2) Electric Connections	35
(c) Maximum Weight and Size	35
(d) Interchangeability	36
(2) Amplifier and Control Racks	36
(a) Definition	36
(b) General Construction	36
(c) Front Servicing	36
(d) Covers and Doors	36
(e) Personnel Protection	36
(f) Mounting of Amplifiers and Signal Generator Assemblies	36
(g) Mounting of Controls	36
(h) Accessibility	36
(i) Terminal Boards	36
(j) Provision for Cable Entrance	36
(k) Ventilation	37
(ℓ) Maximum Weight and Size	37
(m) Other Limiting Dimensions	37
b. Electrical Requirements	37
(1) Amplifiers and Signal Generator Assemblies	37
(a) General	37
(1) Dual Tube Design	37
(2) Power Supply	37
(3) Test Facilities	37
(4) Standby and Ready	37
(a) Standby	37
(b) Ready	37
(c) Time Interval for Ready	38
(d) Tube Life During Standby	38
(5) Amplifier Phasing	38
(b) Amplifier Assemblies, Low and Intermediate Power, Types	38
IC/FCA, IC/FDA and IC/FGA	38
(1) Definition	38
(2) Amplifier Design Characteristics	38
(a) Input Impedance	38
(b) Output Voltage	38
(c) Output Power	38
(d) Amplifier Gain	38
1. Rated	38
2. Reserve	38
(e) Frequency Response	39
(f) Distortion	39
1. At Various Frequencies	39
2. At Various Input levels	39
(g) Noise and Microphonics	39
(h) Output Voltage Regulation	39
(i) Isolation Against Longitudonal Voltages	40
(j) Stability	40
(k) Overload Capacity	40

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Check the source to verify that this is the current version before use.

Paragraph | Page

E. 2. b. (1)(b) (3) Compressor Circuit . . . 40
(a) Cutout Switch . . . 40
(b) Compressor Characteristic . . . 40
(c) Time Constant . . . 41
(d) Compressor Adjustments . . . 41
(e) Effect of Compressor on Amplifier Characteristics . . . 41
(4) Overload Limiter Circuit . . . 41
(a) Overload Limiter Characteristic . . . 41
(b) Time Constant . . . 41
(5) Amplifier Controls . . . 42
(a) Gain Control . . . 42
(b) Compressor Cutout Switch . . . 42
(6) Interstage Input Circuit . . . 42
(a) Input Impedance . . . 42
(b) Input Level . . . 42
(c) Compressor Circuit By-Pass . . . 42
(c) Amplifier Assemblies, High Power, Types IC/FJA and IC/FMA . . . 42
(1) Definition . . . 42
(2) Input Impedance . . . 42
(3) Output Voltage . . . 42
(4) Output Power . . . 42
(5) Amplifier Gain . . . 43
(6) Frequency Response . . . 43
(7) Distortion . . . 43
(8) Noise and Microphonics . . . 43
(9) Output Voltage Regulation . . . 43
(10) Stability . . . 43
(11) Overload Capacity . . . 43
(d) Signal Generators . . . 44
(1) Number and Type . . . 44
(2) Generated Signals . . . 44
(3) Description of Signals . . . 44
(a) General Character of Signals . . . 44
(b) Chemical Attack Alarm . . . 44
(c) General Alarm . . . 44
(d) Collision Alarm . . . 44
(e) Diving Alarm . . . 44
(f) Flight Crash . . . 45
(g) Flight Warning . . . 45
(h) Cease Firing . . . 45
(i) Salvo . . . 45
(4) Duration of Signals . . . 45
(5) Output Level . . . 45
(6) Adjustment of Output Level . . . 45
(7) Noise and Microphonics . . . 45
(2) Amplifier and Control Racks . . . 46
(a) Introduction . . . 46
(1) Amplifier System Classification . . . 46
(a) By size . . . 46
(b) By Circuit Function . . . 46
(2) Amplifier System Accommodations . . . 46
(a) Input Facilities . . . 46
(b) Output Facilities . . . 46
(3) Amplifier System Operating Voltages . . . 46
(a) Loudspeaker Circuits . . . 46
(b) Volume Indicator Circuits . . . 46

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:26Z
Check the source to verify that this is the currentversion before use.

Paragraph			Page
E.2.	b. (2)(b)	Operating Controls	47
		(1) General	47
		(a) Circuit Transfer Switches	47
		(b) Switch Markings	47
		(c) Switch and Relay Circuits	47
		(2) Power Indicator Light	47
		(3) Power Switches	47
		(4) Input Disconnect Switches	47
		(a) Active Switches	47
		(b) Spare Switches	47
		(5) Amplifier Transfer Switches	47
		(a) General	47
		(b) 1-2-3MC, 1-6MC, 1-7MC	47
		(c) 5MC	48
		(d) 17MC	48
		(6) Signal Generator Selector Switches	48
		(a) General	48
		(b) 17MC	48
		(7) Output Disconnect Switches	48
		(a) General	48
		(b) 1-2-3MC and 1-6MC	48
		(c) 17MC	48
		(8) System Volume Controls	48
	(c)	Circuit Protective Features	49
		(1) Power Circuits	49
		(a) Amplifier Main Power Circuit	49
		(b) Relay Power Circuits	49
		(c) Microphone Equipment Power Circuits	49
		(2) Microphone Audio Circuits	49
		(3) Volume Indicator Circuits	49
		(4) Busy Indicator Circuit	49
		(5) Loudspeaker Audio Circuits	49
	(d)	System Adjustments	49
		(1) Output Voltage to Loudspeaker Groups	49
	(e)	Test Facilities	49
		(1) General	49
		(2) Amplifier Start Switches	50
		(3) Signal Generator Start Switches	50
		(4) Volume Indicator Meters and Test Select Switches	50
		(a) Input Metering	50
		(b) Output Metering	50
		(c) Combined Metering	50
		(5) Test Load Resistor and Switch	50
		(6) Test Input Volume Control	50
	(f)	Control Circuits	50
		(1) Microphone Press-to-Talk Switch	50
		(2) Loudspeaker Group Selector Switches	51
		(3) Alarm Contact Makers	51
		(a) General	51
		(b) Additional Functions for Specific Systems	51
		1. 1-2-3MC, 1-6MC	51
		2. 1-7MC	51
		(4) Loudspeaker Group Selection	51

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:26Z
Check the source to verify that this is the currentversion before use.

Paragraph		Page
E.2.	b. (2)(f) (5) System Priorities	52
	(a) Alarm Signal Priority	52
	(b) Speech Priority	52
	(6) Busy Indicator Circuits	52
	(7) Local Loudspeaker Cutout Circuit	53
	(8) Spare Control Relays	53
	3. Loudspeakers	53
	a. Introduction	53
	(1) Low Power Loudspeakers, Types IC/SAA and IC/SAG	53
	(2) High Power Loudspeakers, Types IC/SBA and IC/SBG	53
	(3) Super Power Loudspeakers, Types IC/SCD, IC/SDC, IC/SDD, IC/SDF, IC/SEB, IC/SEC and IC/SFD	54
	(a) Movable	54
	(b) Fixed	54
	(4) Pressure Proof Loudspeaker, Type IC/SHA	54
	(5) Headset Type IC/SJA and Jack Box, Type IC/SKG	54
	(6) Matching Transformer, Type IC/SLA	54
	b. Mechanical Requirements	54
	(1) General Construction	54
	(a) Degree of Enclosure	54
	(1) All Types, Except IC/SHA	55
	(2) Type IC/SHA	55
	(b) Ease of Installation	55
	(c) Serviceability	55
	(d) Protective Screens	55
	(2) Loudspeaker Driver Unit	55
	(a) Magnet Structure	55
	(1) Field	55
	(2) Resistance to Corrosion	55
	(3) Air Gap Clearance	55
	(4) Serviceability	55
	(b) Voice Coil and Diaphragm Assembly	55
	(1) Serviceability	55
	(2) Protection, Positioning and Polarization	56
	(3) Materials	56
	(3) Super Power Loudspeaker	56
	(a) Multi-Unit Design	56
	(b) Facilities for Installation	56
	(c) Type IC/SEC	56
	(d) Type IC/SDC	57
	(4) Maximum Weights and Dimensions	57
	(5) Terminal Tubes	58
	(6) Resistance to Gun Blast	58
	(7) Resistance to Pressure	58
	(8) Resistance to Depth Charge	58
	(9) Endurance Requirement	58
	c. Electrical Requirements	58
	(1) Rated Operating Voltage	58
	(2) Maximum Electrical Input	59
	(3) Minimum Output Sound Pressure	60
	(4) Output Sound Pressure Distribution	61
	(5) Frequency Response Characteristic	61
	(6) Load Characteristic	62
	(7) Loudspeaker Noises	62
	(8) Polarization of Voice Coil Connections	62

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:26Z
Check the source to verify that this is the current version before use.

Paragraph		Page
E. 3.	c. (9) Loudspeaker Transformer	62
	(a) Operation at Reduced Input Line Voltage	62
	(b) Operation at Reduced Output Sound Pressures	62
	(c) Transformer Location	62
	(10) Loudspeaker Volume Control	62
	(11) Individual Loudspeaker Unit Disconnect Switches	62
	(12) Terminal Board	62
	(13) Electrical Circuits	63
	(14) Pressure Proof Loudspeaker, Type IC/SHA	63
	(a) Use as Microphone	63
	(b) Sensitivity	63
	(c) Voice Coil Impedance	63
	(d) Use with Matching Transformer	63
	(15) Jack Box, Type IC/SKG	63
	(16) Matching Transformer	63
	(a) Tap Connections	63
	(b) Frequency Response Characteristic	63
F.	Methods of Sampling, Inspection and Tests	64
1.	Classification	64
2.	General	64
3.	Type or Brand Approval Tests	64
	a. Type or Brand Approval	64
	b. Supplementary Data	65
	c. Manufacturer's Test Engineers	65
	d. Granting of Approval	65
	e. Order of Tests	65
	(1) Microphone Component Tests	65
	(2) Amplifier Component Tests	66
	(3) Loudspeaker Component Tests	67
	f. Description of Tests	68
	(1) General Inspection	68
	(2) Microphone Sensitivity	68
	(3) Microphone Frequency Response	68
	(4) Microphone Induced Voltage	68
	(5) Noise Pickup	69
	(6) Microphone Endurance	69
	(7) Power Supply Voltage and Frequency	69
	(8) Temperature Rise	69
	(9) Maximum Loudspeaker Electrical Input	69
	(10) Minimum Loudspeaker Output Pressure	70
	(11) Loudspeaker Output Sound Pressure Distribution	70
	(12) Loudspeaker Frequency Response	70
	(13) Loudspeaker Load Characteristic	70
	(14) Sensitivity of Type IC/SHA as Microphone	70
	(15) Loudspeaker Endurance Test	70
	(16) Salt Spray Test	71
	(17) Tumbling Test	71
	(18) Gun Blast Test	71
	(19) Depth Charge Test	71
	(20) Dielectric Strength Test	71
4.	Contract Suitability Tests	72
5.	Inspection Tests	72
	a. Production Tests	72
	b. Sample Tests	73
	c. Other Tests	74
	d. Non-Standard Production Test Methods	74

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Check the source to verify that this is the current version before use.

Paragraph Page

G. Packaging and Preparation for Shipment 75
1. Preservation, Packing and Packaging 75
2. Marking 75

H. Notes 75
1. Ordering Information 75
a. Information Normally Required 75
b. Information Required for Deviation From Standard Requirements 76
2. Specifications, Plans, Publications - Where Obtainable 76
3. Qualification (Type or Brand Approval) 76
4. Patent Notice 77
"Restricted" Clause 77

FIGURES

1. Standard Designation System for Switch and Relay Contacts
2. Standard System for Designation of Component Parts on Schematic Wiring Diagrams
3. Frequency Response Limits for Microphone Equipment
4. Frequency Response Limits for Amplifier Equipment
5. Frequency Response Limits for Low Power Loudspeakers
6. Frequency Response Limits for High Power Loudspeakers
7. Frequency Response Limits for Super Power Loudspeakers
8. Frequency Response Limits for Pressure Proof Loudspeakers
9. Standard Circuits and Terminal Markings,Portable Microphone - Type IC/MPM Microphone Handset - Type IC/MPH, Microphone(Flexible Mounting) - Type IC/MFM
10. Standard Circuit and Terminal Markings, Microphone Control Station - Type IC/MSA
11. Standard Circuit and Terminal Markings, Microphone Control Station - Type IC/MSB
12. Standard Circuits and Terminal Markings, Microphone Control Box - Type IC/MBA; Microphone Control Box - Type IC/MBB
13. Standard Circuits and Terminal Markings , Microphone Control Box - Type IC/MBC, Microphone Control Box - Type IC/MBD
14. Standard Terminal Markings, Microphone Control Box - Type IC/MBT
15. Standard Circuits and Terminal Markings, Microphone Jack Box - Type IC/MJB, Microphone Jack Box - Type IC/MJT, Microphone Jack Box (P.P.) - Type IC/MJP
16. Standard Circuits and Terminal Markings, Loudspeakers - Types IC/SAA and IC/SBA

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R17A11(SHIPS)
July 15, 1948
SUPERSEDING
SGS(65)-156a
1 September 1938

BUREAU OF SHIPS SPECIFICATION

ANNOUNCING-EQUIPMENT, SHIPBOARD

A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

A-1. Specifications.- The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

Navy Department specifications:

General Specifications for Inspection of Material.
15C1 - Cables, Electric, (Shipboard Use).
16E4 - Electronic Equipment, Ship and Shore - General Specification For.
16T30 - Transformers and Inductors (Audio and Power), for Use in Electronic and Communication Equipment.
17C13 - Connectors and Lugs, Terminal, Pressure - Grip (Solderless) (for Electric Cable).
17E11 - Enclosures (for Nonrotating Electrical Equipment)(Shipboard Use).
17F2 - Fuses, Electric.
17-I-26 - Instrument-Markings, Self-Luminous (for Electrical Instruments and Other Marking Purposes).
17-I-61 - Interior-Communication-Equipment; General Specification.
17R6 - Relays, Electric (Shipboard Use).
17R10 - Rectifiers, Metallic Cell.
R17T23 - Telephones, Sound Powered.
35B2 - Books, Instruction; Preparation, Contents, and Approval.
40T9 - Tests; Shock, Vibration, and Inclination (for Special Equipment) General Specification.
42N2 - Nameplates, Instruction Plates, and Other Designating Markings for Electrical and Mechanical Equipment (Shipboard Use).
52P18 - Primer, Zinc-Chromate (Formula No. 84).
66S3 - Shockproof-Equipment, Class HI (High-Impact); and Tests for.

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Joint Army-Navy specifications:
JAN-1A - Tubes, Electron, Radio.
JAN-C-5 - Capacitors, Mica-Dielectric, Fixed.
JAN-I-6 - Instruments, Electrical Measuring, Indicating, Basic, Voltmeters, and Ammeters.
JAN-R-11 - Resistors, Fixed, Composition.
JAN-R-19 - Resistors, Variable, Wire-Wound (Low Operating Temperature).
JAN-S-23 - Switches, Toggle (For Electronic and Communications Use).
JAN-C-25 - Capacitors, Direct-Current, Paper-Dielectric, Fixed (Hermetically Sealed in Metallic Cases).
JAN-R-26 - Resistors, Fixed Wire-Wound, Power-Type.
JAN-S-63 - Switches, Sensitive.
JAN-C-76 - Cable (Hook-up Wire), Electric, Insulated, Radio and Instrument.
JAN-R-93 - Resistors, Accurate, Fixed, Wire-Wound.
JAN-R-184 - Resistors, Fixed, Wire-Wound (Low-Power).
JAN-B-233 - Boxes, Spare Parts (Shipboard Use).
JAN-W-583 - Wire, Magnet.

Federal specification:
QQ-S-571 - Solder, Soft (Tin, Tin-Lead, and Lead-Silver).

Bureau of Ships specifications:
Section S1-1 - Plans.
Section S1-5 - Machinery Books (Mechanical, Electrical, Electronic).
Section S62-2 - Electric Power Distribution, Cables.
Section S65-2 - Announcing Systems.
Appendix 6 - Instructions for Painting and Cementing Vessels.

A-2. Other publications.- The following publications of the issue in effect on date of invitation for bids, form a part of this specification:

Joint Army-Navy publications:
JPI-14 - Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electro-Mechanical).
JANP-109 - Manual of Standard Description for Electronic Equipment and Material.[1]
Preferred List of Army-Navy Electron Tubes.

Bureau of Ships publication:
Description and Application of Navy Standard Stuffing Tubes, Terminal Tubes and Cable Clamps.

Bureau of Supplies and Accounts publication:
Catalogue of Navy Material - General Stores Section.

[1]See paragraph H-2c.

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A-3. Drawings.- The following Bureau of Ships drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

9-S-1841-L - Wire Terminals.
9-S-4222-L - Incandescent Lamps, Naval Vessels.
9-S-4401-L - Battle Telephone Straps, Types A and B.
9-S-4718-L - Switch, Rotary Snap, DPST.
9-S-5005-L - Plug, Type T, for Sound Powered Telephones.
9-S-5010-L - Terminal Tubes, Types 21 (Aluminum), 22-B (Brass) and 22-S (Steel).
9-S-5235-L - Terminal Tubes, Steel, Types 51 (Machined Alternate), 52 (Drawn) for Cable to WT, Sheet Steel Enclosure.
9-S-5355-L - Terminal Tubes, Steel with I.P.S. Threaded Fitter for Threaded Assembly, Type 57 (Alternate).
9-S-5401-L - Switch, Rotary Snap, Panel Mounting, DPST, Type BBB.
9-S-5402-L - Switch, Rotary Snap, Panel Mounting, TPST, Type DDD.
9-S-5454-L - Jack Box, WT, for Sound Powered Telephones, Type T-25.
9-S-5457-L - Terminal Tubes, Steel, Type 59 (Alternate), Electric Cables Entering WT, Sheet Steel Enclosure.
9000-S6202-73075 - Switch, MP Rotary, Types 1JR (Single Pole Blades), 2JR (Double Pole Blades), 3JR (Common Blades), Panel mtd.
9000-S6202-73295 - Switch, MP Rotary, Action Cutout, Type 4JR, Panel mtd.
9000-S6503-73436 - Switch, Rotary Selection, 16 Position, Type JA.
9000-S6503-73437 - Switch, Rotary Selection, 30 Position, Type JA.
9000-S6202-73440 - Dead Front Fuse Holder with Blown Fuse Indicator N.W.T.
9000-S6501-73460 - Receptacle, WT, for Battle Announcing Microphones.
9000-S6501-73461 - Plug, WT, for Battle Announcing Microphones.
9000-S6501-73462 - Plug, WT, for Battle Announcing Microphones (Submarine Applications).
9000-S6501-73463 - Receptacle, PP, for Battle Announcing Microphones
9000-S6202-73496 - Switch, Rotary Snap, Panel Mounted, 4 PTT.
9000-S6505-73774 - Typical Nameplates for Interior Communication and Fire Control Equipment.

B. CLASSES AND TYPES

B-1. Classes.- Shipboard announcing equipment covered by this specification shall be of the following classes:

Microphone components
Amplifier components
Loudspeaker components

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B-2. Nomenclature.- The nomenclature employing the symbols IC/ followed by three letters as outlined in the following paragraphs, shall serve to identify each basic type of component covered by this specification under the above three general classifications. Further identification of each basic type as to manufacturer and model shall be accomplished by the addition of two or more symbols following the type designation. The symbols applying to any particular manufacturer's model shall be assigned by the bureau concerned at the time that the plans of the equipment are submitted for bureau approval.

B-3. Microphones.- Microphone components shall be of the following types:

Microphone control stations:
- Type IC/MSA - Three circuits
- Type IC/MSB - Two circuits

Microphone control boxes:
- Type IC/MBA - Basic unit (volume indicator and jack only)
- Type IC/MBB - Loudspeaker selection
- Type IC/MBC - Busy indicator
- Type IC/MBD - Loudspeaker selection and busy indication
- Type IC/MBT - Facilities for talkback circuits

Microphone jack boxes:
- Type IC/MJB - Basic unit (jack only)
- Type IC/MJP - Pressure proof
- Type IC/MJT - Talk-back

Microphones, miscellaneous
- Type IC/MPM - Portable microphone
- Type IC/MPH - Handset
- Type IC/MFM - Microphone with flexible mounting
- Type IC/MBU - Basic microphone unit (assembly)

B-4. Amplifiers.- Amplifier components and assemblies shall be defined by the symbol IC/ followed by the appropriate designation as follows. (It should be noted that in an announcing system whose amplification is obtained by means of one or more component amplifier racks and separate

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control rack(s), individual type designations shall be assigned to each such component rack. It should also be noted that type designations shall be assigned to the basic amplifier assemblies and signal generator assemblies contained within the component amplifier or control rack(s).

First letter	Second letter	Third letter		
Number of amplifier channels; or designation as a control rack or assembly	Amplifier output power rating in watts per channel	Control features: Announcing circuit controls involved	Control features: Type of signal generators	Control features: Number of signal generators
A - One channel	A - 0	A - No controls	None	0
B - Two channels	B -	B - Miscellaneous controls	None	0
C - Three channels	C - 20	C - 1MC (only)	GAC	2
D - Four channels	D - 40	D - 1MC-2MC	GAC	2
E - Control rack	E -	E - 1MC-2MC-3MC	GAC	2
F - Amplifier assembly	F -	F - 1MC-3MC-5MC	GAC	2
G - Signal generator assembly	G - 125	G - 1MC-6MC	GAC	2
	H -	H -		
	I -	I -		
	J - 250	J - 1MC (with other announcing circuits)	GAC	2
	K -	K -		
	L - 500	L - 1MC-7MC	GAL	2
	M -	M -		
	N -	N - 5MC	GAN	2
	O -	O		
	P - 1000	P		
		Q - 17MC	GAQ	*
		R -		

* Number equal to number of amplifier channels.

Illustrative examples.- The following examples of the use of the amplifier nomenclature are presented for purposes of illustration:

Amplifier, type IC/ADA would designate a single channel, 40-watt component amplifier containing no control circuits.

Amplifier, type IC/BGL would designate a dual-channel, 125-watt component amplifier containing necessary control circuits for a combined 1MC-7MC system and including two signal generators capable of generating the proper 1MC alarm signals for submarines.

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Amplifier, type IC/FGA and Signal Generator, type IC/GAL would designate the individual amplifier assembly and signal generator assembly respectively, in the type IC/BGL amplifier described above.

A typical amplifier for an aircraft carrier shipboard announcing system may comprise the following component items:

One amplifier (dual channel) type IC/BMA
One amplifier (single channel) type IC/AMA
One amplifier (Control rack) type IC/EAE

B-5. Loudspeakers.- Loudspeaker components shall be defined by the symbols IC/ followed by the appropriate designation as follows:

First letter	Second letter basic design	Third letter additional requirements
S	A - Low power	A - No additional requirements
	B - High power	B - Circular pattern - fixed
	C - Super power (125 watts)	C - Circular pattern - movable
	D - Super power (250 watts)	D - Eliptical pattern - fixed
	E - Super power (500 watts)	E - Eliptical pattern - movable
	F - Super power (1000 watts)	F - Elongated pattern - fixed
	G -	G - With volume control
	H - Pressure proof	H - Flat radial pattern
	I -	I -
	J[1] - Announcing headset	J - With press-to-talk switch
	K[1] - Announcing jackbox	
	L[1] - Matching transformer	

[1] The announcing headset and jackbox, while not usually considered to be loudspeakers, have been included herein as part of the general category of components which serve to terminate an announcing system and thus, are connected to the amplifier output circuit. Similarly, the matching transformer is included herein, since it is associated directly with loudspeaker equipment.

C. MATERIAL AND WORKMANSHIP

C-1. Material.- It is the intent of this specification that the quality of the material used in the construction of announcing equipment be at least equal to that covered by specifications referenced in section A of this specification and Navy Department Specification 17-I-61. It is not desired that all such material be tested in accordance with referenced specifications in each individual case; however, the Naval inspector may require such material tests whenever in his judgment it is necessary to ascertain that the quality of a material used meets the requirements of the specification. Where material is not in accordance with the requirements of specifications it must be clearly demonstrated that an improvement in operating characteristics or a saving in weight without sacrifice in reliability can be accomplished thereby, and that such substitutions do not preclude the subsequent use of Navy standard material in effecting repairs or replacements. Departures from referenced specifications shall be noted on the working plans specified in paragraph D-11b, and will thus be subject to specific bureau approval.

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C-2. Aluminum alloys.- The use of aluminum alloys shall be in accordance with the requirements of Navy Department Specification 17-I-61.

C-3. Magnesium.- Magnesium or magnesium alloys in any form shall not be used.

C-4. Iron and steel.- The use of iron and steel shall be in accordance with the requirements of Navy Department Specification 17-I-61.

C-5. Protection against corrosion.- The requirements for protection against corrosion shall be in accordance with the requirements of Navy Department Specification 17-I-61.

C-6. Painting.-

C-6a. The exterior and interior surfaces of all boxes, panels, racks, and enclosures shall be prepared, cleaned, and finished in accordance with the requirements of the General Specifications, Appendix 6.

C-6b. One coat of zinc chromate primer shall be applied in accordance with the requirements of Navy Department Specification 52P18.

C-6c. One finishing coat of alkyd-resin base enamel, shall be applied (Munsell color rating 2.5GY 6.0/0.5, light yellow-gray-green). Components to be used in exposed locations shall have a gloss factor of between 75 and 95 units. Components to be used solely in protected locations shall have a gloss factor of between 15 and 35 units.

C-7. Plastic and phenolic materials.- Plastic and phenolic materials shall conform to the requirements for insulating materials of Navy Department Specification 17-I-61.

C-8. Inflammable materials.- Inflammable materials shall be in accordance with the requirements of Navy Department Specification 17-I-61.

C-9. Toxic compounds.- The use of compounds which will readily produce harmful toxic effects under the conditions normally encountered in Naval service will not be permitted.

C-10. Lubricants.- Lubricants shall be in accordance with the requirements of Navy Department Specification 17-I-61.

C-11. Workmanship.- Workmanship shall be of such quality that the equipment will be sturdy, of excellent appearance, and of high standards with respect to the mechanical fits and the arrangement and connection of wiring.

D. REQUIREMENTS COMMON TO ALL CLASSES AND TYPES

D-1. Scope.- The principle purpose of announcing systems installed aboard Naval vessels is the transmission and reproduction of intelligible speech and discernible alarm signals over high noise levels. To meet this end, the basic requirements of the announcing equipment covered by this specification are reliability, sturdiness, simplicity of design, ease of servicing and maintenance, light weight, compactness, and the ability to withstand severe shock, vibration, gunfire and the adverse effects of salt spray, high humidity and extremes of temperature. The basic requirements of the system are (a) flexibility of operation to permit maximum utilization of the facilities provided and (b) facilities for readily isolating a faulty component or circuit so that remaining components and circuits can continue in operation. An additional objective of this specification is to insure designs of major components, such as microphones, microphone components, amplifiers and loudspeakers, which will permit electrical interchangeability and satisfactory operation of these components in any announcing system, regardless of its manufacture.

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D-2. Power supply.- All components covered by this specification shall be designed for operation from a power source of 115 volts, 60 cycles having operating tolerances within the limits of 122 volts at 65 cycles and 108 volts at 55 cycles. All components except power amplifiers as defined in paragraph E-2b(1)c(1) shall be designed for operation from a single phase power source. Power amplifiers shall be designed for operation from a three-phase power source.

D-3. Batteries.- The use of batteries of any type for any purpose in the equipment covered by this specification is prohibited.

D-4. General design requirements (mechanical).-

D-4a. Bulkhead mounting.- Enclosures designed for bulkhead mounting shall be provided with three points of suspension, two above the center of gravity of the enclosure and one below.

D-4b. Deck mounting.- Enclosures and racks designed for deck mounting shall have provision for bracing to a bulkhead at the top or side.

D-4c. Front servicing.- All deck mounted enclosures and racks shall be designed to permit complete servicing from the front.

D-4d. Fabricated or drawn sheet metal enclosures.-

D-4d(1). Preferred design.- Boxes or enclosures of fabricated or drawn sheet brass are preferred to the use of castings. Such designs, however, should take into consideration the possibility that emergency conditions may arise which will force a conversion to the use of alternate materials, such as sheet steel; and the design should permit such a conversion with a minimum of retooling.

D-4d(2). Mounting straps.- Mounting straps shall be secured to fabricated or drawn sheet metal enclosures in a positive manner. Rivets, tack welds or spot welds, when used alone, are not considered adequate for this purpose. All edges of the straps in contact with the case shall be secured by a continuous weld or by brazing.

D-4d(3). Stiffening grooves.- Grooves shall be indented in large flat sheet metal surfaces along the diagonals to lend rigidity.

D-4d(4). Minimum sheet thicknesses for watertight enclosures.- Sheet steel used in watertight enclosures whose longest dimension is 16 inches or below shall be a minimum thickness of 0.062 inches (U.S. Standard Gage No. 16.) In watertight enclosures whose longest dimension is over 16 inches, minimum sheet steel thickness shall be 0.078 inches (U. S Standard Gage No. 14).

D-4e. Watertight enclosures.- Through bolting or threading into watertight enclosures will not be permitted. Suitable bosses shall be provided for this purpose in cast enclosures. Blind tapped welded buttons shall be used in sheet metal enclosures.

D-4f. Gaskets.- Gaskets for enclosures shall be in accordance with the requirements of Navy Department Specification 17-I-61.

D-4g. Bolts, nuts, screws, threads, locking devices, lockwashers, taper pins, etc.- Bolts, nuts, screws, threads, locking devices, lockwashers, taper pins, etc., shall be in accordance with the requirements of Navy Department Specification 17-I-61. The heads of screws or bolts subject to ships service painting (outside or weather deck) shall be of the open slot or nut head type to obviate paint or dirt interference with subsequent removal. Combined slotted nut head screws are preferred, provided they are of such size that they may be replaced by Navy standard stock screws.

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D-4h. Interchangeability.- All similar parts, including spares, of corresponding equipment furnished on the same contract or built to the same plans, shall be interchangeable, mechanically and electrically, without the necessity of further machining or hand fitting of any kind.

D-4i. Standard stock parts.- Standard stock parts and hardware shall be used to as great an extent as practicable. For the purpose of this specification, standard stock is defined as material listed in the General Stores Section of the Catalogue of Navy Material, and includes such items as bearings, grease, oil, lubricants, cable, cord, wire, bolts, screws, nuts, washers, etc.

D-4j. Proprietary parts.- Where Navy standard stock parts are not suited for the purpose intended, parts and materials shall be used which are standard, easily obtainable and produced by reliable manufacturers, as far as possible within the limits of this specification. In order to facilitate the procurement of replacement parts, the design shall not be based upon the use of component parts of special manufacture where suitable units of standard manufacture are available. Similarly the design shall not be based on the use of component parts produced by only one manufacturer when an equivalent design available from several sources of manufacture might be employed.

D-4k. Serviceability.-

D-4k(1). Units shall be constructed with a view toward maximum accessibility for repair and replacement of components. All assembled component parts shall be so mounted in the major assemblies as to permit replacement of those likely to become defective, without extensive disassembly.

D-4k(2). Items which are subject to replacement or servicing shall not be permanently secured by rivets, welding or other means which prohibit ready removal.

D-4k(3). Heads of screws or bolts which hold parts on the back of a panel shall have a different type of head in order to distinguish them from the screw heads used for securing the panel itself.

D-4*l*. Design for production.- The design, layout, and assembly of the units and their component parts shall be such as to facilitate quantity production.

D-4m. Rounded corners and edges.- All edges and corners of external surfaces, normally exposed, shall be rounded to prevent injuring personnel. Sharp surfaces of any kind shall be avoided.

D-4n. Shock and vibration mountings.- Shock and vibration mountings, where employed in the design of the equipment shall be in accordance with the requirements of Navy Department Specification 17-I-61.

D-4o. Resistance to shock.- All equipment covered by this specification except portable components shall be in accordance with the requirements for type A of Navy Department Specification 66S3.

D-4p. Resistance to vibration.- All components covered by this specification shall conform to the requirements and tests for vibration resistance as prescribed in Navy Department Specification 40T9. In addition, the component shall be energized for normal operation for the duration of the test. There shall be no generation of microphonics when measured in accordance with the requirements of paragraphs E-2b(1)b(2)g, E-2b(1)c(8), and E-2b(1)d(7); there shall be no loosening, breakage, or mal-adjustment of screws, bolts, electrical connections or other parts; and there shall be no adverse effect upon the performance of the component.

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D-4q. Resistance to salt spray.- All microphone equipment (except microphone control box, type IC/MBT) and all loudspeaker equipment (except headset, type IC/SJA, jack box, type IC/SKG and transformers, type IC/SLA) shall withstand the salt spray test described in paragraph F-3f(16). Following the salt spray test, close examination of the disassembled equipment shall reveal no corrosive action or damage to parts due to the test. The equipment shall function satisfactorily upon being reassembled.

D-4r. Resistance to moisture.- All insulating materials which are cut or machined shall have such surfaces sealed against moisture absorption before further assembly

D-5 General design requirements (electrical).-

D-5a. Wiring.-

D-5a(1). Hook-up wire.- The individual conductors used in wiring of components shall be of stranded construction, of suitable cross-section to provide ample and safe current-carrying capacity and mechanical strength, and shall be of one of the following types:

Type SRIB, SHFS, DHFW, SRI of Navy Department Specification 15C1;
Type SRIR, SRHV, SRRF of Joint Army-Navy Specification JAN-C-76.

D-5a(2). Lacing.- Wherever practicable, groups of two or more conductors shall be bundled and laced.

D-5a(3). Sleevings or tape.- All cable forms connecting between a fixed and a movable portion of an assembly shall be provided with sufficient slack and shall be covered with a protective sleeving or tape (of fire-resistant, non-hygroscopic material) to eliminate the possibility of chafing or mechanical abrasions on the surface of any individual conductors.

D-5a(4). Cable clamps.- Cable clamps shall be provided at frequent intervals along the length of all cable forms to secure cable runs against vibration or undue movement. Such clamps shall be adequately cushioned to prevent damage to conductor insulation.

D-5a(5). Grommets.- Suitable grommets shall be used in holes through which cable forms or conductors pass.

D-5a(6). Pre-formed cables.- The use of pre-formed cables in the wiring of equipment is preferred to the point-to-point method of wiring.

D-5a(7). Wiring to terminals.- As far as practicable, each connecting wire from an electrical part shall be so connected to a connecting terminal or lug, that other wires will not interfere with the removal of that wire from the terminal.

D-5a(8). Color coding.- Where color coding is used for chassis wiring, it shall be in accordance with the requirements of Navy Department Specification 17-I-61.

D-5b. Mounting of parts.- All small parts, such as resistors and capacitors which are mounted by their leads shall be mounted on terminal or connection boards. Kinks or other slack shall be provided in the mounting leads to permit thermal expansion of the terminal or connection board without stressing the components. Large and medium large components shall be securely fastened by clamps. Long resistors shall be fastened at both ends. Ceramic or composition resistors which are secured by screws shall have pliable washers inserted under the screws to prevent undue stress on the resistors.

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D-5c. Shielding and radio frequency noise interference.-

D-5c(1). Objective.- The equipment design, construction and shielding (particularly of wiring and contacts) shall be such as to:

Shield amplifier input circuits and low level contacts from the effects of stray electromagnetic and/or electrostatic fields.[1]

Prevent radiation or conduction from any part of the equipment of radio-frequency energy in excess of the limits specified in Navy Department Specification 16E4, when measured as described therein.[1]

D-5c(2). Bonding.- Where access doors or cover plates which form a part of the shielding are required, suitable bonding shall be provided. (Braid jumpers and captive screws are considered satisfactory. Hinges are not considered satisfactory conducting means for bonding purposes.)

D-5d. Segregation of low level circuits.-

D-5d(1). Care shall be taken to separate physically amplifier input circuits from power and output circuits.

D-5d(2). Terminal boards for ships cables shall be so located as to provide maximum separation between cables carrying low level circuits and all other cables.

D-5e. Grounds.-

D-5e(1). No electrical circuit which connects directly to external ships wiring shall be grounded to the equipment or to ships hull.

D-5e(2). All internal amplifier chassis grounds shall be isolated from input, output and power circuits by means of transformers.

D-5f. Terminal boards.-

D-5f(1). General.- Terminal boards, satisfactory to the bureau concerned, shall be provided for all external connections to a piece of equipment. Terminal board designs employing screws threaded into the board from the back shall be avoided.

D-5f(2). Mounting.- Terminal boards for receiving ship's cable shall be secured to a fixed portion of a piece of equipment and not to hinged panels or other movable members, which may require flexing of ship's wiring during normal operation, maintenance or servicing.

D-5f(3). Marking.- All terminal boards shall be marked in a clear and permanent manner as shown on figures 9 to 16 inclusive. The markings shall be so located as to be visible from the normal servicing position.

D-5g. Soldering.-

D-5g(1). General.- All soldering shall be effected in a neat and workmanlike manner and the surfaces surrounding the soldered connections shall be free from splattering of resin residues. All soldered connections shall be of such character and quality that the bonding between the soldered items may be determined by visual inspection. There shall be no evidence of "cold soldering", and the use of excessive amounts of solder will not be acceptable.

[1] Ships wiring inter-connecting the installed equipment will be shielded, grounded and segregated in accordance with the requirements of section S62-2 of the General Specifications for Machinery.

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D-5g(2). Strength of soldered joint.- In general, soldering alone shall not be depended upon for a satisfactory connection, but where wires and terminals are joined to be soldered, the wires shall be hooked, wrapped around or otherwise mechanically secured to the terminals prior to soldering.

D-5g(3). Multiple soldered connections.- No soldered connection shall terminate more than three leads including leads from component parts. Multi-ended or equivalent solder type terminals or lugs shall be employed wherever feasible to reduce to a practical minimum the number of leads terminating at any one soldered connection.

D-5g(4). Soldering flux.- Only pure resin or a mixture of alcohol and resin (or equivalent) shall be used as a flux unless the use of other fluxes of a demonstrated suitable nature is satisfactory to the bureau concerned. Acid or corrosive fluxes shall not be employed.

D-5h. Temperature.-

D-5h(1). Operating temperature.- All components that are installed in exposed locations shall be capable of normal operation over an ambient temperature range of - 35°F. to + 180°F. All amplifier components shall be capable of normal operation over an ambient temperature range of +40°F. to +130°F. All loudspeaker components for interior applications shall be designed for normal operation over an ambient temperature range of +40°F. to +180°F.

D-5h(2). Temperature rise.- The temperature rise of any part of the amplifier components, except electronic tubes and resistors shall not exceed 80°F. at any ambient temperature up to 130°F. Resistors shall not exceed a final temperature of +300°F.

D-5i. Clearances.- Clearances between any two electrical circuits or between an electrical circuit and ground shall be such as to meet the requirements of paragraph D-5j for dielectric strength and insulation resistance. For circuits in which the voltage exceeds 150 volts between conductors or between a conductor and ground the clearances shall be not less than 1/4 inch unless separated by a minimum of 1/8 inch of insulating material.

D-5j. Dielectric and insulation characteristics.-

D-5j(1). Dielectric strength.- All electrical circuits, with the exception of microphone and loudspeaker voice coil circuits, shall be capable of withstanding the dielectric test specified in paragraph F-3f(20).

D-5j(2). Insulation resistance.- The insulation resistance of electrical circuits following the dielectric strength test specified in paragraph F-3f(20) shall not be less than 25 megohms at 500 volts D.C. at approximately room temperature (75°F.) and at a relative humidity of approximately 50 percent.

D-5j(3). Loudspeaker voice coil circuits.- Loudspeakers shall be capable of withstanding a 150 volt D.C. voltage between voice coil and magnet structure while a signal voltage of value specified below is applied to the voice coil and warbled from 50 to 5000 cycles per second in not less than 15 seconds. The leakage current shall not exceed 4 milliamperes (0.004 amperes).[1]

D-5j(3)a. For horn type loudspeakers where the lower cut-off frequency is above 50 cycles per second, the voltage applied to the voice coil shall be a value of 1/2 "rated voice coil voltage".

D-5j(3)b. For all other types of loudspeakers, the voltage applied to the voice coil shall be a value of full "rated voice coil voltage".

[1]Leakage current will be limited to a maximum value of 5 milliamperes (0.005 amperes) to prevent destruction of the voice coil.

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D-5j(3) c. "Rated voice coil voltage" for the above purpose shall be defined as that voltage at the voice coil terminals corresponding to rated voltage (see par. E-3c(1)), applied to the normal loudspeaker input terminals.

D-5k. Interchangeability.- In the design of the equipment, due precaution shall be taken to provide for design tolerances sufficient to accommodate, without departure from the specified performance, various parts such as tubes, resistors, capacitors, etc., having the limiting dimensions and nominal characteristics of the applicable specifications for the particular part involved.

D-5*l*. Serviceability.- All parts shall be so placed and secured as to permit removal, or replacement, without disturbing or removing any other part. (This requirement is not applicable to sub-assemblies which may be removed as separate units for servicing, replacement, etc. provided removal or replacement can be effected without damage to other parts or to the wiring). Terminal panels, containing suitably mounted small parts may be considered as unit sub-assemblies and shall comply with the above requirements to the extent that the external wiring to the several terminals shall possess sufficient slack to permit the complete detachment of the panel from the mounting with adequate clearance for servicing or replacement of any or all parts mounted thereon.

D-6. Electrical parts.-

D-6a. General.- Electronic tubes, resistors, capacitors, transformers, inductors, relays, terminal connectors, fuses, voltmeters, ammeters, switches and lamps shall be in accordance with the requirements of the applicable specifications, or drawings listed in section A. The use of standard components, covered by Joint Army-Navy specifications listed in section A, is mandatory wherever those specifications are applicable to the components used.

D-6b. Electronic tubes.-

D-6b(1). Preferred list.- Unless otherwise specifically approved all electronic tubes shall be chosen from the Joint Army-Navy Preferred List of Electron Tubes.

D-6b(2). Mounting.- All electronic tubes shall be mounted in a vertical position, and shall be securely held in their sockets and adequately cushioned to withstand severe shock as well as normal ship vibration.

D-6b(3). Artificial cooling.- Artificially cooled vacuum tubes shall not be used.

D-6c. Electron - Tube sockets.- Tube sockets shall be of the single-unit type. Gang type sockets are not acceptable. (This requirement is not intended to preclude the assembly of several tube sockets on a supporting frame so long as the contact clips for any socket are on insulating material which is not continuous with that of any other socket.) Contact springs shall have positive contact and positive action. Socket contacts shall be silver plated phosphor bronze or beryllium copper. Sockets shall securely retain tubes under all service conditions, or shall be provided with suitable tube clamps.

D-6d. Capacitors.- The use of electrolytic capacitors is not acceptable. The normal working voltage applied to any capacitor shall not exceed 75 percent of the standard rating.

D-6e. Resistors.- All resistors shall be capable of continuously dissipating, without damage, 50 percent more power than the maximum they will dissipate in normal use of the equipment in an ambient temperature of 132°F. or less.

D-6f. Potentiometers.- Potentiometers or variable resistors employed as volume or gain controls shall be of wire-wound construction and shall provide facilities for staking to obviate turning of the part when the shaft is rotated in service.

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D-6g. Transformers.-

D-6g(1). Input transformers.- Amplifier input transformers shall be completely enclosed in a high permeability alloy shield. Primary and secondary windings shall be separated from each other by one or more electrostatic shields, grounded as necessary to minimize the effects of longitudinal voltages.

D-6g(2). D.C. current carrying capacity.- All audio and power transformers and inductors which carry direct current shall be designed to carry in normal use, without damage at least 25 percent more direct current than the maximum they will carry in normal operation of the equipment.

D-6h. Relays.-

D-6h(1). Relays shall be designed for 115 volt, 60 cycles per second A C. operation. Power supply tolerances shall be as specified in paragraph D-2.

D-6h(2). Relays shall conform to the requirements for auxiliary type; service B operation, of Navy Department Specification 17R6.

D-6h(3). Relays shall operate satisfactorily, shall make firm, low-resistance contact, and shall retain their setting in any position of the equipment and under any service condition. The contacts shall be noncorrosive and, where practicable, of the self cleaning type. Relays shall be readily accessible for inspection, cleaning, adjustment or replacement.

D-6h(4). Relays, the contacts of which control power of 25 watts or less, shall be of the multiple contact type and shall be capable of one million operations at the rate of 120 operations per minute without exhibiting any erratic operation of the contacts in making or breaking the electric circuits and without requiring any repair or servicing. Relays, the contacts of which control power in excess of 25 watts and all ratchet type motors and/or other electromagnetic operated devices, shall be capable of 10,000 operations at the highest rate at which they are likely to be operated in service use, but in no case at less than two operations per minute, without requiring any repair or servicing. The foregoing requirements shall be met with the relay contacts loaded as they normally would be used in the equipment.

D-6h(5). When automatic sequencing or coding cycles are required, suitable precautions shall be taken in the design to prevent erratic operation under conditions of shock specified in paragraph D-4o.

D-6i. Terminal boards and terminals.- Terminal boards shall be of the screw type with suitable barriers between terminals. The design shall be such that the terminals cannot rotate in place after they have been secured. Terminal boards in equipment to which two or more external connections per terminal will be required should be fitted with studs in lieu of tapped holes to facilitate stacking of terminals while the terminals are being fastened during installation. (Solderless type terminals are preferred for all connections to terminal boards and for all other connections where wire terminals are used. Wire terminals for permanent connection to external wiring need not be provided with the equipment.) The associated terminal mounting nuts and screws shall not be assembled on the equipment but shall be furnished in bags of a suitable textile material, securely fastened to the equipment.

D-6j. Fuse holders.- Fuse holders shall be in accordance with the requirements of Bureau of Ships Drawing 9000-S6202-73440.

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D-6k. Fuses.- Fuses shall be of the Navy standard cartridge type suitable for use with the fuse holders specified in paragraph D-6j.

D-6ℓ. Switches.-

D-6ℓ(1). The handles or keys of rotary type selector switches shall be provided with mechanical stops to prevent undue strain on the switch handle and shaft, and to prevent rotation of the fixed portion of the switch.

D-6ℓ(2). Handles of lever type switches shall be of metal construction.

D-6ℓ(3). Switch knobs and handles shall be positively secured to the shafts and shall not rely on friction of a set screw for operation. This shall be accomplished by use of a square (or other non-circular) shaft with a mating hole in the knob or handle. Set screws holding the knob or handle to the shaft shall thread directly into the shaft.

D-6ℓ(4). No switch shall rely on the use of a spring for operation. In toggle-type switches, springs may be employed to accelerate the switch action, but operation of the switch shall be possible in the evenf of spring failure.

D-6m. Indicator lamps and lenses.-

D-6m(1). Lamps.- Lamps shall conform to the requirements of Bureau of Ships Drawing 9-S-4222-L. Type VG-7 neon lamps, with suitable shunt resistance to prevent spurious indications due to voltages induced in ship's inter-connecting cables, shall be used as busy indicator lamps in microphone equipment.

D-6m(2). Lenses.- Standard lens colors shall be used as follows:

White - for "power on" indicators
Red - for "busy" indications

D-6n. Volume indicator meters.-

D-6n(1). General.- Volume indicator meters shall indicate proper amplifier output. The meter pointers shall be suitably damped to prevent excessive fluctuations during normal speech transmission.

D-6n(2). Scale calibration.- The scale shall be calibrated in decibels relative to rated amplifier output voltage. The range shall be from -12 db to +6 db. A 1000 cycle per second signal of 1.73 volts r.m.s. applied to the meter terminals shall correspond to rated amplifier output voltage in accordance with paragraph E-2b(2)a(3)b and shall produce a meter reading of "0" decibels.

D-6n(3). Markings.- Dial markings and pointer shall be painted with a suitable self-luminous paint. The dial shall have a white background with black markings. A color band shall be provided below the numerals with black in the area corresponding to -12 db to -3 db and from +3 db to +6 db. The area of the band from -3 db to +3 db shall be white and shall be marked "normal".

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D-6n(4). Meter sensitivity.- Meter sensitivity shall be a minimum of 2000 ohms per volt for full scale deflection.

D-7. Marking of parts.- Parts, such as transformers, capacitors, resistors and relays, shall be identified by markings both on the part and adjacent to the part on the chassis or board on which the part is mounted. The marking shall be that designation assigned in the associated schematic wiring diagram, and shall be clearly and indelibly made.[1]

D-8. Engineering services.-

D-8a. Services required.- When included as an item in the contract or order (see par. H-1a), "engineering services" shall cover the services of a competent engineer to serve in the following capacity:

D-8a(1). Engineering supervision of installation and/or consultation as required by the Navy or shipbuilding activity. This shall include supervision of loudspeaker locations to insure most efficient and effective sound coverage.

D-8a(2). Inspection of the system immediately after installation is completed.

D-8a(3). Engineering services in connection with the trial of the system for a period not to exceed 15 consecutive days within 18 months of commissioning of ship, to observe operation, check loudspeaker ratings with actual shipboard noise levels, and supervise any changes which may be authorized during this period.

D-8b. Report of services.- The contractor shall submit a report to the bureau concerned, for each installation with copy to the cognizant Inspector of Naval material. Payments for engineering services will be authorized upon receipt of this report. The report shall contain the following information:

(1) Services performed.
(2) Details of any changes made in the equipment after delivery, and the reasons for such changes.
(3) Suggestions for any design changes, which if made in later equipments would facilitate future installation and/or operation.
(4) Comments and recommendations for improving the techniques employed by the shipyard in accomplishing the installation.
(5) Certification, by the Officer-in-Charge of the Shipyard, of services performed.

D-8c. Multiple manufacture.- In such cases where the equipment is comprised of components procured from more than one manufacturing source, the responsibility for the services outlined above shall be vested in the manufacturer of the amplifier components.

D-9. Definitions.-

D-9a. Definitions of terminology.-

D-9a(1). Part.- A part is a single item which cannot be further disassembled.

Examples: Capacitors, resistors, electron tubes, bolts.

[1]Where the size of the part does not permit marking, as on small resistors, markings adjacent to the part will be sufficient. Markings on the chassis only will also be sufficient for electronic tubes, and should include the Joint Army-Navy Tube type as well as the schematic designation.

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D-9a(2). Assembly.- An assembly is a mechanical grouping of two or more parts fixed or joined together into one item which performs a limited function subordinate to the functions of the component.

Examples: Low power amplifier, signal generator.

D-9a(3). Component.- A component is a mechanical grouping of parts and/or assemblies and/or accessories fixed or joined together into one item which must be mechanically or electrically connected with one or more components to perform its intended function.

Examples: Loudspeaker, portable microphone, jack box, amplifier rack, control rack.

D-9a(4). Equipment.- An equipment consists of one or more components and/or accessories and/or attachments complete within itself to perform its intended function when properly installed.

Examples: Shipboard announcing equipment including components as follows: portable microphone, microphone jack box, amplifier, loudspeaker.

D-9a(5). System.- Equipment properly installed and interconnected by ships wiring so as to be capable of performing its intended function.

D-9a(6). Accessory.- An accessory is a device, mechanism, or machine which is used with a component or equipment to aid, control, or supplement the functions of that component or equipment or to assist in testing or adjustment.

D-9a(7). Attachment.- An attachment is an item which is attachable as an entity to another item for the purpose of extending the other item's range of functions.

D-9b. Definitions of degree of enclosure.-

D-9b(1). Pressure proof.- A component shall be considered pressure proof when submerged in water under an external hydrostatic pressure of 450 pounds per square inch without the entry of water into the case. Tests shall be conducted in accordance with the methods described in Navy Department Specification 17E11, except that the test pressure shall be 450 in lieu of the 300 pounds per square inch.

D-9b(2). Watertight.- A component shall be considered watertight, when submerged in water under a pressure equivalent to a head of 15 feet for a period of 30 minutes without the entry of water into the case. Tests shall be conducted in accordance with the methods described in Navy Department Specification 17E11.

D-9b(3). Waterproof.- A component shall be considered waterproof when exposed and submerged under 3 feet of sea water (specific gravity 1.025) for a period of one hour, and is capable of normal operation after natural drainage while held in the normally mounted position.

D-9b(4). Dripproof.- A component shall be considered dripproof when exposed to drops of liquid or solid particles falling on it any any angle not greater than 45 degrees from the vertical, without any interference with the operation. All openings shall be constructed so as not to permit the passage of a rod 1/2 inch in diameter through the opening. Tests shall be conducted in accordance with tne methods described in Navy Department Specification 17E11.

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D-10. Nameplates and instruction plates.-

D-10a. Equipment nameplates.- Each component to which a type number is assigned shall be provided with a nameplate, mounted in a conspicuous location on the component. Nameplates shall be type 2 in general accordance with Bureau of Ships Drawing 9000-S6505-73774 and Navy Department Specification 42N2. Nameplate markings shall be as follows:

(1) First line:- "Shipboard Announcing equipment."
(2) Second line:- Name of component and Navy type designation. (To be furnished by the bureau concerned not later than the time of approval of working plans).
(3) Third line:- Bureau of Ships stock number. (This number may require as many as seventeen spaces). To be furnished by the bureau concerned at the time that working plans are approved.
(4) Fourth line:- Salient design characteristics, rating and/or electrical data (as applicable), such as voltage, frequency, number of phases, current, power, impedance.
(5) Fifth line:- "Navy Department - Bureau of Ships".
(6) Sixth line:- Manufacturer's Name
(7) Seventh line:- Contract number; blank space for Naval inspector's official stamp and serial number.

D-10b. Spare part box nameplates.- Spare parts boxes shall be provided with a nameplate conforming to the requirements of paragraph D-10a. Where a set of spare parts consists of two or more separate boxes, all such boxes shall carry identical nameplates, except that the several boxes shall have additional markings on each nameplate to indicate "Box 1 of 3 boxes", "Box 2 of 3 boxes," etc.

D-10c. Designation plates.- Designation plates such as "Power On", "1MC-BUSY" and loudspeaker group identification plates shall be made of nickel-copper alloy in accordance with the requirements of Navy Department Specification 46M7 for any components which may be used in exposed locations. Designation plates for components to be used only in protected locations may be made of laminated phenolic or molded plastic material.

D-10d. Diagrams and instruction sheets.- Wherever practicable, schematic wiring diagrams and/or abridged instruction sheets shall be supplied with and secured to assemblies or components. Such diagrams and instruction sheets shall be of metal or plastic, permanently marked and securely fastened.

D-11. Plans.-

D-11a. Preliminary plans (formerly type A).- The contractor shall submit preliminary plans, if requested by the bureau concerned, for the advance information of the shipbuilding activity or ship design agent pending the submission of working plans. These plans, shall consist of the following:

D-11a(1). A cable diagram showing the cable runs required between components of an equipment, the number of active conductors per cable and the currents and voltages involved.

D-11a(2). Outline plans of the equipment components showing overall dimensions, mounting dimensions and size of mounting holes.

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D-11b. Working plans (formerly type B).- Working plans shall be prepared and submitted in duplicate to the bureau concerned for approval as soon as practicable after award of contract. Any work done by the contractor in advance of working plans approval is at the contractor's own risk.. Where changes are required, copies shall be submitted in duplicate for final approval. Plans shall conform in size, arrangement, title, form, etc. to the requirements of section S1-1 of the General Specifications for Machinery. The following plans and plan information shall be required:

D-11b(1). Overall schematic diagrams.- A complete electrical schematic diagram of the system (when the contract covers a complete equipment or the amplifier portion thereof) shall be drawn to represent clearly the operation of each component and the proper inter-connections between components.

D-11b(1)a. A thin broken line shall be used to represent the boundaries of each component all terminals to which external connections are made shall be shown within these boundaries, with the appropriate numbers and markings.

D-11b(1)b. Relay contacts and switch sections and contacts shall be numbered and lettered to conform to the requirements of the standard system shown on figure 1. Markings of all switch positions shall be shown, and the position of each switch corresponding to that on the plan shall be indicated.

D-11b(1)c. Each part, such as resistors, capacitors, relays, etc., shall be given a unique designation in accordance with the requirements of the system shown on figure 2.

D-11b(1)d. A separate list giving electrical rating and manufacturer's name and Service Part Number for each part shall be included with the schematic for plan checking purposes only. (This list need not appear on the final plan which will be reproduced in the instruction book, inasmuch as the information will be furnished in greater detail as part of the text.)

D-11b(1)e. Assemblies, such as amplifiers and signal generators, for which a separate schematic diagram is supplied, may be represented by a block with terminals shown and appropriately marked.

D-11b(1)f. The schematic wiring of a component or assembly which appears several times in the system need be shown only once; blocks with terminals shown and marked will suffice for the remaining identical components or assemblies.

D-11b(1)g. A system of differentiating between audio and control circuits such as one employing heavier weight lines for audio input and output circuits, shall be employed.

D-11b(1)h. Appropriate notes shall be added to the plan giving the following information:

(1) Power requirements under "ready" and "standby" conditions
(2) Heat dissipation under "ready" and "standby" conditions.

D-11b(2). Schematic diagrams of assemblies.- Separate schematic diagrams shall be drawn for each major assembly such as amplifier assemblies and signal generator assemblies, following the general form specified in paragraph D-11b(1). Additional notes shall be added to the plan giving the following information as applicable:

(a) Rated output power (or voltage in the case of signal generator assemblies) at maximum distortion of 10 percent.
(b) Input and output impedance (at 1000 cps).
(c) Amplifier gain (at 1000 cps).
(d) Frequency response characteristic.

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D-11b(3). Wiring diagrams.- Separate wiring diagrams shall be drawn for each component such as amplifiers and control racks and each major assembly such as amplifier assembly or signal generator assembly. Wiring diagram for components such as loudspeakers or microphones may be included on the respective mechanical assembly plans.

D-11b(3)a. Wiring diagrams shall show all parts in correct relation to physical location, with terminals clearly shown and marked, and shall show all wiring between parts.

D-11b(3)b. The color code of all wires shall be indicated. Appropriate notes shall designate wire size and type.

D-11b(3)c. Designations of parts shall correspond to those assigned in the schematic diagrams.

D-11b(4). Mechanical assembly plans.- Mechanical assembly plans shall be drawn for each component and for each major assembly.

D-11b(4)a. Assembly plans shall contain sufficient views in plan, elevation and cross section drawn to scale to show clearly the details of mechanical design, construction and assembly of the component.

D-11b(4)b. Each part shall be numbered, indicated and listed as follows:

(1) Piece number
(2) Name and brief description of piece, (including such information as gage thickness, thread data, length, etc.).
(3) Quantity required.
(4) Material.
(5) Applicable specifications.
(6) Remarks (containing references to notes specifying surface finish or treatment).
(7) Contractor's Service Part Number.
(8) Manufacturer's name and Service Part Number for parts not manufactured by the contractor, (this information may be omitted for those parts shown on schematic diagrams, since it will be available on the list required by paragraph D-11b(1)d.)

D-11b(4)c. Appropriate notes shall be contained on the plans to cover any special methods of assembly, special treatments, use of anti-sieze compounds, and use of lubricants in accordance with the requirements of paragraph C-10.

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D-11b(4)d. The following data shall also be included:

(1) Weight
(2) Overall dimensions
(3) Mounting dimensions and size of mounting holes
(4) Clearances required for servicing as for example:
When hinged panels or doors are employed.
(5) Accoustical data, such as impedance and sensitivity of microphones; rated output sound pressure and corresponding power input for loudspeakers; frequency response for loudspeakers and microphones; polar distribution of sound pressure for loudspeakers.
(6) Electrical data, such as 1000-cycle impedance, d.c. resistance, number of turns, and size and type of wire for loudspeaker voice coils and microphone windings, operating frequency and voltage for amplifiers and signal generators.
(7) Air gap data for loudspeakers and microphones
(8) Electrical schematic diagrams and wiring diagrams for loudspeakers and microphone components. (These may be supplied as separate plans at the option of the contractor.)

D-11b(4)e. A drawing of each component nameplate shall be included on the plan.

D-11b(4)f. The markings of all designation and instruction plates shall be shown on the plan.

D-11b(5). List of spare parts.- The list of spare parts shall be made on a single plan in the form of separate 8 x 10-1/2 inch blocks, conforming in general with the requirements of Joint Army-Navy Specification JAN-B-233. An additional column shall be provided for reference to the box number in which each piece is located, where more than one spare parts box is required. The reference designations assigned on the schematic diagram shall be included as additional identification for each piece. There shall be included on a separate block, a table of weights and dimensions of each spare parts box.

D-11b(6). Clarity and legibility.- In the preparation of working plans,the clarity and legibility shall be such that the plans may be satisfactorily reduced in size to approximately 11 inches in height for insertion in instruction books.

D-11b(7). Design changes after plan approval.- After working plans have been approved, no significant changes in design or construction shall be made without the approval of the bureau concerned.

D-12. Instruction books.- Instruction books shall be type B(unclassified) in accordance with the requirements of Navy Department Specification 35B2, and General Specification for Machinery Section S1-5, except that the text, instruction matter, plans and drawings shall be presented in the following general form and order:

D-12a. Title page.- The title page shall contain the NAVSHIPS and interior communication instruction book numbers, title of the equipment and systems covered, vessels covered, contract number, manufacturer's name and address.

D-12b. General data.- The following general data shall follow the title page:

D-12b(1). Safety warning.-

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D-12b(2). Contract guarantee.-

D-12b(3). Component list.- This shall consist of a table containing:

a. Quantity per equipment
b. Description of item
c. Navy type designation
d. Bureau of Ships Stock Number
e. Manufacturer's model number (optional)
f. Dimensions
g. Weight (with and without packing)

D-12b(4). Electronic tube complement.- This shall consist of a list containing the following:

a. Tube type
b. Description of component in which used.
c. Type designation of component in which used.
d. Quantity per component.
e. Quantity of components.
f. Total number of tubes.

D-12b(5). Microphone technique.- Brief general instructions on microphone technique shall be given. Two duplicate detachable copies shall be provided and shall be marked "Detach for use in Microphone talker training". For systems employing super-power loudspeakers subject to acoustical feedback under operating conditions, this section shall include additional pertinent advice.

D-12c. Table of contents.- The table of contents shall list all sections and subsections with corresponding page numbers. A list of plans, drawings and photographs with corresponding figure numbers shall also be given.

D-12d. Section 1 introduction.- This section shall include appropriate introductory comments, and definitions of technical terms employed in the text.

D-12e. Section 2 system description.- This section shall describe completely the system and circuit functions. All relay and switching operations under the various possible conditions of use shall be explained. Busy light functions and relative precedences of voice and signal circuits shall be covered. (For complex circuits, tables or charts should be used freely to present for ready reference, switch and relay functions, control panel set-ups, etc.).

D-12f. Section 3 installation.- This section shall include general recommendations and instructions on shielding, grounding, bonding and other good practices to be followed in the installation of low level circuits. Mounting instructions for the equipment components shall be given. General principles of loudspeaker mounting, location and orientation shall be outlined.

D-12g. Section 4 adjustments and tests.- This section shall contain instructions for the adjustment and test of the system and its major components upon initial installation or under other conditions where complete system readjustment may be required.

D-12h. Section 5 operation.- This section shall contain simple, brief, and effective instructions for operation of the system, including normal routines and precautions to be observed in operating the equipment. (The use of tables and charts for the presentation of these instructions is highly desirable.)

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D-12i. Section 6 maintenance.-

D-12i(1). Preventative maintenance.- This section shall cover all maintenance procedures and routine adjustments, which should be performed periodically and regularly for the purpose of preventing failure or impairment of equipment. Included in this section shall be routine maintenance check charts containing the following:

A tabulation of periodic routine mechanical and electrical tests and checks which should be accomplished regularly to insure continuity of service at peak performance.
Arrangement of the table should be such as to indicate what is to be done, and how to do it.
Emphasis should be placed upon the test facilities which are incorporated in the amplifiers and control racks.
Instructions shall be provided on the care, cleaning and lubrication of relays, switches and moving or rotating parts.

D-12i(2). Corrective maintenance.- This section shall cover all information necessary to permit a technician to locate trouble and to make repairs or adjustments to each component, assembly or sub-assembly of the equipment. Included in this section shall be the following:

Trouble shooting guides for the localization and location of faults giving possible sources of trouble, their symptoms, probable causes, and instructions for remedying the faults.
Complete instructions on signal tracing, use of test instruments and other common servicing techniques.
Ample illustrations, photographs, exploded views giving details of mechanical assemblies, and simplified schematic diagrams of the electrical circuits.
Complete description of each system component, and the associated assemblies or sub-assemblies. This shall include adequate information to convey an understanding of the function and operation of the equipment, with specific maintenance and servicing instructions. Descriptions of amplifier and signal generator assemblies should provide an explanation of the electronic circuits. A preferred method of describing amplifier circuits is to present the description in sections, such as amplifier features, power circuits, main audio transmission path and mechanical arrangements. Theory of operation should be included where unusual or unconventional circuits are involved.
Voltage and resistance diagrams or tables for each electronic assembly showing normal voltages (with and without audio signal) and resistances as measured at the terminals of each tube socket and at other significant points in the circuit.

D-12j. Section 7, list of repair parts.- There shall be included a complete list of all repair (replaceable) parts containing the following information in tabular form. This list shall be entirely separate from the detailed parts lists required on mechanical assembly plans and the list of spare parts required in paragraphs D-11b(4)b and D-11b(5) respectively. Any sub-assembly which cannot be readily disassembled in the field, due to pressed fits, jigged assembly, critical adjustment, or similar reason shall be supplied as a single repair part for the purpose of this list: (The provisions of Navy Department Specification 35B2 which require illustrations of each assembly, sub-assembly and their component parts do not apply.)

Reference designation assigned in the schematic diagram.

Name of part and complete description in accordance with the patterns established in Joint Army-Navy Publication JANP-109, except that data required to appear in other columns shall not be duplicated here.

Function - This shall consist of a brief statement of the use, purpose, or function of the part in the component.
Joint Army-Navy type number. -
Actual manufacturer's name.
Actual manufacturer's Service Part Number.
Contractor's Service Part Number.
Navy stock number (blank) unless available).

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D-12k. Section 8, Illustrations - plans, drawings, and photographs.-

D-12k(1). Plans.- Reproductions of all approved working plans shall be contained in this section. Additional block diagrams or explanatory drawings, should be included as necessary to supplement the descriptive matter contained in the text. Wherever feasible, such diagrams and sketches should be inserted into the text as close as possible to that portion of the text to which they apply. The use of different colors for such diagrams and drawings is highly recommended. All plans which will be used for reference purposes while reading the text shall be provided with a blank section the same size as a page at the left hand edge of the plan. This will permit the plan to be withdrawn clear of the book while the text is being studied. Plans shall be reproduced on a page the same height as the printed pages, in order that all folds will be parallel to the bound edge of the book.

D-12k(2). Drawings.- Diagrams of switches and relays used in the system showing the terminal numbering shall be inserted as additional drawings. The standard color codes for resistors and capacitors shall be included.

D-12k(3). Photographs.- Photographs of all components shall be inserted in this section unless they have been included under section 6.

D-12ℓ. Blank pages.- Five blank pages shall be inserted at the end of the book for memorandum purposes.

D-12m. Numbering of sections, pages and figures.-

D-12m(1). Numbering of sections.- Sections shall be numbered as indicated above. Subsections and paragraphs shall be numbered following the decimal system; e.g. under section 2, subsections should be numbered 2.10, 2.20 etc., and main paragraphs under subsection 2.10 should be numbered 2.11, 2.12 etc.

D-12m(2). Numbering of pages.- Pages shall be numbered consecutively.

D-12m(3). Numbering of illustrations.- Plans, drawings, and photographs shall be marked consecutively with figure numbers in addition to the plan numbers and titles. The figure number should appear in the lower right corner of the page when it is folded into the book.

D-12n. Instruction book approval.- Preliminary instruction books shall be submitted to the bureau concerned for approval and assignment of instruction book numbers, in ample time to permit printing prior to the delivery date of the equipment. Except as specified below, equipment will not be accepted for delivery without finished instruction books. In emergencies, and only when specifically approved by the Bureau, basic instructions in typewritten or mimeographed form and blueprint copies of schematic diagrams will be accepted as a temporary substitution for finished instruction books.

D-12o. Instruction books for components.- The above instruction book requirements cover the general case where an entire equipment, comprising microphone, amplifier and loudspeaker components is to be described. In cases where only one component or type of component, such as microphones or loudspeakers, is involved, a separate instruction book prepared in accordance with the applicable requirements above shall be prepared for that type of component. When such an instruction book is prepared covering amplifier components, additional information of a general nature shall be included to cover the principles of operation, installation and maintenance of the microphone components, the loudspeaker components, and the overall aspects of the complete system.

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D-12p. Additional instruction books.- In addition to the quantities of instruction books which may normally be required, instruction books shall be supplied as follows:

(1). One copy of the applicable instruction book shall be packed in each amplifier and control rack so that it will not become detached from the equipment until the amplifier or rack is opened for installation.
(2). One copy of the applicable instruction book shall be packed in each spare parts box. (See par. D-13a(18).)

D-12q. Abridged instruction sheets.- An abridged instruction sheet containing a summary of salient instructions for proper installation and operation of the component, shall be packed with each microphone and loudspeaker component.

D-13. Spare parts sets.-

D-13a. Spare parts.- Spare parts shall be furnished on a per equipment basis as follows. (The quantities of components which comprise an equipment for the purpose of determining the make-up of a spare parts set, shall be specified in the contract or order (see par. H-1a):

(1). Cone or diaphragm assemblies - 100 percent up to 5 of each type; in excess of 5, 1 for each 20 loudspeakers or fraction thereof.
(2). Loudspeaker driver units (consisting of magnet structure and cone or diaphragm assembly) - 1 for each 50 units of each type, or fraction thereof.
(3). Microphone units (complete with microphone case and press-to-talk switch - 1 for each 2 or fraction thereof up to 10 equipment microphones; in excess of 10, 1 for each 10 microphones or fraction thereof.
(4). Electronic tubes - 100 percent.
(5). Amplifier assemblies - (complete with electron tubes) - 1 of each type, where two or more are employed in the equipment.
(6). Signal generator assemblies (complete with electron tubes) - 1 of each type where two or more are employed in the equipment.
(7). Relays - 1 of each type plus 1 for each 25 relays or fraction thereof.
(8). Switches - 1 of each type switch conforming to the requirements of Bureau of Ships Drawings listed in paragraph A-3; 2 of each other type.
(9). Contact springs - 1 of each type when separate from relays and switches.
(10). Condensers, resistors and potentiometers - 2 of each type; resistors furnished for circuit protection - 10 percent.
(11). Transformers and inductors - 1 of each type; loudspeaker transformers - 1 of each type plus 1 for each 25 loudspeakers, or fraction thereof.
(12). Electronic tube sockets - 2 of each type.
(13). Plugs - 2 of each type.
(14). Jacks, receptacles and connectors - 1 of each type.
(15). Meters - 1 of each type.
(16). Lens and windows - 1 of each type.
(17). Loudspeaker blast valves - 1 for each 25 loudspeakers, where blast valve is not a part of the cone or diaphragm assembly.
(18). One copy of the applicable instruction book in each spare parts box.

D-13b. Test headset.- For test purposes there shall be included a headset (Navy magnetic type 49003 or equivalent) with a series capacitor, and fitted with combination test prods and insulated "alligator-type" clips.

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D-13c. Spare parts boxes.- Each set of spare parts shall be furnished in spare parts boxes, conforming to the requirements for type M of Joint Army-Navy Specification JAN-B-233.

D-13d. Marking of spare parts boxes.- All spare parts boxes comprising one set of spare parts shall carry identical nameplates except that the several boxes comprising one set shall have additional markings on each nameplate to indicate "Box 1 of 3 boxes", "Box 2 of 3 boxes", etc.

D-13e. Protection and identification of spare parts.- All spare parts shall be protected, and identified in accordance with the requirements of Army-Navy Joint Packaging Instruction JPI-14.

D-13f. Multiple manufacture.- Where the equipment is comprised of components procured from more than one manufacturing source, separate sets of spare parts will be required from each such source covering the items associated with that source.

D-13g. Bulk spares.- Within 30 days after bureau approval of working plans, a complete list of repair parts, prepared in general accordance with the requirements of paragraph D-12j, shall be submitted to the bureau concerned. This list will be used as a basis of procurement for setting up the initial stock of bulk repair parts in the ships parts supply system. Only by prompt submission of this list, will it be possible to initiate procurement in time to obtain the benefit and economy of quantity purchase and fabrication of such repair parts concurrently with the regular production of the basic equipment.

D-14. Qualification (type or brand approval).- Shipboard announcing equipment furnished under this specification shall be of a brand which has been tested and found suitable for the purpose intended as determined by the tests specified in section F (see also par. H-3).

E. REQUIREMENTS APPLICABLE TO INDIVIDUAL CLASSES AND TYPES

E-1. Microphones.-

E-1a. General descriptions of microphone types.-

E-1a(1). Basic microphone unit, type IC/MBU.- The basic microphone unit shall consist of a microphone element contained in a watertight case, provided with a press-to-talk switch, and fitted with a cable entrance gland. This unit will be used with or as a part of the microphone components described herein.

E-1a(2). Microphone control station, types IC/MSA and IC/MSB.- A microphone control station shall consist of a box designed for bulkhead mounting, with a front panel containing a microphone and all necessary switches, indicator lights and meter. The box shall be provided with a cover hinged at the bottom. When closed, the cover shall form a protective enclosure around the panel and shall restore the microphone to its normally secured position without possibility of damage to the cable or microphone. The cover latch shall prevent the cover from opening under shock. The front panel shall be loosely hinged to the box so that the panel will be supported, when the panel securing bolts are removed for servicing. The panel when in place on the box, shall form a watertight enclosure. The three types of microphone control stations accommodate different numbers of announcing circuits as follows, providing the necessary number of switches and indicator lights for each such application.

Microphone control station	Number of announcing circuits
Type IC/MSA	3
Type IC/MSB	2

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E-1a(3). Microphone control box, types IC/MBA, IC/MBB, IC/MBC, and IC/MBD.- A microphone control box shall consist of a watertight box, designed for bulkhead mounting and containing all necessary switches, indicator lights, meter and microphone jack. The cover shall be loosely hinged to the box so that the cover will be supported, when the box is opened for servicing. The four types of microphone control boxes provide different combinations of switches and indicator lights as follows, to accommodate various circuit applications:

Microphone control box	Number of selector switches	Number of indicator lights
Type IC/MBA	0	0
Type IC/MBB	5	0
Type IC/MBC	0	2
Type IC/MBD	5	2

E-1a(4). Microphone control box, type IC/MBT.- The microphone control box, type IC/MBT, shall consist of a drip-proof box with hinged front cover, designed for bulkhead mounting and containing all necessary components and control facilities to permit complete operation and control of a two-way announcing system, permitting selective outgoing calls to remote stations with return incoming calls from those stations.

E-1a(5). Microphone jack box, types IC/MJB and IC/MJT.- A microphone jack box, types IC/MJB and IC/MJT, shall consist of watertight box designed for bulkhead mounting and containing a microphone jack. The type IC/MJT jack box shall contain in addition an indicator light and a talk-switch to adapt it for talk-back use in two-way announcing systems.

E-1a(6). Microphone jack box, type IC/MJP.- The microphone jack box, type IC/MJP, shall consist of a pressure-proof box designed for bulkhead mounting and containing a microphone jack.

E-1a(7). Portable microphone, type IC/MPM.- The portable microphone, type IC/MPM, shall consist of a basic microphone unit, type IC/MBU, mounted on a chest plate with neck strap, and fitted with twenty feet of cable and a microphone plug.

E-1a(8). Handset, type IC/MPH.- The handset, type IC/MPH, shall consist of a watertight handset assembly containing a microphone element, a receiver element and a press-to-talk switch, and fitted with fifteen feet of cable and a microphone plug.

E-1a(9). Microphone, Type IC/MFM.- The microphone, type IC/MFM, shall consist of a basic microphone unit, type IC/MBU, mounted on the end of a flexible brass arm, 24 inches in length. The other end of the flexible arm shall be securely fastened to the cover of a watertight junction box designed for bulkhead mounting.

E-1b. Mechanical requirements.-

E-1b(1). Basic microphone unit, type IC/MBU.-

E-1b(1)a. Maximum size.- The microphone unit, exclusive of switch and gland shall not exceed the following dimensions:

	Inches
Diameter.	2-3/4
Depth.	2-1/2

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E-1b(1)b. Serviceability.-

E-1b(1)b(1). Microphone element.- The microphone element shall be defined as the microphone unit less the housing and press-to-talk switch.

E-1b(1)b(2). Ready replacement.- The microphone unit shall be so designed that the microphone element may be quickly removed and replaced without disturbing any electrical wiring and without exposing the magnetic assembly to foreign matter.

E-1b(1)b(3). Interchangeability.- All microphone elements shall be interchangeable within the housings of microphone units furnished under the same contract or order.

E-1b(1)b(4). Electrical connections.- Electrical connections to the microphone element shall be made with rolled-on silver or silver-plated contacts (plating to be not less than 0.0005 inch thick) or other materials satisfactory to the bureau concerned. These connections shall be held in contact by means of brass screws.

E-1b(1)b(5). Polarization.- The microphone element and unit housing shall be polarized as necessary to insure against incorrect insertion of the microphone element into the housing, with consequent damage to the contact arrangement.

E-1b(1)c. Air gap clearance - Minimum air gap clearance in magnetic or dynamic type microphones shall be 0.005 inches.

E-1b(1)d. Press-to-talk switch.-

E-1b(1)d(1). Contact arrangement.- The microphone housing shall be provided with a normally-open, double-pole, single-throw, spring return switch, which when operated will complete two separate control circuits, each capable of carrying 1.0 ampere at 115 volts, 60 cycles, a.c. with an inductive load having a power factor of 50 percent.

E-1b(1)d(2). Switch construction.- The switch button shall be insulated from the switch and its electrical circuits. The use of shielding and the extent of shielding shall be such as may be necessary to reduce electrical noise pickup by the microphone circuit to a value within the limit specified in paragraph E-1c(1)d.

E-1b(1)d(3). Switch action.- The switch button action shall be such than an indication of switch closure is given while operating the switch. The spring action shall be such that undue pressure will not be required to hold the button in the closed position.

E-1b(1)d(4). Switch endurance requirement.- The microphone switch shall after 100,000 operations at rated load as specified in paragraph E-1b(1)d(1), be capable of satisfactory operation on both voice and control circuits, and shall show no bending of springs or failure of parts.

E-1b(1)e. Resistance to gun blast.- The microphone unit shall at the conclusion of the gun blast test specified in paragraph F-3f(18), show no damage, and shall not vary in acoustic performance beyond the limits specified in paragraphs E-1c(1)b and E-1c(1)c.

E-1b(1)f. Microphone endurance requirement.- The microphone unit shall be capable of withstanding an endurance test of 100 hours operation when tested as described in paragraph F-3f(6), with a voltage of 0.087 volts (r.m.s.) connected to the voice coil circuit. There shall be no lowering of sensitivity or failure to meet other specification requirements specified herein, as a result of this test.

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E-1b(1)g. Mounting facility.- The design of the microphone unit shall permit mounting it readily and rigidly with its back to the bulkhead. The necessary mounting attachment associated with this feature shall be provided with each microphone. The mounting attachment with the microphone in place shall be capable of meeting the requirements for resistance to shock and vibration as specified in paragraphs D-4o and D-4p.

E-1b(2). Microphone stations and boxes.-

E-1b(2)a. Parts of microphone stations and boxes.-

E-1b(2)a(1). Tabulation of required parts.- The following table lists the required quantities of parts which make up the various types of microphone stations and boxes described in paragraph E-1a.

Parts for microphone stations and boxes

Part	Quantity per component										
Type of component IC/----	MSA	MSB	MBA	MBB	MBC	MBD	MBT	MJB	MJP	MJT
Microphone	1	1								
Volume indicator meter	1	1	1	1	1	1	1			
Busy indicator lights	3	2			2	2				1
Loudspeaker selector switches	8	5		5		5	10			
"Power On" indicator light							1			
Talkback disconnect switches							10			
"1MC Tie-In switch							1			
Fuses							2			
Talk switch	1	1								1
Microphone jack	1	1	1	1	1	1	1	1	1	1
Designation plates	*	*	*	*	*	*	*	-	-	*
Instruction plate	1	1	1	1	1	1	1	1		1

*Quantity as required.

E-1b(2)a(2). Description of parts.-

E-1b(2)a(2)a. Microphone.- The microphone shall be a basic microphone unit type IC/MBU, mounted in a yoke or other suitable adjustable mounting, designed to ensure maximum cable life by preventing strain from being placed upon the cable in any normal use of the microphone.

E-1b(2)a(2)b. Volume indicator meter.- The volume indicator meter shall meet the requirements of paragraph D-6n and shall serve to indicate proper amplifier output. On type IC/MSA and IC/MSB microphone control stations, the meter shall be mounted above the level of the microphone. On other types of microphone control boxes, the meter shall be located above the switches and microphone jack.

E-1b(2)a(2)c. Busy indicator lights.- The busy indicator lights shall meet the requirements of paragraph D-6m and shall serve to indicate that the corresponding circuit is in use.

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E-1b(2)a(2)d. Loudspeaker selector switches.-

E-1b(2)a(2)d(1). Type IC/MBT microphone control box.- Loudspeaker selector switches shall be of the double-pole, single-throw, self-locking type, and shall serve to connect the amplifier output directly to the corresponding remote loudspeakers or loudspeaker groups.

E-1b(2)a(2)d(2). All other types of microphone stations and boxes.- Loudspeaker selector switches shall be of the single-pole, single-throw, self-locking type, and shall serve to operate control circuits connecting the corresponding loudspeakers or loudspeaker groups to the output of the amplifier.

E-1b(2)a(2)e. "Power On" indicator light.- The "power on" indicator light shall meet the requirements of paragraph D-6m, and shall serve to indicate that power has been made available to the microphone control box.

E-1b(2)a(2)f. Talkback disconnect switches.- Talkback disconnect switches shall be of the double-pole, single-throw, self-locking type, and shall serve to disconnect the remote talk-back circuits from the amplifier input.

E-1b(2)a(2)g. "1MC Tie-In" switch.- The "1MC Tie-In" switch shall be of the double-pole, single-throw, self-locking type, and shall serve to connect the output of the "1MC" system to the remote loudspeaker bus.

E-1b(2)a(2)h. Fuses.- The fuses on the type IC/MBT microphone control box shall serve to protect the busy indicator light circuit to the remote type IC/MJT microphone jack boxes.

E-1b(2)a(2)i. Talk switch.- The talk switch on the type IC/MJT microphone control box shall be of the multi-pole, double throw type, and shall perform the following functions:

Transfer the associated loudspeaker or loudspeaker group from the output circuit of the amplifier to the input circuit; energize the control circuits specified in paragraph E-1c(4)a(3) and E-1c(4)a(4).

E-1b(2)a(2)j. Microphone jack.-

E-1b(2)a(2)j(1). Type IC/MJP microphone jack box.- The microphone jack shall be of the pressure-proof type, in accordance with the requirements of Bureau of Ships Drawing 9000-S6501-73463, and shall be mounted on the cover of the box.

E-1b(2)a(2)j(2). Type IC/MSA and IC/MSB microphone control stations.- The microphone jack shall be of the watertight type, in accordance with the requirements of Bureau of Ships Drawing 9000-S6501-73460. The jack shall be mounted on the side or bottom of the box so as to be accessible for use even when the cover is closed.

E-1b(2)a(2)j(3). All other types of microphone boxes.- The microphone jack shall be of the watertight type, in accordance with the requirements of Bureau of Ships Drawing 9000-S6501-73460. The jack shall be conveniently located on the cover.

E-1b(2)a(2)k. Designation plates.- Designation plates shall be provided for all lights, switches, fuses, and microphone jack. The designation markings for busy indicator lights, loudspeaker selector switches and talkback disconnect switches shall be as specified in the order or contract (see par. H-1a). Where no markings are specified in the contract or order, blank plates suitable for engraving shall be supplied.

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E-1b(2)b(2)ℓ. Instruction plate.- An instruction plate shall be provided, containing concise instructions on the use of the station or box and the proper method of talking into the microphone. On microphone control stations, types IC/MSA and IC/MSB, the instruction plate shall be mounted on the inside of the cover. On other types of microphone boxes, the instruction plate shall be mounted on the cover.

E-1b(2)b. Provision for cable entrance.- Provision shall be made as follows for the entrance of all necessary cables in both the top and the bottom of microphone control stations or boxes.

E-1b(2)b(1). Type IC/MBT microphone control box.- One or more removable plates, suitable for drilling, and capable of accommodating a minimum of 5 type C and 2 type G terminal tubes (see Bureau of Ships publication - Description and Application of Navy Standard Stuffing Tubes, Terminal Tubes and Cable Clamps) shall be provided on the control box for cable entrance.

E-1b(2)b(2). All other types of microphone stations and boxes.- Provision shall be made for accommodating a minimum of 2 sets of terminal tubes for cable entrance to the box. Terminal tube sizes for one set of terminal tubes shall be as shown in the following listing. The depth and hinging of the cover on microphone control stations, types IC/MSA and IC/MSB, shall be such that the cover can hang open without interfering with the ships' cables leading into those terminal tubes located at the bottom.

Station or box type	Microphone cable	Control circuit cable
IC/MSA	C	N
IC/MSB	C	L
IC/MBA	C	G
IC/MBB	C	K
IC/MBC	C	J
IC/MBD	C	L
IC/MJB	C	G
IC/MJP	C	--
IC/MJT	C	G

E-1b(2)b(3). Watertight plugs.- All holes drilled for terminal tubes shall be made watertight with removable plugs.

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E-1b(2)c. Maximum weight and dimensions.- The microphone stations and boxes, exclusive of terminal tubes (and with cover closed in the case of microphone control station, types IC/MSA and IC/MSB) shall not exceed the following weights and dimensions:

Station or box type	Weight in pounds	Height in inches	Width in inches	Depth in inches
IC/MSA	50	18-1/2	12-1/2	10-1/2
IC/MSB	50	18-1/2	12-1/2	10-1/2
IC/MBA	15	10	8	5-1/2
IC/MBB	15	10	8	5-1/2
IC/MBC	15	10	8	5-1/2
IC/MBD	15	10	8	5-1/2
IC/MBT	35	22	18	7-1/2
IC/MJB	5	7	7	5
IC/MJP	10	7	8	5
IC/MJT	6	8	8	6

E-1b(2)d. Resistance to pressure.- The microphone jack box, type IC/MJP (pressure-proof), shall be capable of meeting the pressure-proof requirements specified in paragraph D-9b(1).

E-1b(2)e. Resistance to depth charge.- The microphone jack box, type IC/MJP (pressure-proof) shall, at the conclusion of the depth charge test specified in paragraph F-3f(19), show no damage and shall still be capable of meeting the pressure-proof requirements defined in paragraph D-9b(1).

E-1b(3). Portable microphone, type IC/MPM, and handset, type IC/MPH.-

E-1b(3)a. Parts of portable microphone and handset.-

E-1b(3)a(1). Tabulation of required parts.- The following tabulation lists the required parts which make up the portable microphone, type IC/MPM, and handset, type IC/MPH, described in paragraphs E-1a(7) and E-1a(8), respectively.

	Type IC/MPM	Type IC/MPH
Microphone unit	1	--
Microphone element	--	1
Receiver element	--	1
Press-to-talk switch	--	1
Chest plate	1	--
Neck strap	1	--
Microphone cable	20 ft.	15 ft.
Microphone plug	1	1
Handset shell	--	1

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E-1b(3)a(2). Description of parts.-

E-1b(3)a(2)a. Microphone unit.- The microphone unit in the type IC/MPM portable microphone shall be the basic microphone unit, type IC/MBU, described in paragraph E-1b(1).

E-1b(3)a(2)b. Microphone element.- The microphone element in the type IC/MPH handset shall be identical to and interchangeable with that described in paragraph E-1b(1)b(1).

E-1b(3)a(2)c. Receiver element.- The receiver element in the type IC/MPH handset shall be a sound powered telephone receiver, in accordance with the requirements of Navy Department Specification R17T23.

E-1b(3)a(2)d. Press-to-talk switch.-

E-1b(3)a(2)d(1). Description.- The handle of the type IC/MPH handset shall be provided with a normally open, single-pole, single-throw, spring-return switch with suitable provision for locking the switch in the closed position. Operation of the switch shall complete a control circuit capable of carrying 1.0 ampere at 115 volts, 60 cycles, a.c. with an inductive load having a power factor of 50 percent.

E-1b(3)a(2)d(2). Switch construction and action.- The handset switch construction and action shall be as specified in paragraphs E-1b(1)d(2) and E-1b(1)d(3).

E-1b(3)a(2)d(3). Switch endurance requirement.- The handset switch shall, after 100,000 operations at rated load as specified in paragraph E-1b(3)a(2)d(1), be capable of satisfactory operation and shall show no bending of springs or failure of parts.

E-1b(3)a(2)e. Chest plate.- The chest plate of the type IC/MPM portable microphone shall be a suitable type, constructed of sheet brass finished with black rubber japan or equivalent.

E-1b(3)a(2)f. Neck strap.- The neck strap of the type IC/MPM portable microphone shall be type A, in accordance with the requirements of Bureau of Ships Drawing 9-S-4401.

E-1b(3)a(2)g. Microphone cable.- The microphone cable shall be of the following type, in accordance with the requirements of Navy Department Specification 15C1.

Component	Cable type
Type IC/MPM IC/MPH	MCOS-5 MCOS-6

E-1b(3)a(2)h. Microphone plug.- The microphone plug shall be in accordance with the requirements of the following Bureau of Ships' drawings:

Microphone type	BUSHIPS Dwg. No.
IC/MPM IC/MPH	9000-S6501-73461 9000-S6501-73462

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E-1b(3)a(2)i. Microphone handset shell.- The handset shell of the type IC/MPH handset shall be in accordance with the requirements of Navy Department Specification R17T23.

E-1b(3)b. Maximum weight.- The weights of the completely assembled portable microphone, type IC/MPM, and the handset, type IC/MPH, shall not exceed the following:

Microphone type	Max. weight in pounds
IC/MPM	5
IC/MPH	5

E-1b(3)c. Strength of cable assembly.- The cable when properly connected to the microphone (or handset) and plug shall be capable of withstanding a pull of 50 pounds at either the microphone (or handset) or plug end without evidence of the cable pulling loose or disturbing any of the electrical connections.

E-1b(3)d. Resistance to tumble.- The portable microphone and handset shall, at the conclusion of the tumble test specified in paragraph F-3f(17), show no damage and shall not vary in acoustic performance beyond the limit specified in paragraphs E-1c(1)b and E-1c(1)c.

E-1b(3)e. Microphone mounting attachment.- Each portable microphone, type IC/MPM, shall be provided with a mounting attachment, as specified in paragraph E-1b(1)g.

E-1b(4). Microphone, type IC/MFM, (flexible mounting).-

E-1b(4)a. The microphone, type IC/MFM, shall be as described in paragraph E-1a(9).

E-1b(4)b. Provision for cable entrance.- Provision shall be made for accommodating a minimum of 2 type C and 2 type G terminal tubes (see Bureau of Ships publication - Description and Application of Navy Standard Stuffing Tubes, Terminal Tubes and Cable Clamps) at the sides of the junction box.

E-1b(4)c. Maximum weight and dimensions.- The type IC/MFM microphone shall not exceed the following weight and dimensions:

Weight	12 pounds
Height of box	5 inches
Length of flexible arm	24 inches
Width of box	6-1/2 inches
Depth of box	4-1/2 inches

E-1b(4)d. Microphone clip.- A spring clip designed for bulkhead mounting shall be provided with each type IC/MFM microphone to permit securing the flexible arm when the microphone is not in use.

E-1c. Electrical requirements.-

E-1c(1). Basic microphone unit, type IC/MBU.-

E-1c(1)a. Impedance.- The microphone unit shall meet the performance requirements of this specification, when connected to a nominal 150-ohm resistive load.

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E-1c(1)b. Sensitivity.- The voltage developed by the microphone across a 150-ohm resistive load with an input sound pressure of 100 dynes per square centimeter shall be not less than 0.0087 volts when measured as specified in paragraph F-3f(2). Following the gunblast and tumble tests specified in paragraphs E-1b(1)e and E-1b(3)d respectively, the microphone voltage shall not vary in excess of 1.5 decibels from the original measured value.

E-1c(1)c. Frequency response characteristic.- The frequency response characteristic of the microphone shall fall within the limits shown on figure 3, when measured as specified in paragraph F-3f(3). Although the upper and lower limits are separated widely enough to permit normal fluctuations at certain frequencies or over narrow bands of frequencies, the mean of the measured response curves shall have a slope closely following the slope of the desired curve. Following the gunblast and tumble tests specified in paragraphs E-1b(1)e and E-1b(3)d respectively, the response at any frequency shall not vary in excess of 1.5 decibels from the corresponding original measured value.

E-1c(1)d. Induced voltage.- The voltage induced into the microphone circuit as a result of energizing the control circuits shall be not more than a value of 70 decibels below rated microphone output voltage (0.0087 volts) when measured as specified in paragraph F-3f(4).

E-1c(2). Electrical circuits.- The electrical circuits and terminal board markings of all microphone components shall conform to the following applicable figures:

Microphone type	Figure No.
IC/MBU	9
IC/MSA	10
IC/MSB	11
IC/MBA	12
IC/MBB	12
IC/MBC	13
IC/MBD	13
IC/MBT	14
IC/MJB	15
IC/MJT	15
IC/MJP	15
IC/MPM	9
IC/MPH	9
IC/MFM	9

E-1c(3). Noise pickup.- The voltage induced into the microphone voice circuit of all microphone components (except the type IC/MBU basic microphone unit covered in paragraph E-1c(1)d) as a result of fully energizing all power, control, busy and volume indicating circuits shall be not more than a value of 70 decibels below rated microphone output voltage (0.0087 volts) when measured as specified in paragraph F-3f(5).

E-1c(4). Control circuits - microphone control box, type IC/MBT.- Control circuits shall be provided in the type IC/MBT microphone control box to accomplish the following functions:

E-1c(4)a. Local talk switch.- The operation of the press-to-talk switch on a microphone connected into an IC/MBT control box shall:

(1) Transfer amplifier input from remote talk-back circuits to local microphone.
(2) Transfer amplifier output from local loudspeaker to remote loudspeaker bus.

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(3) Place the associated amplifier in the "ready" condition as defined in paragraph E-2b(1)a(4)b.
(4) Energize busy indicator lights at all remote talk-back stations.

E-1c(4)b. Remote talk switch.- The operation of the talk switch at a remote talk-back station (microphone jack box, type IC/ MJT) shall perform the functions specified in paragraphs E-1c(4)a(3) and E-1c(4)a(4).

E-1c(4)c. "1MC" alarm circuit.- The operation of a "1MC" alarm contact maker, as specified in paragraph E-2b(2)f(3)a(6) shall cause the "1MC" system output to be connected to the local loudspeaker and one remote loudspeaker group.

E-2. Amplifiers and controls.-

E-2a. Mechanical requirements.-

E-2a(1). Amplifier assemblies and signal generator assemblies.-

E-2a(1)a. Definition.- An amplifier assembly or signal generator assembly shall consist of a basic amplifier or signal generator unit complete with chassis and all parts, intended for mounting within an amplifier rack or enclosure or within a control rack.

E-2a(1)b. Construction.-

E-2a(1)b(1). Accessibility.- The construction shall be such that all wiring, terminals and electrical connections shall be accessible for servicing and test purposes without requiring the removal of an amplifier assembly or signal generator assembly from the rack or enclosure in which it is mounted. (The use of hinged or pivoted chassis secured by readily accessible thumb screws will be acceptable for this purpose).

E-2a(1)b(2). Electrical connections.- All electrical connections to an amplifier assembly or signal generator assembly shall be made by plug-and-jack connectors held securely in place while connected and permitting disconnect without the use of any tools.

E-2a(1)c. Maximum weight and size.- The maximum weight and cubic volume of amplifier assemblies and signal generator assemblies shall not exceed the following values. Maximum volume is defined as the volume of the smallest rectangular parallelepiped that will enclose the assembly. (Every effort should be exerted to keep weight and size to a practical minimum consistent with other requirements of this specification.)

Assembly	Weight in pounds	Volume in cubic inches
Amplifier, type IC/FCA (20 watts)	50	2000
Amplifier, type IC/FDA (40 watts)	100	3500
Amplifier, type IC/FGA (125 watts)	140	6000
Amplifier, type IC/FJA (250 watts)	180	6000
Amplifier, type IC/FMA (500 watts)	200	6000
Signal generator, type IC/GA	40	1400

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E-2a(1)d. Interchangeability.- Any amplifier or signal generator assembly shall be interchangeable with all others of the same design and manufactured from the same set of approved plans.

E-2a(2). Amplifier and control racks.-

E-2a(2)a. Definition.- An amplifier or control rack shall consist of an assembly of one or more amplifier or signal generator assemblies and/or control facilities, such as switches and relays, mounted in a single rack or enclosure.

E-2a(2)b. General construction.- The amplifier rack or enclosure or control rack shall be of drip-proof construction designed for deck or bulkhead mounting. Any rack or enclosure exceeding 250 pounds in weight or 30 inches in height shall be designed for deck mounting with provision for top bracing to a bulkhead at the back of the enclosure. All other racks or enclosures shall be designed for bulkhead mounting.

E-2a(2)c. Front servicing.- All racks or enclosures shall be designed to permit complete servicing from the front.

E-2a(2)d. Covers and doors.- Racks or enclosures shall be provided with covers or doors which may be readily removed or opened for servicing or inspection. Covers shall be secured by knurled or wing nuts of the captive type capable of being unfastened by hand. Where independent amplifier channels are contained in the same rack or enclosure, sectional doors or covers are preferred to permit servicing of a disabled channel without danger of personnel coming in contact with circuits in an operating channel.

E-2a(2)e. Personnel protection.- Provision shall be made wherever possible to prevent personnel from accidentally coming in contact with electrically energized parts. In the case of assemblies in which voltages in excess of 240 volts are employed, a red warning nameplate shall be mounted in a conspicious location and a manually operated switch shall be provided to de-energize such circuits within the assembly during servicing.

E-2a(2)f. Mounting of amplifier and signal generator assemblies.- Dowels or studs designed to match with bushings or holes in the assembly, or shelves, or supports shall be provided to sustain the weight of the assembly while the assembly is being mounted in or removed from the rack or enclosure. Shelves or angle supports shall be used where the weight of the assembly exceeds 50 pounds.

E-2a(2)g. Mounting of controls.- All switches, except those specified in paragraph E-2a(2)e, operating controls, test controls, and meters shall be mounted on the front of the control rack or panel, and shall be visible and accessible without requiring the opening of covers or doors.

E-2a(2)h. Accessibility.- All terminals, and electrical connections within the rack or enclosure shall be accessible for servicing and test purposes without requiring the removal of assembly or sub-assembly from the rack or enclosure.

E-2a(2)i. Terminal boards.- Terminal boards shall be provided in a convenient location (preferably at the bottom of the rack or enclosure but not less than 12 inches from the bottom of deck mounted racks) for connection to all ship's cables entering the enclosure. Amplifier racks or enclosures designed for use with a separate control rack or other amplifier racks, shall have provision for inter-rack connections, but all external ship's cables shall be connected in the control rack.

E-2a(2)j. Provision for cable entrance.- Removable gasketed plates, suitable for drilling to accommodate terminal tubes for cable entrance, shall be provided at the top and bottom of the rack or enclosure.

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E-2a(2)k. Ventilation.- Unless otherwise specified, only natural ventilation shall be employed.

E-2a(2)l. Maximum weight and size.- Amplifier and control racks shall be of the minimum practical size and weight consistent with other requirements of this specification. The dimensions shall be proportioned to utilize space most economically. No single rack shall exceed the following maximum dimensions and weight.

	For surface vessels	For submarines
Maximum height	72 inches	62 inches
Maximum width	30 inches	20 inches
Maximum depth	20 inches	15 inches
Maximum weight	900 pounds	750 pounds

E-2a(2)m. Other limiting dimensions.- The lowest part of any rack on which apparatus is mounted shall not be closer than 4 inches to the deck. Meters and other indicating apparatus shall be mounted not closer than 12 inches to the deck.

E-2b. Electrical requirements.-

E-2b(1). Amplifier and signal generator assemblies.-

E-2b(1)a. General.-

E-2b(1)a(1). Dual-tube design.- At least 2 tubes operating in parallel shall be used in every stage so that the removal of one-half the tubes per stage shall not reduce the gain or the output power of an amplifier by more than 4 decibels, shall not reduce the output level of a signal generator by more than 1 decibel, and shall not cause the frequency response of an amplifier to fall outside of the limits of this specification. Wherever possible, separate parts shall be associated with each tube to minimize the effect of short circuits within tubes. "Push-pull" operation of tubes will be acceptable in lieu of parallel tubes in output stages only.

E-2b(1)a(2). Power supply.- Each amplifier and signal generator assembly shall contain its own tube plate and filament power supply.

E-2b(1)a(3). Test facilities.- A plate current meter with associated switch or switches shall be provided on each amplifier and signal generator assembly to permit measurement of plate current in each tube stage. Meter markings and the measuring circuits shall be such that a ready indication of proper operating currents can be obtained without reference to charts or tables.

E-2b(1)a(4). "Standby" and "ready" conditions.-

E-2b(1)a(4)a. "Standby" condition.- An amplifier or signal generator assembly shall be defined to be in the "standby" condition, when power has been made available, but the equipment is not transmitting or generating any signal.

E-2b(1)a(4)b. "Ready" condition.- An amplifier or signal generator assembly shall be defined to be in the "ready" condition, when the amplifier is fully energized and is capable of transmitting speech or other signal at full output power, or when the signal generator is fully energized and is generating a signal at full output voltage.

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E-2b(1)a(4)c. Time interval between "standby" and "ready" conditions.- Starting from a "standby" condition, the amplifier or signal generator assembly shall reach the ready condition within 2 seconds of the time the amplifier start circuit or a signal generator control circuit is energized.

E-2b(1)a(4)d. Tube life during "standby" condition.- While in the "standby" condition, power shall be consumed in such a way as to insure maximum useful life of electronic tubes and other parts consistent with the time interval requirement of paragraph E-2b(1)a(4)c.

E-2b(1)a(5). Amplifier phasing.- To insure proper operation in the event that amplifiers are operated in parallel, one of the input terminals and one of the output terminals of each amplifier assembly shall be marked with a plus (+). Each amplifier shall be so wired that like-marked input and output terminals will be in corresponding electrical phase relationship whenever a signal is being transmitted through the amplifier.

E-2b(1)b. Amplifier assemblies, low and intermediate power, types IC/FCA, IC/FDA and IC/FGA.-

E-2b(1)b(1). Definition.- A low or intermediate power amplifier assembly is one designed to deliver the rated output power when driven by any of the microphone components described in paragraph E-1.

E-2b(1)b(2). Amplifier design characteristics.-

E-2b(1)b(2)a. Input impedance.- Input impedance shall be 150 ohms (+20 ohms/ - 0 ohms) resistive at any frequency between 200 and 8000 cycles per second.

E-2b(1)b(2)b. Output voltage.- Amplifier output voltage when delivering rated output power shall be 70 volts (± 4 volts) r.m.s.

E-2b(1)b(2)c. Output power.- Rated output power for the following amplifier types shall be as follows. Rated output power for other amplifier types shall be specified in the order or contract (see par. H-1b).

TYPE	WATTS (min.)
IC/FCA	20
IC/FDA	40
IC/FGA	125

E-2b(1)b(2)d. Amplifier gain.-

E-2b(1)b(2)d(1). Rated gain.- With a 1000 cycle per second sine-wave signal of 8.7 millivolts r.m.s. applied to the amplifier input terminals, the amplifier shall deliver rated output power at rated voltage with a maximum total harmonic distortion of 5 percent. The amplifier gain corresponding to the gain control setting under the above condition shall be referred to as the "rated gain" of the amplifier.

E-2b(1)b(2)d(2). Reserve gain.- The amplifier gain at the "full gain" position of the gain control shall be a minimum of 15 decibels greater than the "rated gain".

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E-2b(1)b(2)e. Frequency response.- At "rated gain" and with a constant voltage sine wave signal of 8.7 millivolts r.m.s. applied to the amplifier input terminals and varied continuously from 200 to 8000 cycles per second, the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall fall within the limits shown on figure 4. The measured response curve shall have a slope closely following the slope of the desired curve. The above requirement shall also apply at any setting of the gain control from 20 decibels below "rated gain" to 10 decibels above.

E-2b(1)b(2)f. Distortion.-

E-2b(1)b(2)f(1). Distortion at various frequencies.- At "rated gain" and with a sine wave voltage applied to the amplifier input terminals and varied from 200 to 4000 cycles per second, the total harmonic distortion of the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the following limits. The input voltage shall be varied to produce the indicated power at each frequency.

Output power (Db relative to "rated-power")	Frequency range (Cycles per second)	Maximum distortion (percent)
-6	200-400	15
-3	400-700	10
0	700-4000	5

E-2b(1)b(2)f(2). Distortion at various input levels.- At "rated gain" and with any 1000 cycle per second sine-wave signal voltage up to 34.8 millivolts r.m.s. applied to the amplifier input terminals, the total harmonic distortion of the output voltage across a resistive load (of such a value as to draw rated output power at 70 volts) shall not exceed the following limits:

Input voltage (millivolts)	Maximum distortion (percent)
0-8.7	5
8.7-17.4	10
17.4-34.8	15

E-2b(1)b(2)g. Noise and microphonics.- At "full gain" and with a 150 ohm resistor connected to the amplifier input terminals, the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the following values when measured with an unweighted full range voltmeter. The measurement shall be made under two conditions of test - Once while the component is held fixed and again while the component is under vibration in accordance with the requirements of paragraph D-4p.

(1). Component stationary.- Maximum measured voltage shall be 0.22 volts r.m.s. (50 decibels below rated output voltage of 70 volts).
(2). Component under vibration.- Maximum measured voltage shall be 0.31 volts r.m.s. (47 decibels below rated output voltage of 70 volts).

E-2b(1)b(2)h. Output voltage regulation.- At "rated gain" and with a steady state sine-wave signal of 8.7 millivolts r.m.s. applied to the amplifier input terminals, a variation of amplifier output load from full rated load (that resistive load which will draw rated output power at 70 volts) to no load (open circuit) shall not cause the output voltage to exceed 99 volts r.m.s. (3 decibels above rated output voltage of 70 volts) at any frequency between 200 and 8000 cycles per second.

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E-2b(1)b(2)i. Isolation against longitudinal voltages.- Longitudinal voltages shall be applied to the amplifier input terminals in the manner described below and shall bear a ratio of not less than 50 decibels to the corresponding voltages required across the normal amplifier input terminals to produce the same amplifier output voltage. Sine-wave voltages at 1000 cycles per second shall be employed for this purpose. The longitudinal voltages shall be applied and measured.

(1) Between both input terminals and ground (input terminals short-circuited).
(2) Between the midpoint of a 150 ohm resistor connected across the input terminals and ground.
(3) Between either input terminal and ground (input terminals open-circuited).

The ratio thus obtained at 1000 cycles should vary with frequency at the general rate of 6 decibels per octave.

E-2b(1)b(2)j. Stability.- The amplifier shall show no evidences of instability or oscillation when operated at "full gain" with amplifier input either short or open-circuited and with amplifier output either short or open-circuited.

E-2b(1)b(2)k. Overload capacity.- At "rated gain" and with a 1000 cycle per second sine-wave signal of 8.7 millivolts r.m.s. applied to the amplifier input terminals the amplifier shall be capable of operating continuously for a period of 30 minutes into a short circuit load across the output terminals without lowering the amplifier gain, (change in gain to be determined after normal load conditions have been reestablished), affecting ability to produce rated output power, or producing other deleterious effects.

E-2b(1)b(3). Compressor circuit.- Each low or intermediate power amplifier shall include circuits which will provide compression characteristics as follows:

E-2b(1)b(3)a. Compressor cut-out switch.- A double-throw switch shall be provided on the amplifier assembly with positions marked "compression in" and "compression out". In the "compression out" position, the compression characteristics shall be disconnected or rendered ineffective and the amplifier characteristic shall revert essentially to a "straight line" within the rated operating range. In the "compression in" position, the amplifier characteristics shall be as specified in paragraphs E-2b(1)b(3)b to E-2b(1)b(3)d, inclusive. The compressor cut-out switch shall have no effect on the overload limiting features described in paragraph E-2b(1)b(4).

E-2b(1)b(3)b. Compressor characteristic.- At any setting of the amplifier gain control from 20 decibels below "rated gain" to 10 decibels above, and with a 500 cycle per second sine-wave signal applied to the amplifier input terminals, the decrease in amplifier output voltage with corresponding decrease in input voltage shall fall within the following limits. Reference output voltage level (0 decibels) is 70 volts at full rated output power. Reference input voltage level (0 decibles) is that input voltage required at the particular amplifier gain setting to produce an output voltage of 70 volts. The above requirement shall also apply at signal frequencies of 1000, 2000 and 4000 cycles per second.

Relative amplifier input voltage (decibels)	Relative amplifier output voltage (decibels)
0	0
-5	-2 ± 2
-10	-4 ± 3
-15	-6.5 ± 3.5
-30	-15 ± 4.5
-40	-22 ± 6

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E-2b(1)b(3)c. Time constant.- Under the conditions described in paragraph E-2b(1)b(3)b with a relative amplifier input voltage of minus 40 decibels, the amplifier output voltage shall return to the rated value (0 decibels relative level) within 0.025 (± 0.005) seconds after the sudden application of corresponding input voltage (0 decibels relative level), at signal frequencies of 500, 1000, 2000 and 4000 cycles per second.

E-2b(1)b(3)d. Compressor adjustments.- Means shall be provided for adjusting the compressor characteristic so that at any fixed amplifier gain, the "cross-over point" of the load characteristic curves (amplifier input voltage versus amplifier output voltage) taken with "compression in" and "compression out" can be made to occur at any output voltage from 2 decibels above rated output voltage (70 volts) to 6 decibels below. When adjusted so that the "cross-over point" is at rated output voltage, the operation of the compressor cut-out switch from "compression in" to "compression out" shall not cause the output voltage to vary in excess of ± 1 decibel when delivering full rated output power at fixed gain, with fixed input and at any frequency between 500 and 5000 cycles per second. The output voltage at the "cross-over point", corresponding to any particular compressor adjustment, shall be independent of the setting of the amplifier gain control.

E-2b(1)b(3)e. Effect of compressor on amplifier characteristics.- The specified amplifier characteristics described in paragraph E-2b(1)b(2), shall be met with "compression in" as well as with "compression out".

E-2b(1)b(4). Overload limiter circuit.- Each low or intermediate power amplifier shall include circuits to provide the overload limiter characteristic described as follows:

E-2b(1)b(4)a. Overload limiter characteristic.- At any setting of the amplifier gain control from "rated gain" to 12 decibels above, and with a 500 cycle per second sine-wave signal applied to the amplifier input terminals, the increase in amplifier output voltage with corresponding increase in input voltage shall fall within the following limits. Reference output voltage level (0 decibels) is 70 volts at full rated output power. Reference input voltage level (0 decibels) is that input voltage required at the particular amplifier gain setting to produce an output voltage of 70 volts. The above requirement shall also apply at signal frequencies of 1000, 2000 and 4000 cycles per second.

Relative amplifier input voltage (decibels)	Relative amplifier output voltage (decibels)
+6	0 ± 1.5
+12	0 ± 2

E-2b(1)b(4)b. Time constant.- Under the conditions described in paragraph E-2b(1)b(4)a, the amplifier output voltage shall return to the specified value (0 ± 2 decibels relative level) within 0.001 ± 0.0005 seconds after the sudden increase of input level from a reference level of 0 decibels to plus 12 decibels at signal frequencies of 500, 1000, 2000 and 4000 cycles per second.

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E-2b(1)b(5). Amplifier controls.- Amplifier controls shall be provided as follows:

E-2b(1)b(5)a. Gain control.- There shall be provided an amplifier gain control, fitted with a suitable knob and readily operable from the front of the amplifier. Gain control setting shall be indicated by a pointer or index on the knob, and a fixed scale calibrated in decibels below full gain. The gain control shall be either of the continuously variable type (in which case only wire-wound potentiometers shall be used) or of the fixed step type (in which case a minimum of 10 steps shall be provided, each step introducing an attenuation of approximately 3 decibels.) Operation of the gain control shall not introduce noise into the amplifier circuit. The end of the gain control shaft shall be slotted to permit screw-driver adjustment, should it be desirable in use to remove the gain control knob.

E-2b(1)b(5)b. Compressor cut-out switch.- A compressor cut-out switch shall be provided as specified in paragraph E-2b(1)b(3)a.

E-2b(1)b(6). Interstage input circuit.- Facilities shall be provided for the interstage injection of signals into the amplifier circuit. The characteristics of the circuit shall be as follows:

E-2b(1)b(6)a. Input impedance.- The input impedance of the interstage circuit shall be 1000 ohms resistive.

E-2b(1)b(6)b. Input level.- With a 1000-cycle per second signal of 6 volts r.m.s. applied to the interstage input terminals, the amplifier shall deliver rated output power at rated voltage independent of the amplifier gain control setting.

E-2b(1)b(6)c. Compressor circuit by-pass.- The compressor circuit shall have no effect on signals introduced at the interstage input circuit, regardless of the compressor cut-out switch setting.

E-2b(1)c. Amplifier assemblies, high power, type IC/FJA and IC/FMA.-

E-2b(1)c(1). Definition.- A high power amplifier assembly is one designed to deliver its rated output power when driven by a low power amplifier assembly, type IC/FDA.

E-2b(1)c(2). Input impedance.- Amplifier input impedance shall be 125 ohms (+ 15/ -0 ohms) at any frequency between 200 and 8000 cycles. Amplifiers whose rated output power is 500 watts or greater, shall be provided with an alternate input circuit having an input impedance of 250 ohms (+25/-0 ohms), over the same frequency range.

E-2b(1)c(3). Output voltage.- Amplifier output voltage when delivering rated output power shall be 70 volts (± 4 volts) r.m.s.

E-2b(1)c(4). Output power.- Rated amplifier power for the following amplifier types shall be as follows. Rated output power for other amplifier types shall be specified in the order or contract (see par. H-1b).

TYPE	WATTS (min.)
IC/FJA	250
IC/FMA	500

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E-2b(1)c(5). Amplifier gain.- With a 1000 cycle per second signal of 70 volts r.m.s. applied to the amplifier input terminals (principle or alternate) the amplifier shall deliver rated output power at rated voltage with a maximum total harmonic distortion of 10 percent.

E-2b(1)c(6). Frequency response.- With a constant voltage sine-wave signal of 70 volts r.m.s. applied to the amplifier input terminals and varied continuously from 200 to 8000 cycles per second, the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall fall within the limits shown on figure 4. The measured response curve shall have a slope closely following the slope of the desired curve.

E-2b(1)c(7). Distortion.- With a sine wave voltage applied to the amplifier input terminals and varied from 200 to 4000 cycles per second the total harmonic distortion of the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the following limits. The input voltage shall be varied to produce the indicated power at each frequency.

Output power (Db relative to "rated-power")	Frequency range (cycles per second)	Maximum distortion (percent)
-6	200-400	20
-3	400-700	15
0	700-4000	10

E-2b(1)c(8). Noise and microphonics.- With a 125 ohm resistor connected to the principle amplifier input terminals, the output voltate across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the following values when measured with an unweighted full range voltmeter. The measurement shall be made under two conditions of test - once while the component is held fixed, and again while the component is under vibration in accordance with the requirements of paragraph D-4p.

Component stationary.- Maximum measured voltage shall be 0.22 volts r.m.s. (50 decibels below rated output voltage of 70 volts).
Component under vibration.- Maximum measured voltage shall be 0.31 volts r.m.s. (47 decibels below rated output voltage of 70 volts).

E-2b(1)c(9). Output voltage regulation.- With a fixed sine-wave signal voltage of 70 volts r.m.s. applied to the amplifier input terminals, a variation of amplifier output load from full rated load (that resistive load which will draw rated output power at 70 volts) to no load (open circuit) shall not cause the output voltage to exceed 99 volts r.m.s. (3 decibels rise above rated output voltage of 70 volts) at any frequency between 200 and 8000 cycles per second.

E-2b(1)c(10). Stability.- The amplifier shall show no evidences of instability or oscillation when operated with amplifier input either short or open-circuited and with amplifier output either short or open-circuited.

E-2b(1)c(11). Overload capacity.- With a 1000 cycle per second sine-wave signal of 70 volts r.m.s. applied to the amplifier input terminals, the amplifier shall be capable of operating continuously for a period of 30 minutes into a short circuit load across its output terminals without lowering amplifier gain, affecting ability to produce rated output power, or producing other deleterious effects. The change in gain will be determined after normal load conditions have been reestablished.

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E-2b(1)d. Signal generators.-

E-2b(1)d(1). Number and type.- Signal generators for any specified amplifier or control rack shall be provided in the type and quantity indicated in paragraph B-4, or as specified in the contract or order (see par. H-1b).

E-2b(1)d(2). Generated signals.- Closure of the appropriate control circuit shall cause signals to be generated as follows:

Signal generator type	Associated shipboard announcing circuit	Generated signals
IC/GAC	1MC (surface vessels)	chemical attack alarm general alarm
IC/GAL	1MC (submarines)	collision alarm diving alarm general alarm
IC/GAN	5 MC	flight crash flight warning
IC/GAQ	17 MC	cease firing salvo

E-2b(1)d(3). Description of signals.-

E-2b(1)d(3)a. General character of signals.- The simulated signals specified in paragraphs E-2b(1)d(3)c, E-2b(1)d(3)d, E-2b(1)d(3)e, E-2b(1)d(3)h, and E-2b(1)d(3)i, shall be designed to be raucous in nature for the purpose of making them as attention arresting as possible. For the same purpose, the generation of signals having a pure sine-wave form should be avoided in the design of the remaining signals, specified in paragraphs E-2b(1)d(3)b, E-2b(1)d(3)f and E-2b(1)d(3)g.

E-2b(1)d(3)b. Chemical attack alarm.- Chemical attack alarm shall consist of a steady tone signal of 1000 (±100) cycles per second.

E-2b(1)d(3)c. General alarm.- General alarm shall consist of a simulated single-stroke gong tone signal, striking at the rate of 90 strokes per minute.

E-2b(1)d(3)d. Collision alarm.- Collision alarm shall consist of a simulated siren tone signal, whose fundamental frequency shall sweep back and forth recurrently over the approximate frequency range of 750 to 1750 cycles per second at the approximate rate of 3 seconds for one full sweep.

E-2b(1)d(3)e. Diving alarm.- Diving alarm shall consist of a simulated motor-operated horn type signal.

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E-2b(1)d(3)f. Flight crash.- Flight crash shall consist of a steady tone signal of 1000 (±100) cycles per second.

E-2b(1)d(3)g. Flight warning.- Flight warning shall consist of a "jump" tone signal alternating between 600 (± 100) and 1000 (± 100) cycles per second at the approximate rate of 90 complete cycles per minute.

E-2b(1)d(3)h. Cease firing.- Cease firing shall consist of a simulated siren tone signal as described in paragraph E-2b(1)d(3)d.

E-2b(1)d(3)i. Salvo.- Salvo shall consist of a simulated motor-operated horn type signal.

E-2b(1)d(4). Duration of signals.- All signals, with the exception of general alarm, shall be generated continuously for as long a period as the correspondi.. control circuit is closed. Momentary closure of the general alarm control circuit shall cause the gener.. alarm signal to be generated for the following periods of time:

Signal generator type	Duration of signal
IC/GAC	15 seconds
IC/GAL	10 seconds

E-2b(1)d(5). Output level.- The signal generator shall be capable of delivering signals at each of the following output levels:

Voltage	Output resistive load	Function
10 (±1.5) volts	500	Amplifier interstage input
0.06 (± 0.01) volts	150	Amplifier test input

E-2b(1)d(6). Adjustment of output level.- A screw-driver type adjustment shall be provided in the signal generator or in combination with the associated control circuits to permit independent adjustment of the signal generator output level for the alarm signal into the amplifier interstage input.

E-2b(1)d(7). Noise and microphonics.- The character of the signals shall not be adversely affected by the presence of any audible hum or other extraneous noises in the generated signals, when those signals are amplified to full output level by any of the amplifier assemblies specified in this specification. This requirement shall be met both with the signal generator stationary and with the generator under vibration in accordance with the requirements of paragraph D-4p.

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E-2b(2). Amplifier and control racks.-

E-2b(2)a. Introduction.- Amplifier and control racks are defined in paragraph E-2a(2)a, which covers mechanical requirements for these components. This section shall cover the electrical facilities which are required for proper functioning of an amplifier system made up of the specified numbers and types of amplifier and signal generator assemblies.

E-2b(2)a(1). Amplifier system classification.- In setting down general requirements for system controls, adjustments and test facilities, it will be necessary to distinguish in many cases between various amplifier systems by the size or by the circuit functions as follows:

E-2b(2)a(1)a. Amplifier system classification by size.-

Large - An amplifier system employing several racks, in which one or more of the racks are employed as control racks and contain all necessary controls.
Small - An amplifier system contained entirely within a single rack, a portion of which is employed as a control panel containing all necessary controls.

E-2b(2)a(1)b. Amplifier system classification by circuit functions.-

(1) 1-2-3 MC - An amplifier system designed primarily for general Announcing (Circuit 1MC), with or without additional facilities for other circuits such as Engineer's Announcing (Circuit 2 MC) and Aviator's Announcing (Circuit 3 MC).
(2) 1-6- MC - An amplifier system designed for General Announcing (Circuit 1MC) and Intership Announcing (Circuit 6MC).
(3) 1-7 MC - An amplifier system designed for General Announcing (Circuit 1MC) and Submarine Control Announcing (Circuit 7 MC).
(4) 5 MC - An amplifier system designed for Flight Deck Announcing (Circuit 5 MC).
(5) 17 MC - An amplifier system designed for Double-Purpose Battery Announcing (Circuit 17MC).

E-2b(2)a(2). Amplifier system accommodations.-

E-2b(2)a(2)a. Input facilities.- There shall be specified in the contract or order, (see par. H-1a) the number, type and location of all microphone stations or boxes associated with each announcing circuit.

E-2b(2)a(2)b. Output facilities.- There shall be specified in the contract or order (see par. H-1a), the number and designation of each loudspeaker output group.

E-2b(2)a(3). Amplifier system operating voltages.-

E-2b(2)a(3)a. System output voltage for loudspeaker circuits.- Rated system output voltage for loudspeaker circuits shall be 70 volts (±4 volts) r.m.s.

E-2b(2)a(3)b. System output voltage for volume indicator circuits.- The voltage available at the terminals of volume indicator circuits to microphone stations or boxes, corresponding to rated system output voltage, shall be 1.73 volts (± 0.08 volts) r.m.s.

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E-2b(2)b. Operating controls.- There shall be mounted on the front of the control panel or control rack/s, operating controls as follows:

E-2b(2)b(1). General.-

E-2b(2)b(1)a. Circuit transfer switches.- All switches employed for circuit transfer, such as input disconnect switches, amplifier transfer switches and signal generator selector switches shall be of the JR Type as covered by Bureau of Ships Plans referred to in paragraph A-3. Wherever practicable, it will be satisfactory to combine two or more switching functions on a single switch sub-assembly.

E-2b(2)b(1)b. Switch markings.- All switches and operating controls shall be clearly marked to indicate the switch or control function, and to indicate all active positions or throws.

E-2b(2)b(1)c. Switch and relay circuits.- Disconnect, transfer and relay switching of all power, input and output circuits shall be accomplished by opening both sides or all phases of the circuits. Single line switching will be acceptable for control circuits only.

E-2b(2)b(2). Power indicator light.- A power indicator (pilot) light, covered by a white lens shall be provided to indicate that electric power is available at the amplifier.

E-2b(2)b(3). Power switches.- Separate power switches shall be provided as follows:

Each amplifier channel
Each signal generator
Relay circuits

E-2b(2)b(4). Input disconnect switches.-

E-2b(2)b(4)a. Active switches.- When the amplifier design is such that more than one microphone station or box can be connected to a common amplifier channel, an input disconnect switch shall be provided for each such station or box, and shall open all circuits to that station or box. This requirement does not apply to microphone audio circuits, if such circuits are connected to the open contacts of relays, which relays are disconnected by the action of the associated input disconnect switches.

E-2b(2)b(4)b. Spare switches.- In large amplifier systems, as classified in paragraph E-2b(2)a(1)a, an additional input disconnect switch shall be provided for each announcing circuit (such as 1MC, 2MC, 3MC, 5MC) to permit future system expansion. These switches shall be connected to appropriate terminal boards and shall be wired into the amplifier and control circuits.

E-2b(2)b(5). Amplifier transfer switches.-

E-2b(2)b(5)a. General.- When the amplifier design is such that more than one amplifier channel is employed in a system, amplifier transfer switches shall be provided to disconnect both input and output circuits of a faulty channel (or one being set up for test) and to transfer the combined system functions to the remaining channel or channels. Separate switches, having corresponding positions, may be used for input and output circuits. Those amplifier channels not selected for use shall be disconnected from all operating circuits and shall be available for test purposes, in accordance with the requirements of paragraph E-2b(2)e. For specific amplifier systems, as classified in paragraph E-2b(2)a(1)b, the following shall apply:

E-2b(2)b(5)b. 1-2-3MC, 1-6MC and 1-7MC systems.- Amplifier transfer switches shall permit each circuit to be connected to its own amplifier channel for normal operation. For other than normal operation, it shall be possible to combine any two or more circuits and connect them to either amplifier channel. The signal generator output circuit shall always be associated with the 1MC circuit for amplifier transfer purposes.

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E-2b(2)b(5)c. 5MC systems.- Amplifier channels shall be arranged in groups of two. Amplifier transfer switches shall permit each channel to drive its associated loudspeaker or loudspeaker groups for normal operation. For other than normal operation, it shall be possible to combine both loudspeakers or loudspeaker groups of a pair and connect them to the output of either amplifier channel of the pair.

E-2b(2)b(5)d. 17MC systems.- Unless otherwise specified, each channel of a 17MC system shall be independent of all other channels, so that amplifier transfer switches will not be required.

E-2b(2)b(6). Signal generator selector switches.-

E-2b(2)b(6)a. General.- Except as noted in paragraph E-2b(2)b(6)b, each amplifier system is provided with two signal generators, one serving as a standby for the other. Signal generator selector switches shall be provided to permit the selection of either signal generator for use with the system. The signal generator not selected for use shall be disconnected from all operating circuits and shall be available for test purposes, in accordance with the requirements of paragraph E-2b(2)e.

E-2b(2)b(6)b. 17MC systems.- Unless otherwise specified, each signal generator of a 17MC system shall be associated with its own amplifier channel, so that signal generator selector switches will not be required.

E-2b(2)b(7). Output disconnect switches.-

E-2b(2)b(7)a. General.- When the amplifier design is such that more than one loudspeaker group can be operated from the output of a common amplifier channel, an output disconnect switch shall be provided for each loudspeaker group. The following additional requirements apply for specific amplifier systems, as classified in paragraph E-2b(2)a(1)b:

E-2b(2)b(7)b. 1-2-3MC and 1-6MC systems.- Additional output disconnect switches shall be provided as follows:

(1) One spare output disconnect switch for each six (or fraction thereof) loudspeakers or loudspeaker groups. These switches shall be connected to appropriate terminal boards, and shall be wired into the amplifier and control circuits to permit future system expansion.
(2) One output disconnect switch for 17MC loudspeakers.
(3) Six output disconnect switches for circuits 11-16MC loudspeakers.
(This applies to amplifier systems designed for General Announcing (Circuit 1MC) and Engineers Announcing (Circuit 2MC) only.)

E-2b(2)b(7)c. 17MC systems.- Unless otherwise specified, output switching of 17MC loudspeaker groups shall be accomplished at the fire control switchboard, so that output disconnect switches will not be required.

E-2b(2)b(8). System volume controls.- Where the amplifier design is such that more than one announcing circuit (such as 1MC, 2MC and 3MC) is incorporated in the same amplifier system, a system volume control shall be provided for each circuit. This control shall permit adjustment of the system output voltage in a minimum of 6 steps, each step introducing a decrease of approximately 3 decibels from rated output voltage. Transfer of a circuit from one amplifier channel to another shall be possible without requiring re-adjustment of the system volume control for that circuit. Operation of the system volume control shall not affect either the amplifier compressor characteristics or the indications of the system volume indicator meters.

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E-2b(2)c. Circuit protective features.-

E-2b(2)c(1). Power circuits.-

E-2b(2)c(1)a. Amplifier main power circuits.- Both sides or all phases of the power circuits feeding each power switch specified in paragraph E-2b(2)b(3) shall be independently fused.

E-2b(2)c(1)b. Relay power circuits.- Power to relay and control circuits, which extend beyond the amplifier enclosure (such as circuits to microphone stations or external contact makers) shall be fed through an isolation transformer to minimize the possibility of grounding the ship's power supply as a result of an accidental ground on such circuits. This requirement shall not apply to the external visual alarm circuits specified in paragraphs E-2b(2)f(3)b(1) and E-2b(2)f(3)b(2).

E-2b(2)c(1)c. Microphone equipment power circuits.- Where both sides of the power circuit connect to a microphone station or box, each such circuit shall be independently fused.

E-2b(2)c(2). Microphone audio circuits.- When the amplifier design is such that more than one microphone station or box can be connected to a common amplifier channel, individual 39-ohm protective resistors shall be wired in series with each side of each microphone audio circuit. A 150 ohm resistor shall be shunted across the audio input circuit to the amplifier. This arrangement provides isolation for each microphone and provides proper loading for each microphone.

E-2b(2)c(3). Volume indicator circuits.- Volume indicator circuits to each microphone station or box shall be protected, so that short-circuiting the terminals of any volume indicator circuit shall not effect the amplifier output voltage, and shall not change the voltage at other volume indicator circuit terminals by more than 1 decibel.

E-2b(2)c(4). Busy indicator circuits.- Busy indicator circuits to microphone stations or boxes shall be protected by series resistors so that short-circuiting the terminal of any busy indicator circuit shall not effect the relay power-circuit nor any of the other busy indicator circuits.

E-2b(2)c(5). Loudspeaker audio circuits.- Where specified in the contract or order (see par. H-1a), series protective resistors shall be provided separately in the size and quantity required for all loudspeaker circuits designated as requiring protection. Such resistors shall reduce the amplifier output voltage from its rated value of 70 volts to a lesser value of 50 volts for each loudspeaker circuit so designated. (These resistors will be mounted externally to the amplifier by the installing activity.)

E-2b(2)d. System adjustments.-

E-2b(2)d(1). Output voltage to loudspeaker groups.- For 1-2-3 MC and 1-6MC systems, as classified in paragraph E-2b(2)a(1)b, provision shall be made for adjusting independently the output voltage of the individual loudspeaker groups. Adjustment shall be possible in a minimum of 3 steps, each step introducing a decrease of approximately 6 decibles from rated output voltage.

E-2b(2)e. Test facilities.- There shall be mounted on the front of the control panel or control rack/s, test facilities as follows:

E-2b(2)e(1). General.- Test facilities shall be provided for the purpose of checking the operation of any amplifier or signal generator assembly as an aid to system servicing and maintenance. Test circuits shall be independent of the active audio and control circuits, and shall be wired through the amplifier and signal generator transfer switches in such a way that it shall be possible to test an amplifier or signal generator assembly only if it has been disconnected from the system. In no case shall it be possible to transmit test signals to any connected loudspeaker circuits.

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E-2b(2)e(2). Amplifier start switches.- An amplifier start switch shall be provided for each amplifier channel, and when operated shall place that channel in the "ready" condition, as defined in paragraph E-2b(1)a(4)b. These switches shall not energize the busy indicator circuits, nor interfere with normal operation from any microphone station or box.

E-2b(2)e(3). Signal generator start switches.- Non-locking type signal generator start switches shall be provided, and shall make it possible to start either signal generator generating any of its specified signals.

E-2b(2)e(4). Volume indicator meters and test selector switches.-

E-2b(2)e(4)a. Input metering.- An input volume indicator meter and its associated input selector switch or switches shall be provided, and shall make it possible to measure the output of either signal generator selected for test, and to connect the output of this signal generator to the input of any amplifier channel selected for test. For 17MC systems, wherein each signal generator is associated only with one amplifier channel, it shall only be required that a signal generator be connected to its associated amplifier channel.

E-2b(2)e(4)b. Output metering.- An output volume indicator meter, and its associated output selector switch or switches shall be provided, and shall permit measuring the output of any amplifier assembly selected for test.

E-2b(2)e(4)c. Combined metering.- For small amplifier systems, as classified in paragraph E-2b(2)a(1)a, it will be permissable to combine the metering facilities specified in paragraphs E-2b(2)e(4)a and E-2b(2)e(4)b by utilizing a common meter for both input and output volume indication.

E-2b(2)e(5). Test load resistor and switch.- A non-locking type switch and associated test load resistor (of such value as to draw rated output power at 70 volts) shall be provided to permit connecting rated load to the output of the amplifier channel being tested.

E-2b(2)e(6). Test input volume control.- A test input volume control shall be provided for controlling the test signal level of the signal generator output (or amplifier channel input) to assist in establishing reference test input levels and to aid in making compressor adjustments. The setting of this control shall be indicated on a fixed scale, suitably marked. Operation of this control shall not affect the system input circuits.

E-2b(2)f. Control circuits.- Control circuits shall be provided to accomplish the following system functions:

E-2b(2)f(1). Microphone press-to-talk switches.- The operation of a microphone press-to-talk switch at a microphone station or box shall:

(a) Connect the microphone audio circuit at that station or box to the input of the selected amplifier channel.
(b) Apply power to the loudspeaker selector switches at that station or box.
(c) Energize appropriate busy indicator circuits.
(d) Actuate appropriate system priority circuits.
(e) Place the connected amplifier channel in the "ready" condition, as defined in paragraph E-2b(1)a(4)b.
(f) Connect the output of the selected amplifier channel to the selected loudspeaker groups.
(g) Disconnect the adjacent loudspeakers on the local loudspeaker circuit associated with that transmitter or box, as specified in paragraph E-2b(2)f(7).

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E-2b(2)f(2). Loudspeaker group selector switches.- The operation of a loudspeaker group selector switch at a microphone station or box shall connect the associated loudspeaker group to the output of the selected amplifier channel, only if the microphone press-to-talk switch is operated at that station or box.

E-2b(2)f(3). Alarm contact makers.-

E-2b(2)f(3)a. General.- The closure of an external alarm contact maker (consisting of a single-pole, single throw switch furnished by the installing activity) shall:

(1) Energize appropriate busy indicator circuits;
(2) Actuate appropriate system priority circuits;
(3) Place the selected signal generator in the "ready" condition, as defined in paragraph E-2b(1)a(4)b, causing the appropriate alarm signal to be generated.
(4) Connect the output of the selected signal generator to the interstage input of the selected amplifier channel(s);
(5) Place the selected amplifier channel(s) in the "ready" condition, as defined in paragraph E-2b(1)a(4)b.
(6) Connect the output of the selected amplifier channel(s) to all connected loudspeakers (those loudspeaker groups not actually disconnected by the loudspeaker disconnect switches, specified in paragraph E-2b(2)b(7)) including the 11-16MC and 17 MC loudspeakers, where applicable;
(7) Apply full rated output voltage to all loudspeaker groups, regardless of the setting of the system volume controls specified in paragraph E-2b(2)b(8).

E-2b(2)f(3)b. Additional functions for specific systems.- For specific amplifier systems, as classified by paragraph E-2b(2)a(1)b, the following additional functions shall be performed by the closure of an alarm contact maker:

E-2b(2)f(3)b(1). 1-2-3 MC and 1-6 MC systems.- Closure of a chemical alarm contact maker shall also energize an external visual alarm circuit, capable of making and breaking a resistive load of 15 amperes at 115 volts, 60 cycles, single phase. Momentary closure of a general alarm contact maker shall also cause this same circuit to be energized intermittently at the approximate rate of 1/3-second on, 1/3-second off for a period of 15 seconds.

E-2b(2)f(3)b(2). 1-7 MC systems.- Momentary closure of a general alarm contact maker shall also energize an external visual alarm circuit intermittently at the approximate rate of 1/3-second on, 1/3-second off for a period of 10 seconds. The circuit shall be capable of making and breaking a resistive load of 5 amperes at 115 volts, 60 cycles, single phase.

E-2b(2)f(4). Loudspeaker group selection.- In general, it shall be possible for each microphone station or box (provided with loudspeaker group selector switches) on a particular announcing circuit to select any of the loudspeaker groups associated with that circuit. In the case of 1-2-3 MC amplifier systems, as classified in paragraph E-2b(2)a(1)b, it shall also be possible for microphone stations on one circuit to select loudspeaker groups on other circuits outlined as follows: Any other special combinations of loudspeaker groups shall be specified in the contract or order (see par. H-1b).

E-2b(2)f(4)a. Circuit 1MC microphone stations shall be able to select circuit 2 MC and 3 MC loudspeaker groups, in addition to circuit 1 MC loudspeaker groups.

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E-2b(2)f(4)b. Circuit 2 MC microphone stations shall be able to select all "crew" loudspeaker groups on circuit 1 MC, in addition to circuit 2 MC loudspeaker groups.

E-2b(2)f(5). System priorities.-

E-2b(2)f(5)a. Alarm signal priorities.- Alarm signals shall have priority over all other control features of the system and over all speech from microphone stations or boxes. In effecting priority of alarm signals over speech, it will be satisfactory to superimpose the generated signal on the speech output, provided the design of the amplifier system is such that the clarity and intensity of the generated signal remains satisfactory. Alarm signals shall take priority over other alarm signals in the same system in the following order, those listed first having highest priority:

(1) Collision alarm
(2) Diving alarm
(3) Chemical attack alarm
(4) General alarm
(5) Flight crash
(6) Flight warning
(7) Cease firing
(8) Salvo

E-2b(2)f(5)b. Speech priorities.- Unless otherwise specified, there shall be no priority of speech between microphone stations or boxes on the same announcing circuit. However, when two or more circuits are connected to a common amplifier channel, or when two or more microphone stations or boxes on different circuits are attempting to select the same loudspeaker group, priority between announcing circuits shall take effect in the following order, those circuits listed first having highest priority:

(1) 1 MC
(2) 3 MC
(3) 2 MC
(4) 5 MC
(5) 6 MC
(6) 7 MC
(7) 17 MC

E-2b(2)f(6). Busy indicator circuits.- Busy indicator circuits shall be provided for all amplifier systems, where it is possible to connect more than one microphone station or box to a single amplifier channel. Where more than one announcing circuit is involved, each microphone station or box shall be provided with one busy indicator circuit for each announcing circuit. These busy indicator circuits shall be energized in accordance with the following general principles:

E-2b(2)f(6)a. A microphone station or box shall receive a busy indication when the amplifier channel to which it is connected is in use, or when a loudspeaker group which can be selected by that station or box is in use.

E-2b(2)f(6)b. When more than one announcing circuit is connected to a single amplifier channel, the busy indicator circuits corresponding to these announcing circuits shall be paralleled and shall be energized simultaneously.

E-2b(2)f(6)c. When an alarm signal is being transmitted, all busy indicator circuits involved shall be energized.

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ck connections, but all external ship's cables shall be connected at the front of rack.

E-2b(2)f(7). Local loudspeaker cut-out circuit.- A local loudspeaker cut-out circuit shall be provided in the amplifier for each microphone station or box on 1-2-3 MC, 1-6 MC and 1-7 MC amplifier systems, as classified in paragraph E-2b(2)a(1)b, to permit automatic cut-out of loudspeakers adjacent to that station or box when it is in use.

E-2b(2)f(8). Spare control relays.- Additional relays shall be provided in the amplifier on the basis of one relay of each type for each 25 (or fraction thereof) of that type employed in the amplifier system, provided a minimum of five of that type are employed. These spare relays shall be for the purpose of facilitating future system revisions or modifications, shall be mounted adjacent to the active relays of corresponding type, and shall be left unconnected.

E-3. Loudspeakers.-

E-3a. Introduction.- Loudspeakers covered by this specification shall be of the following general types, with type designations assigned in accordance with paragraph B-5.

E-3a(1). Low power loudspeakers, types IC/SAA and IC/SAG.- Low power loudspeakers are intended for installation aboard ship in compartments and areas having comparatively low ambient noise levels, such as ward rooms, mess rooms and living spaces. Although speech intelligibility and clarity of alarm signal transmission shall be the primary design objective, strong secondary emphasis shall be placed upon obtaining naturalness of speech reproduction. The two types of low power loudspeakers are distinguished as follows:

Type	Feature
IC/SAA	without volume control
IC/SAG	with volume control

E-3a(2). High power loudspeakers, types IC/SBA and IC/SBG.- High power loudspeakers are intended for installation aboard ship in compartments and areas where high ambient noise levels are likely to be encountered, such as machinery spaces, turrets, gun mounts and weatherdeck areas. High speech intelligibility and clarity of alarm signal transmission are of paramount importance. The two types of high power loudspeakers are distinguished as follows:

Type	Feature
IC/SBA	without volume control
IC/SBG	with volume control

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E-3a(3), Super power loudspeakers, types IC/SCD, IC/SDC, IC/SDD, IC/SDF, IC/SEB, IC/SEC and IC/SFD.- Super power loudspeakers are intended primarily for installation aboard ship in exposed locations where high intensity sound output is required to carry over long distances or to override extremely high noise levels. High speech intelligibility, clarity of alarm signal transmission and high directionality of the sound distribution pattern are the design objectives. The various types of super power loudspeakers are distinguished as follows:

E-3a(3)a. Movable types.-

Type	Power rating - watts	Sound pattern
IC/SEC	500	Circular
IC/SDC	250	Circular

E-3a(3)b. Fixed types.-

Type	Power rating - watts	Sound pattern
IC/SFD	1000	Elliptical
IC/SEB	500	Circular
IC/SDD	250	Elliptical
IC/SDF	250	Elongated
IC/SCD	125	Elliptical

E-3a(4). Pressure proof loudspeaker, type IC/SHA.- Pressure proof loudspeakers, type IC/SHA are intended for topside installation aboard submarines, where they may be subjected to submergence for long periods at considerable depths without requiring protective covers or external enclosures.

E-3a(5). Headset, type IC/SJA and jack box, type IC/SKG.- Headsets, type IC/SJA, and their associated jack boxes, type IC/SKG, are intended for use in compartments or locations aboard ship where only one person is required to hear a message with minimum disturbance to others in the area. The headset shall be of the sound powered telephone type; and the headset receivers, headband, cable and plug shall conform to the requirements of Navy Department Specification R-17T23. The jack box shall be in accordance with the requirements of Bureau of Ships plan 9-S-5454-L (Type T-2S), modified to provided a volume control in place of one of the jack elements.

E-3a(6). Matching transformer, type IC/SLA.- Matching transformer, type IC/SLA, is intended for use with the pressure proof loudspeaker type IC/SHA, for use both as an input and an output matching transformer. The transformer shall be mounted in a watertight enclosure designed for bulkhead mounting.

E-3b. Mechanical requirements.-

E-3b(1). General construction.-

E-3b(1)a. Degree of enclosure.-

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E-3b(1)a(1). All types, except IC/SHA.- All loudspeaker equipment with the exception of pressure-proof loudspeakers (type IC/SHA), shall be of waterproof construction as defined in paragraph D-9b(3). The loudspeaker design shall ensure drainage for all throats, chambers and recesses to prevent the retention of any trapped moisture or condensation which may enter the loudspeaker enclosure while mounted with the sound axis in any position between zero degrees and minus 45 degrees elevation relative to a horizontal position.

E-3b(1)a(2). Type IC/SHA.- Pressure-proof loudspeakers, type IC/SHA, shall be of pressure-proof construction as defined in paragraph D-9b(1). Drainage shall be provided to insure against the retention of trapped water in any recesses external to the pressure proof enclosure, while mounted with its sound axis in any position between zero degrees and minus 45 degrees elevation relative to a horizontal position.

E-3b(1)b. Ease of installation.- Loudspeakers shall be so designed as to permit ease of installation. It shall be possible to tighten mounting bolts with a socket wrench from the front.

E-3b(1)c. Serviceability.- Any servicing or replacement of repair parts shall not require removal of the loudspeaker from its mounting.

E-3b(1)d. Protective screens.- Protective screens shall be provided to protect exposed portions of cones, and to exclude foreign objects from the acoustic path.

E-3b(2). Loudspeaker driver unit.- The loudspeaker driver unit is defined as the magnet structure together with the voice coil and diaphragm assembly, exclusive of horn or housing.

E-3b(2)a. Magnet structure.-

E-3b(2)a(1). Permanent magnet field.- All loudspeaker fields shall be of the permanent magnet type.

E-3b(2)a(2). Resistance to corrosion.- The magnet structure shall be treated to provide resistance to corrosion, and to prevent the formation of corrosion products in the air gap. The following finishing treatment, or its equivalent, shall be employed:

(a) Basic corrosion resistive treatment as specified in Navy Department Specification 17-I-61, followed by:
(b) A phosphating treatment, followed by:
(c) Two coats of zinc chromate primer (N. D. Spec. 52P18).

E-3b(2)a(3). Air gap clearance.- Minimum air gap clearance on either side of the voice coil assembly shall be 0.007 inches.

E-3b(2)a(4). Serviceability.- Removal of the magnet structure as a unit together with the voice coil and diaphragm assembly shall not expose the magnetic gap, nor require removal of such parts as the transformer or terminal board.

E-3b(2)b. Voice coil and diaphragm assembly.-

E-3b(2)b(1). Serviceability.- All voice coil and diaphragm (or cone) assemblies shall be replaceable readily without necessitating cementing or soldering.

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E-3b(2)b(2). Protection, positioning and polarization.- The voice coil and diaphragm assembly shall be provided with accurate guides to perform the following functions:

(a) Permit accurate and positive positioning of the voice coil in the air gap.
(b) Prevent damage to the voice coil while being inserted into or removed from the magnet structure.
(c) Prevent damage to the voice coil if assembly is laid on a flat surface.
(d) Permit insertion into the air gap in only one position.

E-3b(2)b(3). Materials.-

E-3b(2)b(3)a. Diaphragms and cones.- Diaphragms and cones of all loudspeakers, except type IC/SHA, shall be of molded phenolic. Other materials will be considered by the bureau concerned if it can be demonstrated that they will give equal or superior performance.

E-3b(2)b(3)b. Resistance to corrosion.- Material used for gaskets, cone supports and voice coil supports shall be impregnated to render them impervious to moisture and to provide resistance tc corrosion.

E-3b(2)b(3)c. Dissimilar metals.- The use of dissimilar metals for the voice coil winding and the voice coil lead-in wires shall be avoided.

E-3b(2)b(3)d. Rubber.- No portion of a cone shall contain exposed rubber or other material which in the opinion of the bureau concerned will tend to change its characteristics when placed in an atmosphere containing salt, moisture or oil.

E-3b(3). Super power loudspeakers.-

E-3b(3)a. Multi-unit design.- Super power loudspeakers shall be of multi-unit design employing multiple driver units suitably coupled to individual horns or to a common horn assembly. Individual driver units shall be readily removable for servicing.

E-3b(3)b. Facilities for installation.- Mounting brackets and lifting rings shall be provided.

E-3b(3)c. Loudspeaker, type IC/SEC.-

E-3b(3)c(1). Loudspeaker mount.- Super power loudspeaker, type IC/SEC, shall be provided with a mount which will permit manual adjustment of the loudspeaker sound axis in both train and elevation.

E-3b(3)c(2). Limits of adjustment.- Adjustment of the loudspeaker sound axis shall be possible within limits of 360 degrees in train (azimuth), and within limits of +60 degrees (above horizontal) and -30 degrees (below horizontal) in elevation.

E-3b(3)c(3). Locking devices.- A locking type brake shall be provided for holding the loudspeaker in any position of train or elevation. In addition, a snaplock device shall be provided to permit setting the loudspeaker accurately, and locking it securely in a fixed position predetermined at the time of installation.

E-3b(3)c(4). Position indication.- Calibrated scales shall be provided to indicate in degrees, the angles of train and elevation.

E-3b(3)c(5). Mounting axes.- Both the horizontal and vertical mounting axes shall pass through the approximate center of gravity of the loudspeaker in order to minimize moments resulting from motion of the vessel on which mounted.

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E-3b(3)c(6). Bearings.- All training and elevating mechanisms shall be fitted with ball bearings (listed in the Catalogue of Naval Materials).

E-3b(3)c(7). Operator's position.- The design of the training and elevating mechanism shall be such that the operator will not be required to stand within an arc 120 degrees each side of the loudspeaker sound axis measured in the horizontal plane.

E-3b(3)d. Loudspeaker, type IC/SDC.-

E-3b(3)d(1). Loudspeaker mount.- Super power loudspeaker, type IC/SDC, shall be provided with a mount which will permit adjustment of the loudspeaker sound axis within limits of 270 degrees in train (azimuth).

E-3b(3)d(2). Operating facility.- The rotating shaft shall be fitted with a wheel or crank, which will permit insertion of an extension shaft or flexible coupling by the installing activity for remote operation.

E-3b(3)d(3). Locking device.- Facilities for locking the loudspeaker in any position within its limit of rotation shall be included.

E-3b(3)d(4). Position indication.- A calibrated scale shall be provided to indicate in degrees the angle of train.

E-3b(3)d(5). Mounting axis.- The mounting axis shall pass through the approximate center of gravity of the loudspeaker in order to minimize moments resulting from motion of the vessel on which mounted.

E-3b(3)d(6). Bearings.- The training mechanism shall be fitted with ball bearings (listed in the Catalogue of Navy Material).

E-3b(4). Maximum weight and dimensions.- Loudspeaker weights and dimensions shall not exceed the following values: Every effort shall be made in the design of the equipment to hold equipment weight and size to a practicable minimum.

Loudspeaker type	Weight (pounds)	Height (inches)	Width (inches)	Depth (inches)
IC/SAA	15	11	11	8-1/2
IC/SAG	15	11	11	8-1/2
IC/SBA	25	14	14	10
IC/SBG	25	14	14	10
IC/SCD	150	38	12	16
IC/SDC	300	36	32	22
IC/SDD	225	36	20	16
IC/SDF	250	72	12	16
IC/SEB	375	36	36	16
IC/SEC	1000	62	54	30
IC/SFD	750	90	40	16
IC/SHA	60	10	10	9

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E-3b(5). Terminal tubes.- Provision shall be made for accommodating a minimum of two terminal tubes (see Bureau of Ships publication - Description and Application of Navy Standard Stuffing Tubes, Terminal Tubes and Cable Clamps), at the bottom or sides of each loudspeaker. Terminal tube sizes shall be as follows:

Low power loudspeaker	C
High power loudspeaker	C
Super power loudspeaker	G
Pressure proof loudspeaker	C

E-3b(6). Resistance to gun blast.- All loudspeakers, except types IC/SAA and IC/SAG, shall, at the conclusion of the blast test specified in paragraph F-3f(18), show no damage and shall not vary in acoustic performance beyond the limits specified in paragraphs E-3c(3) and E-3c(5). Damage as used above shall be interpreted as covering evidence of rupture, fracture or tearing of any part, or mechanical deformation such as would impair the operation of the device or the installation of spare parts.

E-3b(7). Resistance to pressure.- Pressure-proof loudspeakers shall, after exposure to submergence as specified in paragraph D-9b(1), show no evidence of physical damage, or of water leaking into the housing, and shall not vary in acoustic performance beyond the limits specified in paragraph E-3c(3) and E-3c(5). Damage as used above shall be interpreted as covering evidence of rupture, fracture or permanent distortion of any part such as would impair the operation of the device or the installation of replacement parts.

E-3b(8). Resistance to depth charge.- Pressure-proof loudspeakers, type IC/SHA, shall at the conclusion of the depth charge test described in paragraph F-3f(19), show no damage, shall not vary in acoustic performance beyond the limits specified in paragraphs E-3c(3) and E-3c(5), and shall still be capable of meeting the pressure proof requirement of paragraph D-9b(1). Damage as used above shall be interpreted as covering evidence of rupture, fracture or tearing of any part such as would impair the operation of the device or the installation of replacement spare parts.

E-3b(9). Endurance.- Loudspeakers shall be capable of withstanding an endurance test of 100 hours operation at rated operating voltage (70 volts r.m.s.) followed in the case of all except super power loudspeakers by one hour's operation at 99 volts r.m.s. (3 decibels overvoltage), when tested at an ambient temperature of 132°F. in accordance with paragraph F-3f(15). Loudspeakers shall show no deleterious effects, nor shall there be any lowering of acoustic output or failure to meet the other requirements of this specification as a result of this endurance test.

E-3c. Electrical requirements.-

E-3c(1). Rated operating voltage.- The rated operating voltage of all loudspeaker equipment shall be 70 volts r.m.s.

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E-3c(2). Maximum electrical input.- The volt-ampere electrical input of the various loudspeakers when connected to a 70 volt power source as described in paragraph F-3f(9) shall not exceed the following values:

Loudspeaker type	Maximum volt-amperes
IC/SAA	1.0
IC/SAG	1.0
IC/SBA	7.5
IC/SBG	7.5
IC/SCD	125.
IC/SDC	250.
IC/SDD	250.
IC/SDF	250.
IC/SEB	500.
IC/SEC	500.
IC/SFD	1000.
IC/SHA	5.0

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E-3c(3). Minimum output sound pressure.- The loudspeakers shall be capable of producing minimum output sound pressures as follows, measured on the sound axis at the indicated distance and as described in paragraph F-3f(10). Following gun blast, pressure and depth charge tests, specified in paragraphs E-3b(6), E-3b(7) and E-3b(8), respectively, the measured output sound pressures shall not vary in excess of 2.0 decibels from their original values.

Loudspeaker type	Distance feet	320-500 cycle warble		500-800 cycle warble		800-1250 cycle warble		1250-2000 cycle warble		2000-3200 cycle warble		3200-5000 cycle warble	
		dynes/cm²	db*	dynes/cm²	db*	dynes/cm²	db*	dynes/cm²	db*	dynes/cm²	db*	dynes/cm²	db*
IC/SAA	10	3.5	85	5.0	88	5.0	88	5.0	88	5.0	88	5.0	88
IC/SAG	10	3.5	85	5.0	88	5.0	88	5.0	88	5.0	88	5.0	88
IC/SBA	10	--	--	44.5	107	50	108	56.1	109	63	110	63	110
IC/SBG	10	--	--	44.5	107	50	108	56.1	109	63	110	63	110
IC/SCD	30	--	--	71	111	80	112	80	112	71	111	40	106
IC/SDC	30	--	--	142	117	160	118	160	118	142	117	80	112
IC/SDD	30	--	--	142	117	160	118	160	118	142	117	80	112
IC/SDF	30	--	--	142	117	160	118	160	118	142	117	80	112
IC/SEB	30	--	--	285	123	320	124	320	124	285	123	142	117
IC/SEC	30	--	--	285	123	320	124	320	124	285	123	142	117
IC/SFD	30	--	--	565	129	634	130	634	130	565	129	252	122
IC/SHA	10	--	--	10	94	10	94	10	94	10	94	6.9	93

*Decibels relative to 0.0002 dynes/cm².

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E-3c(4). Output sound pressure distribution.- When measured as described in paragraph F-3f(11), the minimum off-axis sound pressures for a 700-1300 cycle warble signal shall be as follows:

Loudspeaker type	Distance from loudspeaker	Minimum sound pressure in decibels[1] Off-axis angle in degrees					
		5	10	15	20	30	45
	Feet						
IC/SAA	10			-0.5		-1.5	-3.0
IC/SAG	10			-0.5		-1.5	-3.0
IC/SBA	10		-1.5		-3.0	-6.0	
IC/SBG	10		-1.5		-3.0	-6.0	
IC/SDC	30	-0.5	-3.0	-6.0			
IC/SEB	30	-1.0	-5.0	-9.0			
IC/SEC	30	-1.0	-5.0	-9.0			
IC/SHA	10			-0.5		-1.5	-3.0

		Off-axis angle in degrees								
		Horizontal plane						Vertical plane		
		5	10	15	20	30	45	5	10	15
IC/SCD	30			-2.0		-4.0	-7.0	-1.0	-4.0	-8.0
IC/SDD	30		-1.0		-5.0	-9.0		-1.0	-4.0	-8.0
IC/SDF	30			-1.0		-3.0	-6.0	-3.0	-10.0	
IC/SFD	30	-2.0	-6.0	-10.0				-4.0	-11.0	

[1]Decibels relative to measured on-axis sound pressures.

E-3c(5). Frequency response characteristic.- When measured as described in paragraph F-3f(12) the frequency response characteristics of the various loudspeakers shall fall within the limits shown on figures 5, 6, 7 and 8. The mean of the measured response curve shall have a slope closely following the slope of the desired curve except that sharp peaks and dips may extend beyond these limits provided the width at the limit is not greater than one tenth octave (1/10) and provided that there is no more than one such extension in any octave interval. Following the gunblast, pressure, and depth charge tests required in paragraphs E-3b(6), E-3b(7) and E-3b(8), respectively, the response at any frequency shall not vary in excess of two decibels (2 db) from its corresponding original value.

Loudspeaker type	Figure no.
Low power	5
High power	6
Super power	7
Pressure proof	8

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E-3c(6). Load characteristic.- The loudspeaker load characteristic (sound pressure output versus volt ampere input) when determined as described in paragraph F-3f(13) and plotted on logarithmic scales, shall be essentially a straight line. The maximum deviation from the straight line portion of the curve at the point corresponding to operation at 70 volts (r.m.s.) shall be 1.0 decibels.

E-3c(7). Loudspeaker noises.- The loudspeaker design shall be such that natural resonances of the lead-in wires and other mechanical structures shall occur at frequencies outside of the useful range of the loudspeaker. In any event loudspeaker noises such as buzzes, rattles and other spurious sounds should be low enough in intensity so as not to impair the articulation or quality of speech and signal transmission.

E-3c(8). Polarization of voice coil connections.- The mechanical polarization of the voice coil sub-assembly specified in paragraph E-3b(2)b(2)d and the electrical connections thereto shall insure that an outward excursion of the diaphragm will result when a positive potential is applied to the terminal marked "Line + " on the primary winding of the loudspeaker transformer.

E-3c(9). Loudspeaker transformer.- Each loudspeaker shall be provided with a transformer for matching the voice coil circuit to the incoming audio lines. Transformer taps shall be available to permit normal operation at reduced line voltage and to permit operation at reduced output sound pressures, as follows:

E-3c(9)a. Operation at reduced input line voltage.- For low and high power loudspeakers only, an auxiliary transformer tap connection shall be provided, which will deliver the same output voltage to the voice coil circuit from a 50-volt input line, as the normal transformer connection delivers from a 70 volt input line.

E-3c(9)b. Operation at reduced output sound pressures.- For low power and high power loudspeakers only, transformer tap connections shall be provided to reduce the output voltage to the voice coil circuit by 6, 12 and 18 decibels from its full value, while employing either the rated or reduced input voltage taps.

E-3c(9)c. Transformer location.- The loudspeaker transformer shall be incorporated as an integral part of the loudspeaker for low power, high power, and type IC/SEC loudspeakers. For other super power types, the transformer may be mounted in a separate watertight enclosure. (See par. E-3c(16) for requirements covering matching transformer, type IC/SLA to be used with type IC/SHA pressure-proof loudspeaker.)

E-3c(10). Loudspeaker volume control.- Loudspeakers, types IC/SAG and IC/SBG, shall be provided with volume controls which can be operated readily from outside of the loudspeaker. Such volume controls shall be marked and shall permit a 24 decibel reduction in the voice coil voltage from its value at any given loudspeaker transformer tap connection, such reduction to be accomplished in a minimum of 4 steps.

E-3c(11). Individual loudspeaker unit disconnect switches.- Super power loudspeakers shall be provided with disconnect switches to permit opening the electrical circuit to the individual driver units. These switches shall be mounted in the base of the mount for type IC/SEC loudspeakers. For other types of super power loudspeakers, these switches shall be mounted in the enclosure which houses the loudspeaker transformer.

E-3c(12). Terminal board.- Each loudspeaker shall be provided with a terminal board to accommodate all transformer tap connections, as well as to receive incoming audio lines. Adjustment of loudspeaker volume by changing transformer taps shall be accomplished at the terminal board without necessitating soldering. Loudspeaker servicing and replacement of spare parts shall not require disconnection of the ship's wiring.

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E-3c(13). Electrical circuits.- The electrical circuits and terminal board markings shall conform to the requirements of figure 16.

E-3c(14). Pressure proof loudspeaker, type IC/SHA.-

E-3c(14)a. Use as a microphone.- In addition to meeting the loudspeaker requirements of this specification, the type IC/SHA loudspeaker shall be designed for alternate use as a microphone. For this function, the unit shall have satisfactory intelligibility with the residual background noise due to wind reduced to a minimum.

E-3c(14)b. Sensitivity.- The voltage developed by the loudspeaker when used as a microphone across a 150-ohm resistive load with an input sound pressure of 4 dynes/cm^2 shall be not less than 7.0 millivolts, when measured as specified in paragraph F-3f(14).

E-3c(14)c. Voice coil impedance.- The nominal voice coil impedance measured at a frequency of 1000 cycles per second shall be 8 ohms ($\pm$ 2 ohms).

E-3c(14)d. Use with matching transformer.- The requirements of paragraphs E-3c(1) to E-3c(6), inclusive, shall be met with the pressure-proof loudspeaker operating in conjunction with a matching transformer having the electrical characteristics of the type IC/SLA transformer described in paragraph E-3c(16). The -6 db reduced power output tap of the transformer shall be used for this purpose. Similarly, the requirement of paragraph E-3c(14)b shall be met using the type IC/SLA transformer.

E-3c(15). Jack box, type IC/SKG.- The jack box, type IC/SKG, shall be provided with an attenuator circuit, which in combination with the volume control specified in paragraph E-3a(5), and when connected to a 70 volt (r.m.s.) audio line, shall deliver to a 600 ohm load a voltage which may be varied from 5 volts to 0.5 volts (r.m.s.).

E-3c(16). Matching transformer, type IC/SLA.-

E-3c(16)a. Tap connections.- The matching transformer, type IC/SLA, shall be designed as a combination input and output transformer, one winding of which shall match the voice coil impedance (8 ohms nominal at 1000 cycles per second) of the pressure proof loudspeaker, type IC/SHA. The other winding shall be provided with taps as follows:

(1) Amplifier input tap.- This tap shall match to an amplifier input impedance of 150 ohms.
(2) Amplifier output tap.- This tap shall permit the transformer, when terminated by an eight ohm resistive load, to draw 20 watts at 1000 cycles per second from a 70 volt (r.m.s.) power source.
(3) Reduced power output taps.- Additional taps shall be provided to permit reducing the power drawn from a 70 volt (r.m.s.) power source under the conditions specified in paragraph E-3c(16)a(2) by fixed amounts of 6 and 12 decibels below 20 volt-amperes.

E-3c(16)b. Frequency response characteristic.- The type IC/SLA matching transformer shall have a frequency response characteristic that is flat within $\pm$ 1.5 decibels over the frequency range of from 200 to 6000 cycles per second when delivering 20 watts of electrical power to an 8-ohm resistive load.

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F. METHODS OF SAMPLING, INSPECTION AND TESTS

F-1. Classification.- The methods of sampling, inspection and tests of announcing equipment shall fall within the following classifications:

Type or brand approval tests (at Material Laboratory, New York Naval Shipyard).
Contract suitability tests (at Material Laboratory, New York Naval Shipyard).
Inspection tests (at place of manufacture).

F-2. General.- The following applies to tests conducted at the Material Laboratory, New York Naval Shipyard.

F-2a. The standards and methods of tests employed by the Material Laboratory shall be considered as final.

F-2b. The manufacturer shall conduct such tests as may be required to give reasonable assurance that the components submitted for laboratory test will meet the specification requirements before submission is made.

F-2c. Adequate spare parts shall be provided with each component submitted for laboratory test to insure the completion of the tests.

F-2d. Components submitted for laboratory test shall be accompanied by plans showing details of construction and containing a list of the materials and finishes used, and shall include all circuit schematic wiring diagrams.

F-3. Qualification (type or brand approval) tests.-

F-3a. Qualification (type or brand approval).- Type or brand approval tests shall demonstrate that a manufacturer is qualified to produce components suitable for Naval use in accordance with the requirements of this specification. Type or brand approval tests may be conducted on individual components, combinations of components or complete equipments.

F-3a(1). Individual components.- Tests of individual components such as a particular type of microphone, amplifier, or loudspeaker shall demonstrate that the manufacturer is qualified to produce that particular component.

F-3a(2). Class of components.- Tests of several representative types of a particular class of components such as microphone components, amplifier components or loudspeaker components shall be considered as demonstrating that the manufacturer is qualified to produce all items within that particular class of components. As an example: A manufacturer could be approved as a supplier for all types of loudspeaker components if he has presented and had approved each representative type of loudspeaker, such as one low power, one high power, one super-power and one pressure-proof loudspeaker.

F-3a(3). Complete equipment.- Tests of a complete equipment of one manufacture consisting of representative types of each class of components shall be considered as demonstrating that the manufacturer is qualified to produce the equipment covered by this specification.

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F-3b. Supplementary data.- Components submitted for laboratory tests shall be accompanied by copies of the manufacturer's test data and curves similar to that required to be taken by the laboratory in testing to determine compliance with the requirements of this specification.

F-3c. Manufacturer's test engineers.- The manufacturer's test engineers shall be available during the course of the tests to offer advice, data on characteristics and assistance as needed.

F-3d. Granting of approval.- Where the laboratory's report of tests indicates a need for changes in the component tested in order for it to be acceptable for Naval use and to comply with specification requirements, the bureau concerned, may grant approval at its own discretion, without requiring complete retest, if the defects are minor ones, easily correctable. In such cases however, approval will be withheld until the manufacturer has completed the modifications, and incorporated the changes in the plans and data submitted for the bureau files.

F-3e. Order of tests.- Tests shall be conducted as applicable in the following order. Listed opposite each test are the corresponding referenced paragraphs which give the specified performance requirements and the associated test description. The absence of a referenced paragraph under the test description column indicates either that the test required to establish conformance with the specification is apparent from the referenced performance requirement, or that the test consists of a direct measurement.

F-3e(1). Microphone component tests.-

Test	Performance Requirements	Test Description
(a) General Inspection	C, D, E-1	F-3f(1)
(b) Microphone Sensitivity	E-1c(1)b	F-3f(2)
(c) Frequency Response	E-1c(1)c	F-3f(3)
(d) Induced Voltage	E-1c(1)d	F-3f(4)
(e) Noise Pickup	E-1c(3)	F-3f(5)
(f) Dielectric Strength	D-5j(1)	F-3f(20)
(g) Insulation Resistance	D-5j(2)	
(h) Radio Frequency Noise	D-5c	
(i) Microphone Endurance	E-1b(1)f	F-3f(6)
(j) Switch Endurance	E-1b(1)d(4) E-1b(3)a(2)d(3)	
(k) Strength of Cable Assembly	E-1b(3)c	
(ℓ) Dripproof Test	D-9b(4), E-1a(4)	
(m) Watertight Test	D-9b(2), E-1a(1), E-1a(2), E-1a(3), E-1a(5), E-1a(8), E-1a(9)	
(n) Pressureproof Test	D-9b(1), E-1b(2)d	
(o) Salt-Spray Test	D-4s	F-3f(16)
(p) Vibration Test	D-4r	
(q) Tumbling Test	E-1b(3)d	F-3f(17)
(r) Gun Blast Test	E-1b(1)e	F-3f(18)
(s) Microphone Sensitivity	E-1c(1)b	F-3f(2)
(t) Frequency Response	E-1c(1)c	F-3f(3)
(u) Depth Charge Test	E-1b(2)e	F-3f(19)
(v) Pressureproof Test	D-9b(1), E-1b(2)d	
(w) Shock Test	D-4q	

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F-3e(2). Amplifier component tests.-

Test	Performance Requirement	Test Description
(a) General Inspection	C, D, E-2	F-3f(1)
(b) Power Supply Voltage and Frequency	D-2	F-3f(7)
(c) Input Impedance	E-2b(1)b(2)a, E-2b(1)c(2)	
(d) Output Voltage and Power	E-2b(1)b(2)b, E-2b(1)b(2)c	E-2b(1)b(2)d(1), F-3f(7)a
	E-2b(1)c(3), E-2b(1)c(4)	E-2b(1)c(5), F-3f(7)a
(e) Reserve Gain	E-2b(1)b(2)d(2)	E-2b(1)b(2)d(2), F-3f(7)a
(f) Frequency Response	E-2b(1)b(2)e E-2b(1)c(6)	
(g) Distortion	E-2b(1)b(2)f(1) E-2b(1)b(2)f(2) E-2b(1)c(7)	
(h) Noise and Microphonics	E-2b(1)b(2)g E-2b(1)c(8), D-4r	
(i) Output Voltage Regulation	E-2b(1)b(2)h E-2b(1)c(9)	
(j) Isolation Against Longitudinal Voltages	E-2b(1)b(2)i	
(k) Stability	E-2b(1)b(2)j E-2b(1)c(10)	
(ℓ) Overload Capacity	E-2b(1)b(2)k	E-2b(1)b(2)k, F-3f(7)b
	E-2b(1)c(11)	E-2b(1)c(11), F-3f(7)b
(m) Amplifier Phasing	E-2b(1)a(5)	
(n) Compressor Characteristic	E-2b(1)b(3)b	
(o) Compressor Time Constant	E-2b(1)b(3)c	
(p) Compressor Adjustment	E-2b(1)b(3)d	
(q) Compressor Effect on Amplifier Characteristic	E-2b(1)b(3)e	
(r) Limiter Characteristic	E-2b(1)b(4)a	
(s) Limiter Time Constant	E-2b(1)b(4)b	
(t) Gain Control	E-2b(1)b(5)a	
(u) Interstage Input Impedance	E-2b(1)b(6)a	
(v) Interstage Input Level	E-2b(1)b(6)b	
(w) Generated Signals	E-2b(1)d(3)	
(x) Duration of Generated Signals	E-2b(1)d(4)	
(y) Signal Generator Output Level	E-2b(1)d(5)	E-2b(1)d(5), F-3f(7)a
(z) Signal Generator Noise and Microphonics	E-2b(1)d(7), D-4r	

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Test	Performance Requirement	Test Description
(aa) Time Interval Between "Standby" & "Ready"	E-2b(1)a(4)c	
(bb) VI Circuit Voltage	E-2b(2)a(3)b	
(cc) System Volume Controls	E-2b(2)b(8)	
(dd) VI Circuit Protection	E-2b(2)c(3)	
(ee) Busy Indicator Circuit Protection	E-2b(2)c(4)	
(ff) Loudspeaker Audio Circuit Protection	E-2b(2)c(5)	
(gg) Loudspeaker Group Output Voltage Adjustment	E-2b(2)d(1)	
(hh) External Alarm Circuits	E-2b(2)f(3)b(1) E-2b(2)f(3)b(2)	
(ii) Dielectric Strength Test	D-5j(1)	F-3f(20)
(jj) Insulation Resistance	D-5j(2)	
(kk) Radio Frequency Noise	D-5c(1)b	
(ll) Temperature Rise	D-5h(2)	F-3f(8)
(mm) Dripproof Test	D-9b(4), E-2a(2)b	
(nn) Vibration Test	D-4r	
(oo) Shock Test	D-4q	

F-3e(3). Loudspeaker component tests.-

Test	Performance Requirement	Test Description
(a) General Inspection	C, D, E-3	F-3f(1)
(b) Maximum Electrical Input	E-3c(2), E-3c(14)d	F-3f(9)
(c) Minimum Output Sound Pressure	E-3c(3), E-3c(14)d	F-3f(10)
(d) Output Sound Pressure Distribution	E-3c(4), E-3c(14)d	F-3f(11)
(e) Frequency Response	E-3c(5), E-3c(14)d	F-3f(12)
(f) Load Characteristic	E-3c(6), E-3c(14)d	F-3f(13)
(g) Loudspeaker Noises	E-3c(7)	
(h) Polarization of Voice Coil Connections	E-3c(8)	
(i) Operation at Reduced Input Voltage	E-3c(9)a	
(j) Operation at Reduced Output Sound Pressure	E-3c(9)b	
(k) Loudspeaker Volume Control	E-3c(10)	
(l) Sensitivity of IC/SHA Used as Microphone	E-3c(14)b	F-3f(14)
(m) Voice Coil Impedance	E-3c(14)c	
(n) Jack Box Volume Control	E-3c(15)	
(o) Transformer Tap Connections	E-3c(16)a	
(p) Transformer Frequency Response	E-3c(16)b	
(q) Dielectric Strength Test	D-5j(1) D-5j(3)	F-3f(20)
(r) Insulation Resistance	D-5j(2)	

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Test	Performance Requirement	Test Description
(s) Endurance Test	E-3b(9)	F-3f(15)
(t) Waterproof Test	D-9b(3) E-3b(1)a(1)	
(u) Watertight Test	D-9b(2) E-3a(6)	
(v) Pressureproof Test	D-9b(1), E-3b(1)a(2) E-3b(7)	
(w) Salt-Spray Test	D-4s	F-3f(16)
(x) Vibration Test	D-4r	
(y) Gun Blast Test	E-3b(6)	F-3f(18)
(z) Depth Charge Test	E-3b(8)	F-3f(19)
(aa) Minimum Output Sound Pressure	E-3c(3)	F-3f(10)
(bb) Frequency Response	E-3c(5)	F-3f(12)
(cc) Pressureproof Test	D-9b(1) E-3b(1)a(2), E-3b(7)	
(dd) Shock Test	D-4q	

F-3f. Description of tests.-

F-3f(1). General inspection.- The equipment shall be given a thorough visual inspection to determine that the equipment conforms to the requirements of sections C, D and E of this specification. Each unit of the equipment, each subassembly and each part shall be examined as necessary to determine conformance with the requirements of this specification with respect to material, finish, workmanship, fit, construction, interchangeability, clearances, dimensions, weight, identification markings, nameplates and instructions. This general inspection shall include a check of all operating controls, circuit functions, test facilities and adjustments, as applicable.

F-3f(2). Microphone sensitivity.- The sound source for this test shall be the Navy standard artificial mouth, adjusted so as to deliver a sound pressure of 100 dynes per square centimeter at a warble frequency of 800 to 1250 cycles per second under free-field (ff) conditions at a point on the axis and 3/4 inch in front of the ring on the artificial mouth. The microphone to be tested shall be set up facing the artificial mouth and so located and positioned that the microphone mouthpiece will be co-axial with the ring on the artificial mouth and in a plane 3/4 inch in front of the ring. The microphone terminals shall be connected to a 150 ohm resistive load. The voltage developed across the resistor shall be measured.

F-3f(3). Microphone frequency response.- The artificial mouth shall be adjusted to deliver a constant free-field sound pressure of 100 dynes per square centimeter 3/4 inch in front of and on the axis of the ring, at a frequency which shall be varied continuously from 200 to 8000 cycles per second. The microphone to be tested shall be set up as described in paragraph F-3f(2), and shall be connected to a 150 ohm resistive load. The voltage developed across the resistor shall be measured.

F-3f(4). Microphone induced voltage.-

F-3f(4)a. Electromagnetically induced voltage.- The microphone to be tested shall be wired to a 20 foot length of Type MCOS-5 cable in accordance with the electrical circuit shown on figure 9. A 150 ohm non-inductive resistor shall be connected to the microphone circuit at the end of the cable. The press-to-talk switch shall be closed and 1 ampere of 60 c.p.s. sinusoidal current shall be made to flow simultaneously through each pair of control circuit contacts of the switch assembly. The induced voltage developed across the resistor shall be measured.

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F-3f(4)b. Electrostatically induced voltage.- The test described in paragraph F-3f(4)a shall be repeated except that the press-to-talk switch shall be opened and a sinusoidal voltage of 115 volts, 60 c.p.s. shall be applied at the end of the 20 foot cable to the control circuits of the switch assembly. The induced voltage developed across the resistor shall be measured.

F-3f(5). Noise pickup.- All power, control, output, busy and volume indicating circuits shall be fully energized to simulate rated operating conditions. The signal frequency on output and volume indicating circuits shall be 1000 c.p.s. sinusoidal. The microphone voice circuit shall be connected to a 150 ohm non-inductive resistor. The induced voltage developed across the resistor shall be measured.

F-3f(6). Microphone endurance.- A warble frequency signal of 800 to 1250 cycles per second at a voltage of 0.087 volts (r.m.s.) shall be impressed across the microphone voice coil circuit for an uninterrupted period of 100 hours at a nominal ambient room temperature of 75°F.

F-3f(7). Power supply voltage and frequency.-

F-3f(7)a. Under voltage and corresponding frequency.- The following tests shall be conducted at the lower voltage and frequency limits of 108 volts and 55 cycles to demonstrate satisfactory operation at reduced voltage and frequency:

Test	Reference paragraph
Output voltage and power	F-3e(2)d
Reserve gain	F-3e(2)e
Signal generator output level	F-3e(2)x

F-3f(7)b. Overvoltage and corresponding frequency.- The following test shall be conducted at the upper voltage and frequency limits of 122 volts and 65 cycles to demonstrate satisfactory operation at elevated voltage and frequency:

Test	Reference paragraph
Overload capacity	F-3e(2)ℓ

F-3f(8). Temperature rise.- The equipment shall be energized at rated voltage (115 volts r.m.s.) and at rated frequency (60 c.p.s.). All power consuming components such as amplifiers, signal generators, relays and lights shall be energized as necessary to simulate actual operating conditions corresponding to maximum heat generation. For this purpose, operating amplifiers shall be driven to rated output employing a 1000 cycle per second signal. Separate tests shall be conducted at ambient temperatures of 40°F, 75°F and 130°F. In each case, temperature readings of "hot-spots" and component parts shall be taken after thermal equilibrium has been reached.

F-3f(9). Maximum loudspeaker electrical input.- A warble frequency signal of 500-2000 cycles per second at a voltage of 70 volts (r.m.s.) shall be impressed across the normal input terminals of the loudspeaker. The current drawn by the loudspeaker shall be measured to determine the volt-ampere electrical input.

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F-3f(10). Minimum loudspeaker sound output pressure.- The loudspeaker shall be set up under free-field conditions and a 70 volt (r.m.s.) warble signal shall be impressed across the normal loudspeaker input terminals. The output sound pressure shall be measured on the loudspeaker sound axis at the distance specified in paragraph E-3c(3). Separate tests shall be conducted employing signals warbling over the following frequency bands:

320	to	500	cycles per second
500	to	800	cycles per second
800	to	1250	cycles per second
1250	to	2000	cycles per second
2000	to	3200	cycles per second
3200	to	5000	cycles per second

The signals shall be warbled through the above bands with a linear change of frequency on a logarithmic scale with respect to time.

F-3f(11). Loudspeaker output sound pressure distribution.- The loudspeaker shall be set up under free-field conditions and a 70 volt (r.m.s.) warble frequency signal of 800 to 1250 cycles per second shall be impressed across the normal loudspeaker input terminals. The output sound pressure shall be measured at four points (two in a horizontal plane on either side of the sound axis, and two in a vertical plane above and below the sound axis) at the distance and at the angle off the sound axis specified in paragraph E-3c(4). For loudspeaker types other than types IC/SCD, I C/SDD, IC/SDF and IC/SFD, the arithmetical average of the four measurements shall be used as the measured off-axis pressure. For loudspeaker types IC/SCD, IC/SDD, IC/SDF and IC/SFD, the two measurements in the horizontal plane and the two measurements in the vertical plane shall be averaged separately and use as the measured horizontal and vertical off-axis pressures, respectively.

F-3f(12). Loudspeaker frequency response.- The loudspeaker shall be set up under free-field conditions and a 70 volt (r.m.s.) signal shall be impressed across the normal loudspeaker input terminals at a frequency which shall be varied continuously from 200 to 8000 cycles per second. The output sound pressure shall be measured on the loudspeaker sound axis at the distance specified in paragraph E-3c(3).

F-3f(13). Loudspeaker load characteristic.- The loudspeaker shall be set up under free-field conditions and a 800 to 1250 cycles per second warble frequency signal shall be impressed across the normal loudspeaker input terminals at a voltage which shall be varied from zero to 99 volts (r.m.s.), except for superpower loudspeakers where the upper test voltage limit shall be 70 volts (r.m.s.) in lieu of 99 volts. The output sound pressure at each value of test voltage shall be measured on the loudspeaker sound axis at the distance specified in paragraph E-3c(3). The current drawn by the loudspeaker at each value of test voltage shall be measured to determine the volt-ampere electrical input.

F-3f(14). Sensitivity of type IC/SHA loudspeaker used as a microphone.- The sound source for this test shall be the Navy standard artificial mouth, adjusted so as to deliver a sound pressure of 4 dynes per square centimeter at a warble frequency of 800 to 1250 cycles per second under free-field conditions at a point on the axis and 18 inches in front of the ring on the artificial mouth. The loudspeaker to be tested shall be set up 18 inches in front of, on the axis of and facing the ring on the artificial mouth. The normal loudspeaker input terminals shall be connected to a 150 ohm resistive load. The voltage developed across the resistor shall be measured.

F-3f(15). Loudspeaker endurance test.- A warble frequency signal of 800 to 1250 cycles per second at a voltage of 70 volts (r.m.s.) shall be impressed across the normal input terminals of the loudspeaker for a continuous period of 100 hours at an ambient temperature of 130°F. For all except superpower loudspeakers, immediately following this test, the impressed voltage shall be raised to 99 volts (r.m.s.) and the test continued for one additional hour.

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F-3f(16). Salt-spray test.- The microphone or loudspeaker component shall be completely assembled and mounted in its normal operating position. Tests on microphone control stations, types IC/MSA, and IC/MSB shall be conducted with the cover open. The component shall then be subjected to a 20 percent salt-spray at 130°F. for a period of 3 minutes, followed by an air blast at 130°F. for a period of 3 minutes. The cycle shall be repeated continuously under ultra-violet light for a total period of 100 hours, at the end of which period the component shall be disassembled, washed with fresh water, dried, examined and reassembled.

F-3f(17). Tumbling test.- The microphone shall be subjected to a tumbling test of 500 revolutions in a standard rotary tester with a 16 inch drop. The floors of the rotary tester shall be of 3/16 thick masonite sheet (Brinell hardness 11 to 13) backed by wood.

F-3f(18). Gun blast test.-

F-3f(18)a. Simulated gun blast test.- The microphone or loudspeaker shall be mounted on the carriage of the U.S. Navy Simulated Gun Blast Equipment. The front edge of the component shall be positioned in the test plane, and its axis shall be coincident with that of the explosion chamber. The component being tested shall be subjected to 30 rounds of blast at a peak pressure of 9.5 pounds per square inch. For super-power loudspeakers, where the mouth area exceeds the face area of the explosion chamber, but which employ individual horn assemblies the area of each of which is less then the face area of the explosion chamber, the loudspeaker shall be so positioned that the axis of one of the individual horns is coincident with that of the explosion chamber.

F-3f(18)b. Gun blast test at Naval proving ground.- For super-power loudspeakers, in which the mouth area of the loudspeaker or of any of its individual horn assemblies exceeds the face area of the explosion chamber, the loudspeaker shall be exposed to thirty rounds of service charge blast from a 5-inch 38-caliber gun at the Naval Proving Ground, Dahlgren, Virginia. The loudspeaker shall be mounted in a plane normal to the trajectory 8 feet 4 inches from the gun muzzle at a point 12 feet 8 inches from the trajectory. The loudspeaker shall be at least 6 feet above the ground and so positioned as to face the gun muzzle.

F-3f(19). Depth charge test.- The component shall be mounted in the center of a steel plate having minimum dimensions of 3 feet in height, 3 feet in width and 1/4 inch in thickness. The plate shall be submerged and suspended vertically at a depth of 30 feet below the surface of the water. A 55 pound standard T.N.T. charge shall be submerged and suspended at a depth of 30 feet below the surface of the water at a distance of 30 feet from the plate, on the same side of the plate as the component being tested, and on a line perpendicular to the face of the plate at its center. The total depth of water shall be a minimum of 60 feet in the area of the test. The test consisting of the detonation of the charge shall be made a total number of 4 times.

F-3f(20). Dielectric strength test.- A 60 cycle per second sinusoidal wave form alternating potential shall be applied to the circuits. Voltage shall be applied between circuits and from circuits to ground. The value of the voltage shall be raised slowly to an R.M.S. value as follows, shall be held at that value for a period of one minute (except for production tests) and shall then be gradually reduced:

Working voltage of circuit tested	Dielectric test voltage
Up to 60	600
From 60 to 120	1200
From 120 to 240	1500
From 240 to 480	2000
From 480 up	Twice working voltage plus 1000

During production tests, the test period shall be 5 seconds. The power source of the test voltage shall have a minimum reserve capacity of 1 kilowatt.

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F-4. Contract suitability tests.- Contract suitability tests shall be conducted on any part or all of the equipment being furnished on the contract or order, if in the opinion of the bureau, further test seems warrented to determine the continued excellence of manufacture, or the suitability of changes in design, construction or production methods.

F-5. Inspection tests.- Inspection tests shall be conducted as follows at the place of manufacture by the Naval inspector.

F-5a. Tests applicable to each item.- The following tests shall be conducted on each component submitted for acceptance under these specifications on a given contract or order:

F-5a(1). Microphone component production tests.-

Test	Performance Requirements	Test Description
(a) General Inspection	C, D, E-1	F-3f(1)
(b) Microphone Sensitivity	E-1c(1)b	F-3f(2)
(c) Frequency Response	E-1c(1)c	F-3f(3)
(d) Dielectric Strength	D-5j(1)	F-3f(20)
(e) Insulation Resistance	D-5j(2)	
(f) Pressure Proof Test	D-9b(1), E-1b(2)d	

F-5a(2). Amplifier component production tests.-

Test	Performance Requirements	Test Description
(a) General Inspection	C, D, E-2	F-3f(1)
(b) Output Voltage and Power	E-2b(1)b(2)b E-2b(1)b(2)c	E-2b(1)b(2)d(1) F-3f(7)a
	E-2b(1)c(3) E-2b(1)c(4)	E-2b(1)c(5) F-3f(7)a
(c) Reserve Gain	E-2b(1)b(2)d(2)	E-2b(1)b(2)d(2) F-3f(7)a
(d) Frequency Response	E-2b(1)b(2)e E-2b(1)c(6)	
(e) Noise and Microphonics	E-2b(1)b(2)g E-2b(1)c(8), D-4r	
(f) Output Voltage Regulation	E-2b(1)b(2)h E-2b(1)c(9)	
(g) Stability	E-2b(1)b(2)j, E-2b(1)c(10)	
(h) Amplifier Phasing	E-2b(1)a(5)	
(i) Compressor Characteristic (Signal Frequency of 1000 c.p.s. only)	E-2b(1)b(3)b	
(j) Compressor Effect on Amplifier Characteristic	E-2b(1)b(3)e	
(k) Limiter Characteristic (Signal Frequency of 1000 c.p.s. only)	E-2b(1)b(4)a	
(l) Generated Signals	E-2b(1)d(3)	

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Test	Performance Requirements	Test Description
(m) Duration of Generated Signals	E-2b(1)d(4)	
(n) Signal Generator Output Level	E-2b(1)d(5)	E-2b(1)d(5) F-3f(7)a
(o) Signal Generator Noise and Microphonics	E-2b(1)d(7), D-4r	
(p) Time Interval Between "Standby" and "Ready"	E-2b(1)a(4)c	
(q) V.I. Circuit Voltage	E-2b(2)a(3)b	
(r) Dielectric Strength Test	D-5j(1)	F-3f(20)
(s) Insulation Resistance	D-5j(2)	

F-5a(3). Loudspeaker component production tests.-

Test	Performance Requirements	Test Description
(a) General Inspection	C, D, E-3	F-3f(1)
(b) Frequency Response	E-3c(5), E-3c(14)d	F-3f(12)
(c) Loudspeaker Noises	E-3c(7)	
(d) Polarization of Voice Coil Connections	E-3c(8)	
(e) Dielectric Strength Test	D-5j(1) D-5j(3)	D-5j(1)
(f) Insulation Resistance	D-5j(2)	
(g) Pressure-proof Test	D-9b(1), E-3b(1)a(2), E-3b(7)	

F-5b. Tests applicable to representative samples.- The following additional tests shall be conducted on representative samples (in quantity as determined by the Naval inspector) selected by the Naval inspector at random from the production line under each contract or order. If a given contract or order is completed in more than one continuous production run, these tests shall be conducted on separate random samples, selected from each such run.

F-5b(1). Microphone component sample tests.-

Test	Performance Requirement	Test Description
(a) Microphone Endurance	E-1b(1)f	F-3f(6)
(b) Strength of Cable Assembly	E-1b(3)c	
(c) Watertight Test	D-9b(2), E-1a(1), E-1a(2), E-1a(3), E-1a(5), E-1a(8), E-1a(9)	

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F-5b(2). Amplifier component sample tests.-

Test	Performance Requirement
(a) Distortion	E-2b(1)b(2)f(1) E-2b(1)b(2)f(2) E-2b(1)c(7)
(b) Compressor Characteristic	E-2b(1)b(3)b
(c) Compressor Time Constant	E-2b(1)b(3)c
(d) Compressor Adjustment	E-2b(1)b(3)d
(e) Limiter Characteristic	E-2b(1)b(4)a
(f) Limiter Time Constant	E-2b(1)b(4)b
(g) Interstage Input Level	E-2b(1)b(6)b

F-5b(3). Loudspeaker component sample tests.-

Test	Performance Requirement	Test Description
(a) Maximum Electrical Input	E-3c(2), E-3c(14)d	F-3f(9)
(b) Minimum Output Sound Pressure	E-3c(3), E-3c(14)d	F-3f(10)
(c) Output Sound Pressure Distribution	E-3c(4), E-3c(14)d	F-3f(11)
(d) Load Characteristic	E-3c(6), E-3c(14)d	F-3f(13)
(e) Loudspeaker Volume Control	E-3c(10)	
(f) Sensitivity of IC/SHA Usedas Microphone	E-3c(14)b	F-3f(14)
(g) Transformer Frequency Response	E-3c(16)b	
(h) Endurance Test	E-3b(9)	F-3f(15)
(i) Watertight Test	D-9b(2), E-3a(6)	

F-5c. Other tests.- Such other type or brand approval tests shall be made in addition or test samples as may be considered necessary by the Naval inspector to insure that specification requirements are being met by the materials and methods employed in production.

F-5d. Non-standard production test methods.- To facilitate production testing and to coordinate wherever possible Navy test requirements with established industrial and commercial shop procedures, the bureau will permit the use of non-standard test methods and instruments, provided it can be demonstrated to the Naval inspector that the results so obtained are correlatable with the results obtained by the use of standard test methods as specified in this specification. Where production test methods are employed utilizing visual observations, as with the use of an oscilloscope, the requirement for maintaining permanent test records may be dispensed with at the discretion of the Naval inspector.

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G. PACKAGING AND PREPARATION FOR SHIPMENT

G-1. Preservation, packing and packaging.- The equipment shall be preserved, packaged and packed in accordance with the requirements of Army-Navy Joint Packaging Instruction JPI-14. The contract or order shall specify whether packing shall be for domestic or overseas shipment (see par. H-1a).

G-1a. All dessicant containers used for Method II packaging shall be of one of the following types:

(1) Metal cans having glass fibre filters.
(2) Cloth bags having a fibrous linen and all closures effected by a double seam with at least one fold of material.
(3) Fibrous web bags having all closures effected by adhesive or heat sealing or both.

G-1b. Unless otherwise specified, only one stock numbered item shall be packed in one shipping container.

G-2. Marking.-

G-2a. All shipments shall be marked as required by the Navy Shipment Marking Handbook. The applicable, Bureau of Ships Stock Number shall be included in the marking to appear on the outside of each (inside and outside) container.

G-2b. When the packaging of a single stock numbered item requires the use of two or more shipping containers, such shipping containers shall be identically marked except that each shipping container shall have additional marking to indicate "Box 1 of 3 boxes", "Box 2 of 3 boxes", etc. This requirement applies also to spare parts sets when furnished in more than one shipping container per set.

G-2c. When exception is made to the requirement of paragraph G-1b, and the packing of two or more identical items in one shipping container is satisfactory, the total quantity contained therein shall be marked on each inside and outside shipping container.

H. NOTES

H-1. Ordering information.- Invitation for bids, and contracts or orders should specify the following:

H-1a. Information normally required.-

(1) Title, number and date of this specification.
(2) Engineering services.- Provision for engineering services, if required (see par. D-8)
(3) Spare parts.- List and quantities of components comprising an equipment for the purpose of determining spare parts required to make up a set. (See par. D-13a.)
(4) Designation plate markings.- Markings of designation plates for microphone stations and boxes. (See par. E-1b(2)a(2)k.)
(5) Input facilities.- Quantity, type, location and associated announcing circuit of all microphone stations and boxes. (See par. E-2b(2)a(2)a.)
(6) Output facilities.- Total quantity, individual designations and associated announcing circuits of loudspeaker output groups. (See par. E-2b(2)a(2)b.)
(7) Series protective resistors.- Quantity and size of series protective resistors, if required. (See par. E-2b(2)c(5).)
(8) Packing.- Packing requirement, whether for domestic or overseas shipment. (See par. G-1).

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H-1b. Information required for deviation from standard requirements.-

(1) Rated amplifier output power for other than standard amplifier assemblies, types IC/FCA, IC/FDA, IC/FGA, IC/FJA, IC/FMA and IC/FPA. (See pars. E-2b(1)b(2)c and E-2b(1)c(4).)
(2) Quantity and type of signal generators, if other than standard per paragraph B-4a. (See par. E-2b(1)d(1).)
(3) Amplifier switching for 17MC systems, if other than standard. (See par. E-2b(2)b(5)d.)
(4) Signal generator switching for 17MC systems, if other than standard. (See par. E-2b(2)b(6)b.)
(5) Special combinations of loudspeaker group selection from microphone stations, if other than standard selections are required. (See par. E-2b(2)f(4).)
(6) Special system priorities for speech from microphone stations, if other than standard. (See par. E-2b(2)f(5)b.)
(7) Provision for circuit 17MC output disconnect switches, if required. (See par. E-2b(2)b(7)c.)
(8) Any other special controls, circuit requirements or deviations from standards specified, as may be required.

H-2. Specifications, other publications and drawings - where obtainable.-

H-2a. Copies of Navy Department specifications, Joint Army-Navy specifications, and Joint Packaging Instructions may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. When requesting specifications and packaging instructions, refer to both title and number (or symbol).

H-2b. Copies of Bureau of Ships specifications, publications and drawings may be obtained only upon application to the Bureau of Ships, Navy Department, Washington 25, D. C. When requesting specifications, publications and drawings refer to both title and number (or symbol).

H-2c. Copies of Joint Army-Navy Publication JANP-109 - Manual of Standard Descriptions may be obtained upon application to the Navy Catalog Office, 111 East 16th Street, New York, N.Y., except that Naval activities should make application to the nearest issuing office of the registered publication section. (This publication should not be confused with Joint Army-Navy Specification JAN-P-109 - Packaging and Packing for Overseas Shipment: Barrels, Tight).

H-3. QUALIFICATION (TYPE OR BRAND APPROVAL).- IN THE PROCUREMENT OF MATERIALS REQUIRING QUALIFICATION (TYPE OR BRAND APPROVAL) THE RIGHT IS RESERVED TO REJECT ANY BIDS ON BRANDS OR TYPES OF MATERIAL WHICH HAVE NOT BEEN SUBJECTED TO THE REQUIRED TESTS AND FOUND SATISFACTORY FOR INCLUSION ON THE NAVY DEPARTMENT ACCEPTABLE LIST OF APPROVED MATERIALS. THE ATTENTION OF SUPPLIERS IS CALLED TO THIS REQUIREMENT AND MANUFACTURERS ARE URGED TO COMMUNICATE WITH THE BUREAU OF SUPPLIES AND ACCOUNTS AND ARRANGE TO HAVE THE SAMPLES OF THE TYPE OR BRANDS OF MATERIALS THEY PROPOSE TO OFFER TO THE NAVY IN THE FUTURE TESTED FOR APPROVAL IN ORDER THAT THEY MAY BE ELIGIBLE TO BE AWARDED CONTRACTS OR ORDERS.

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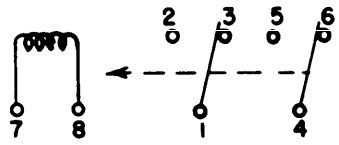
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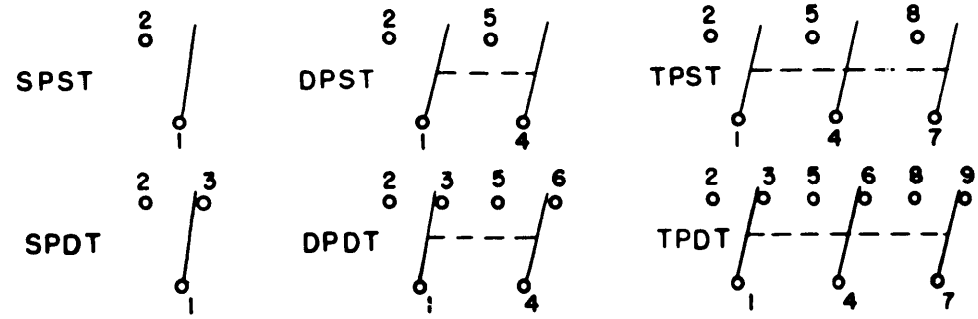
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STANDARD DESIGNATION SYSTEM FOR SWITCH & RELAY CONTACTS

A. RELAYS:



B. SWITCHES:



C. SWITCHES (TYPES JA & JR), ROTORY:

(1) INDIVIDUAL CONTACTS SHOULD BE LETTERED AS INDICATED ON APPLICABLE BUREAU OF SHIPS DRAWINGS. (SEE SECTION A)

(2) SWITCH SECTIONS SHOULD BE NUMBERED CONSECUTIVELY STARTING FROM THE HANDLE END OF THE SWITCH.

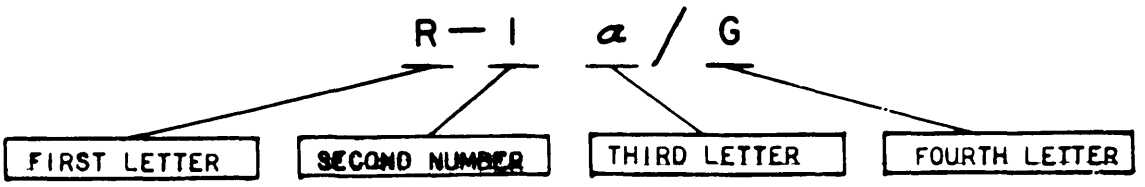
FIGURE 1

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R17A11(SHIPS)

STANDARD SYSTEM FOR DESIGNATION OF COMPONENT PARTS ON SCHEMATIC WIRING DIAGRAMS

EACH ELECTRICAL COMPONENT PART SHALL BEAR A UNIQUE DESIGNATION CONFORMING TO THE FOLLOWING SYSTEM. AS AN EXAMPLE,



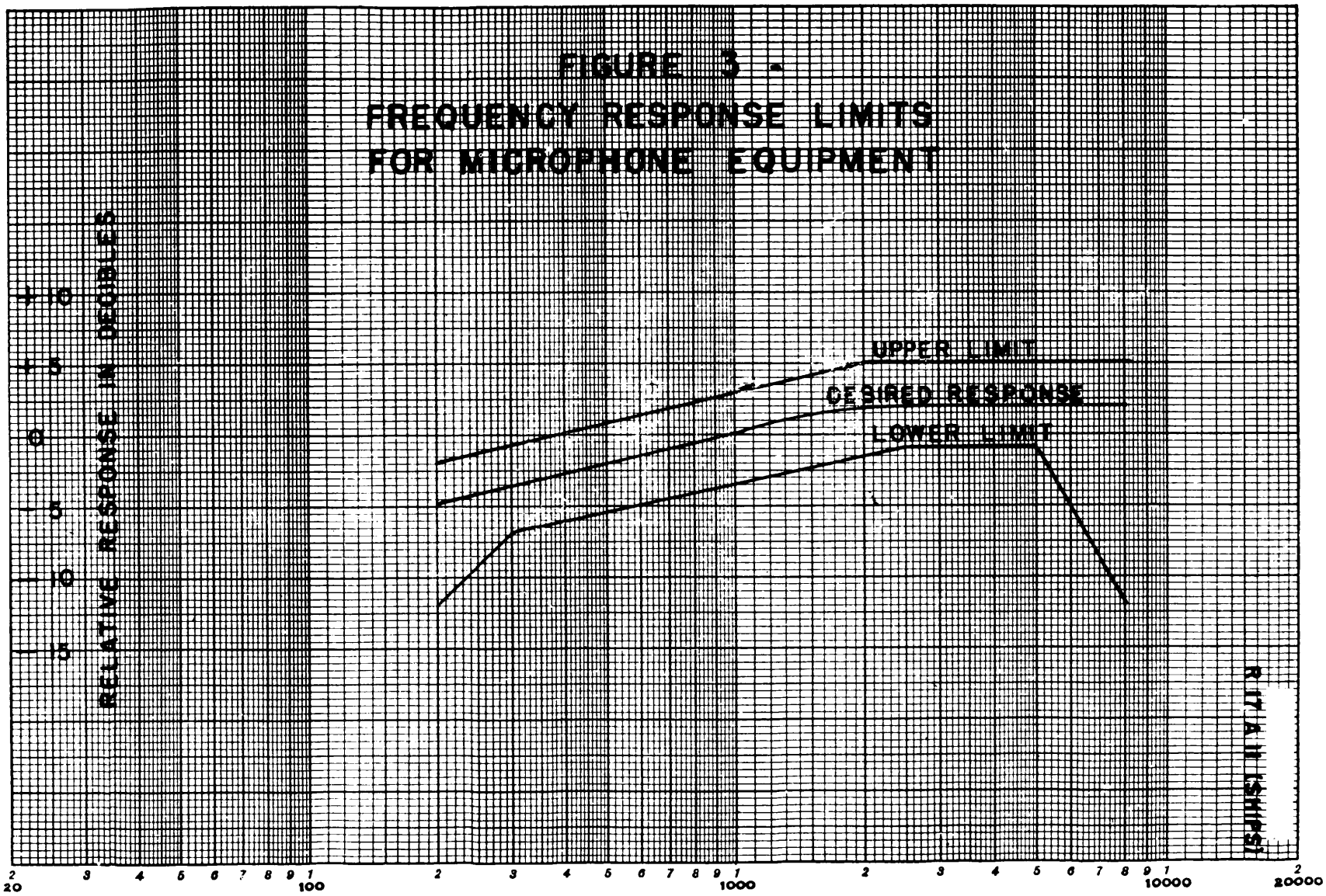
FIRST LETTER	SECOND NUMBER	THIRD LETTER	FOURTH LETTER
DENOTES TYPE OF COMPONENT, SUCH AS RESISTOR, CAPACITOR, ETC., IN ACCORDANCE WITH THE FOLLOWING TABLE.	ASSIGNED CONSECUTIVELY TO EACH DIFFERENT SIZE OR RATING OF A PARTICULAR COMPONENT NOTE-ALL COMPONENTS OF SAME ELECTRICAL RATING AND PHYSICALLY INTERCHANGEABLE SHALL BEAR SAME BASIC NUMBER.	ASSIGNED SUCCESSIVELY TO EACH COMPONENT OF THE SAME TYPE, SIZE AND RATING, WHICH IS USED IN THE PARTICULAR CIRCUIT OR PIECE OF EQUIPMENT BEING REPRESENTED SCHEMATICALLY	OPTIONAL-MAY BE USED IF DESIRED TO INDICATE THE PIECE OF EQUIPMENT IN WHICH THE PART IS USED; E.G. SIGNAL GENERATOR. CODE USED SHOULD BE INDICATED ON DRAWING.

CODE FOR COMPONENT TYPE LETTERS

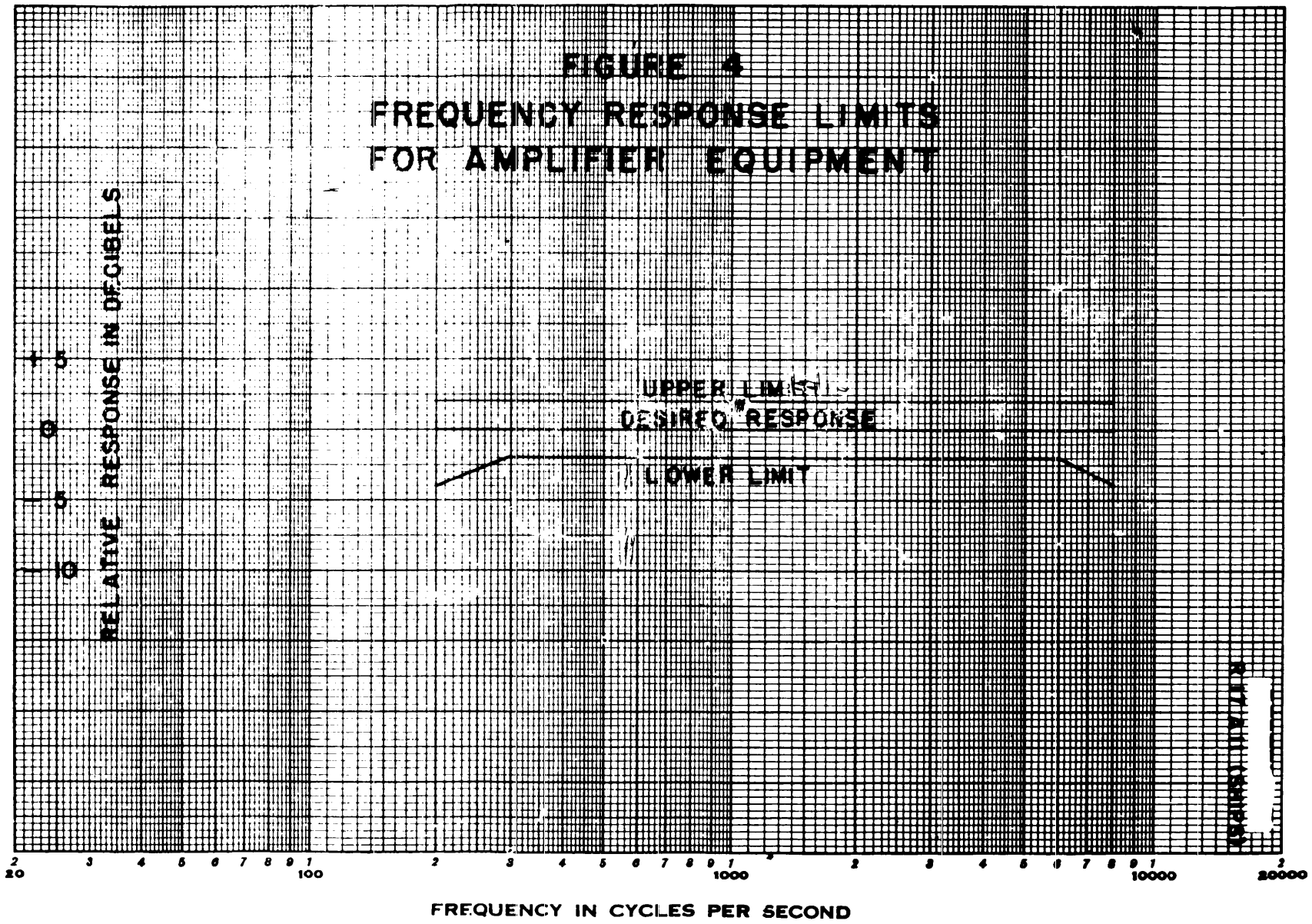
CAPACITOR	-C	REACTOR	-L
FUSE	-F	RELAY	-K
JACK	-J	RESISTOR	-R
LAMP	-I	SWITCH	-S
METER	-M	TRANSFORMER	-T
MOTOR	-B	TUBE, ELECTRONIC	-V
PLUG	-P		

FIGURE 2

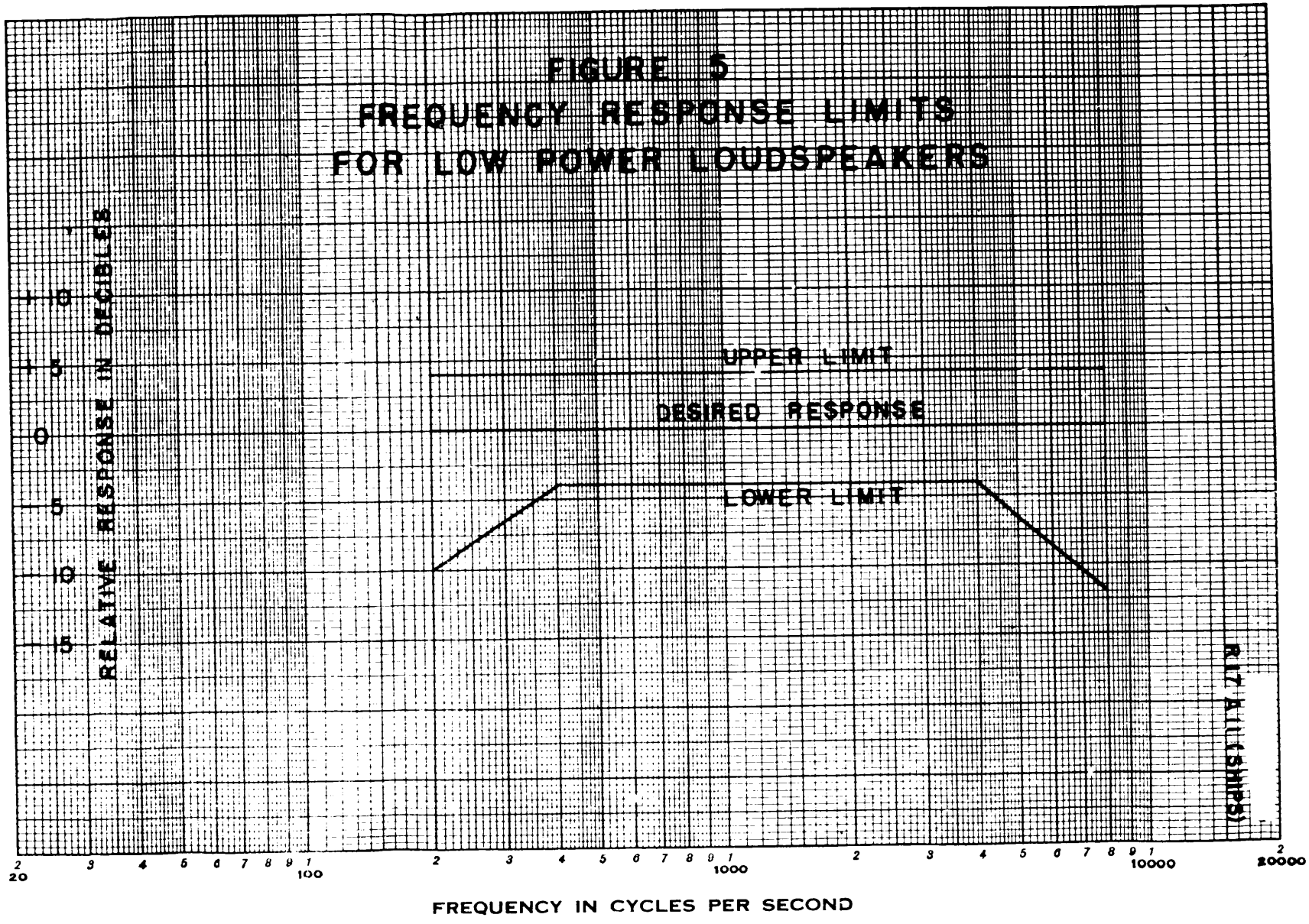
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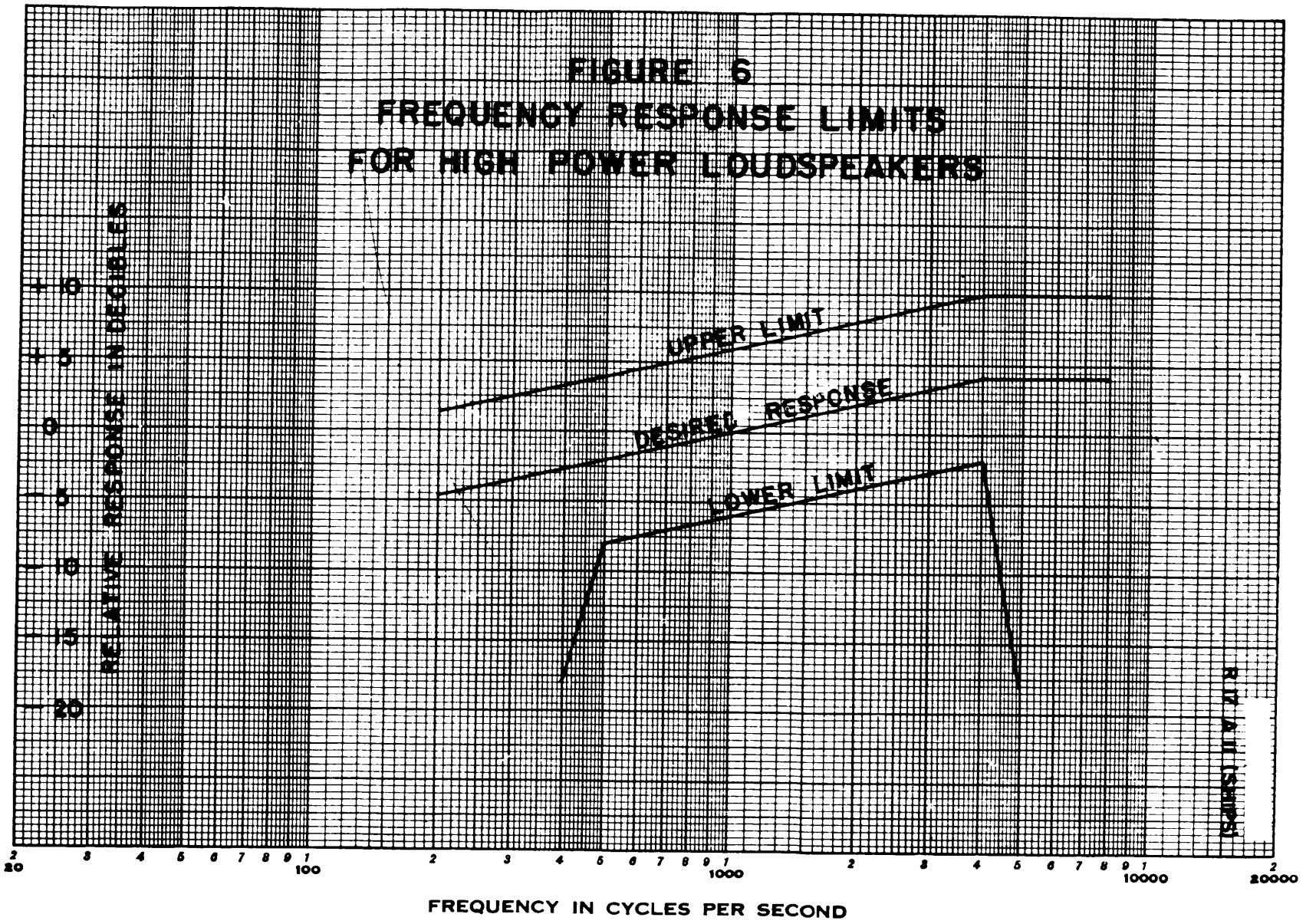
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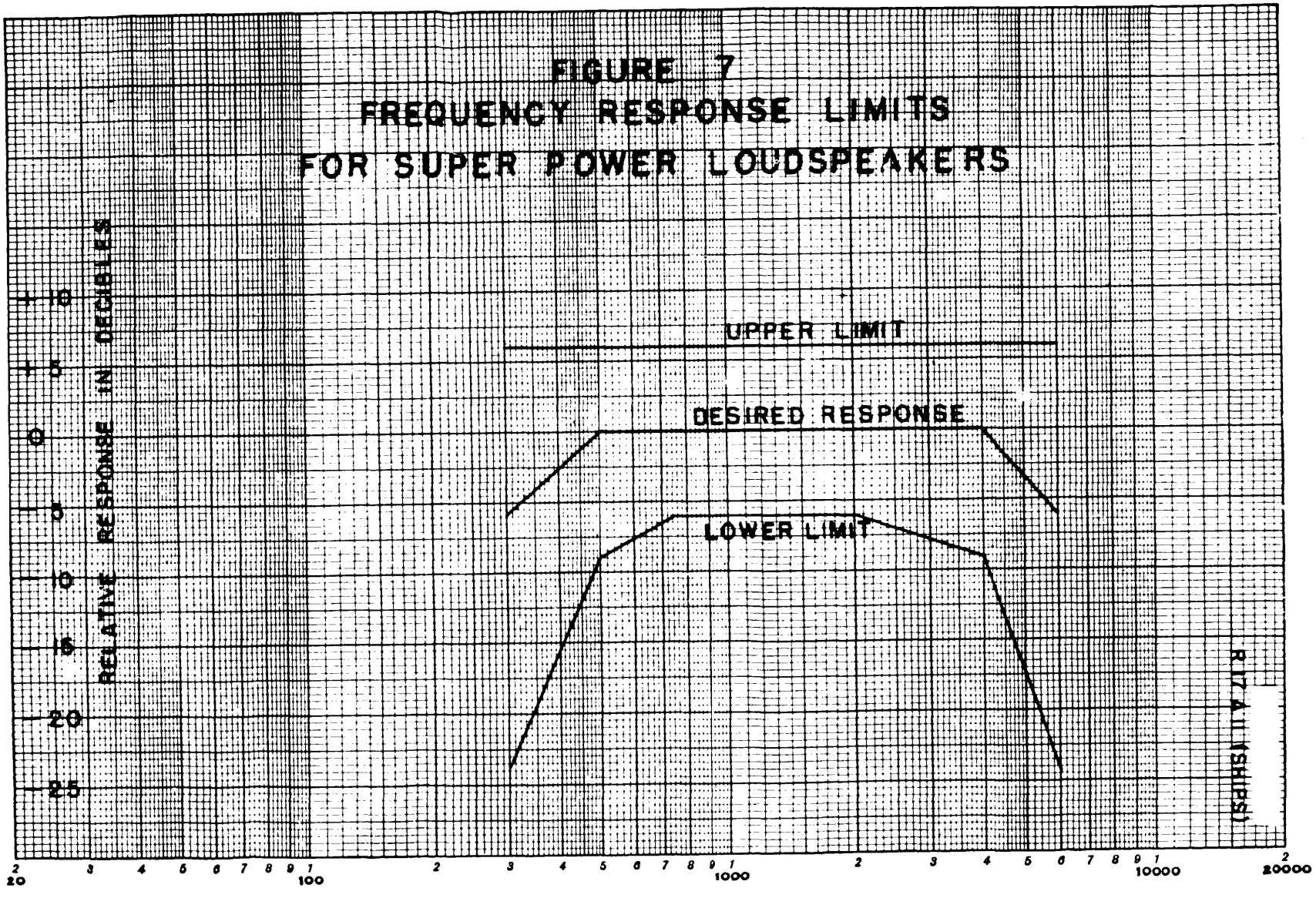
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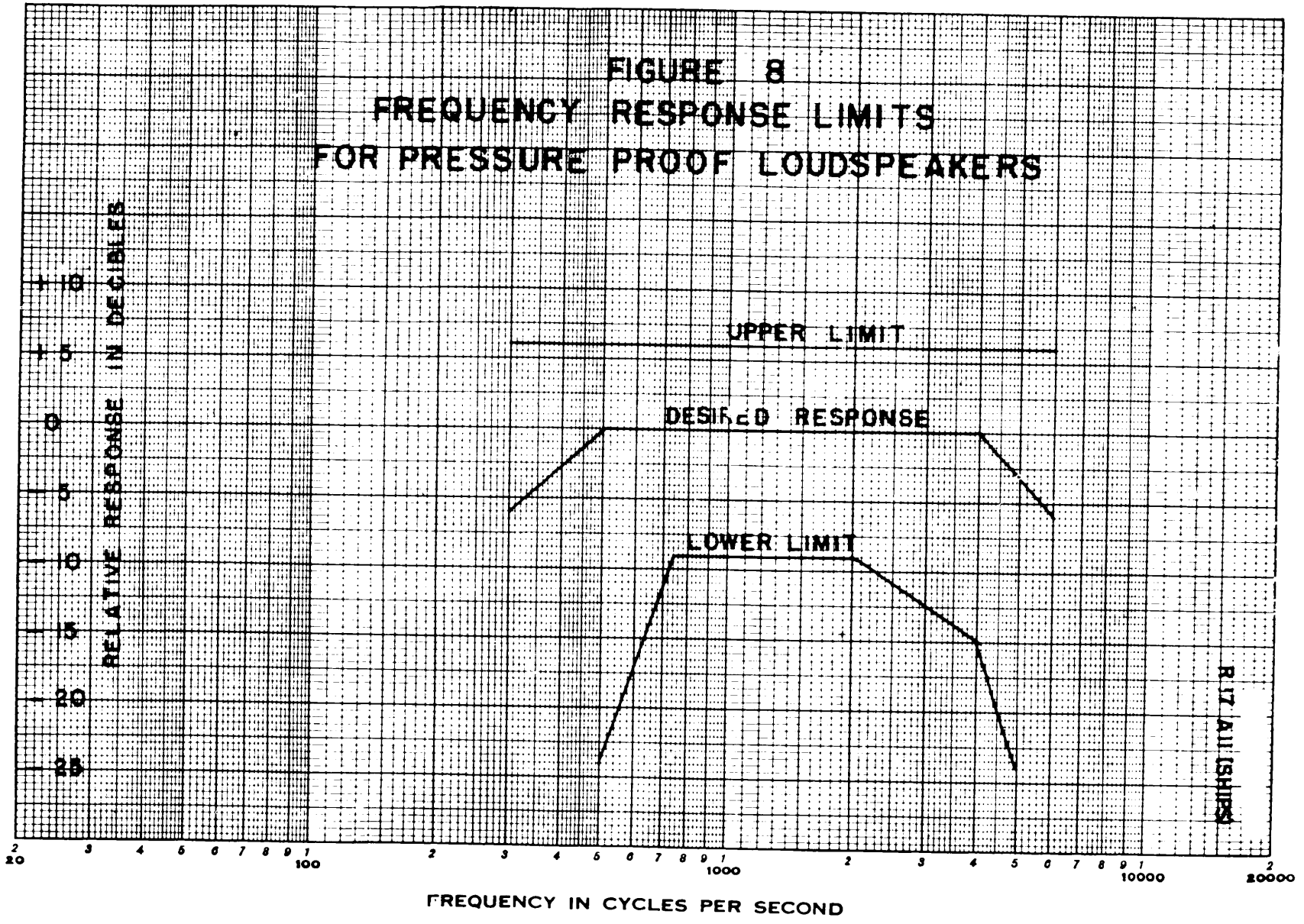
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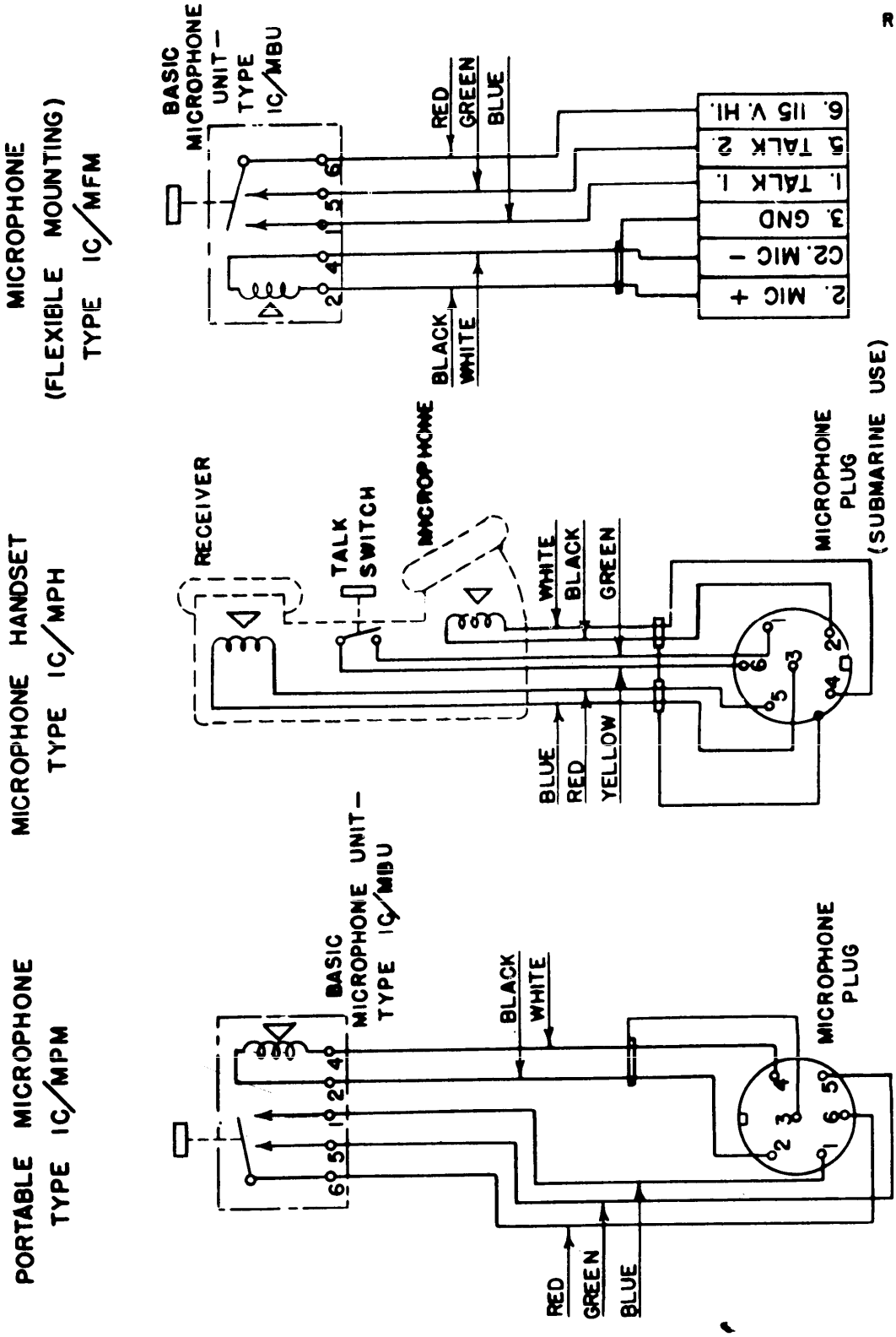


FIGURE 9
STANDARD CIRCUITS AND TERMINAL MARKINGS

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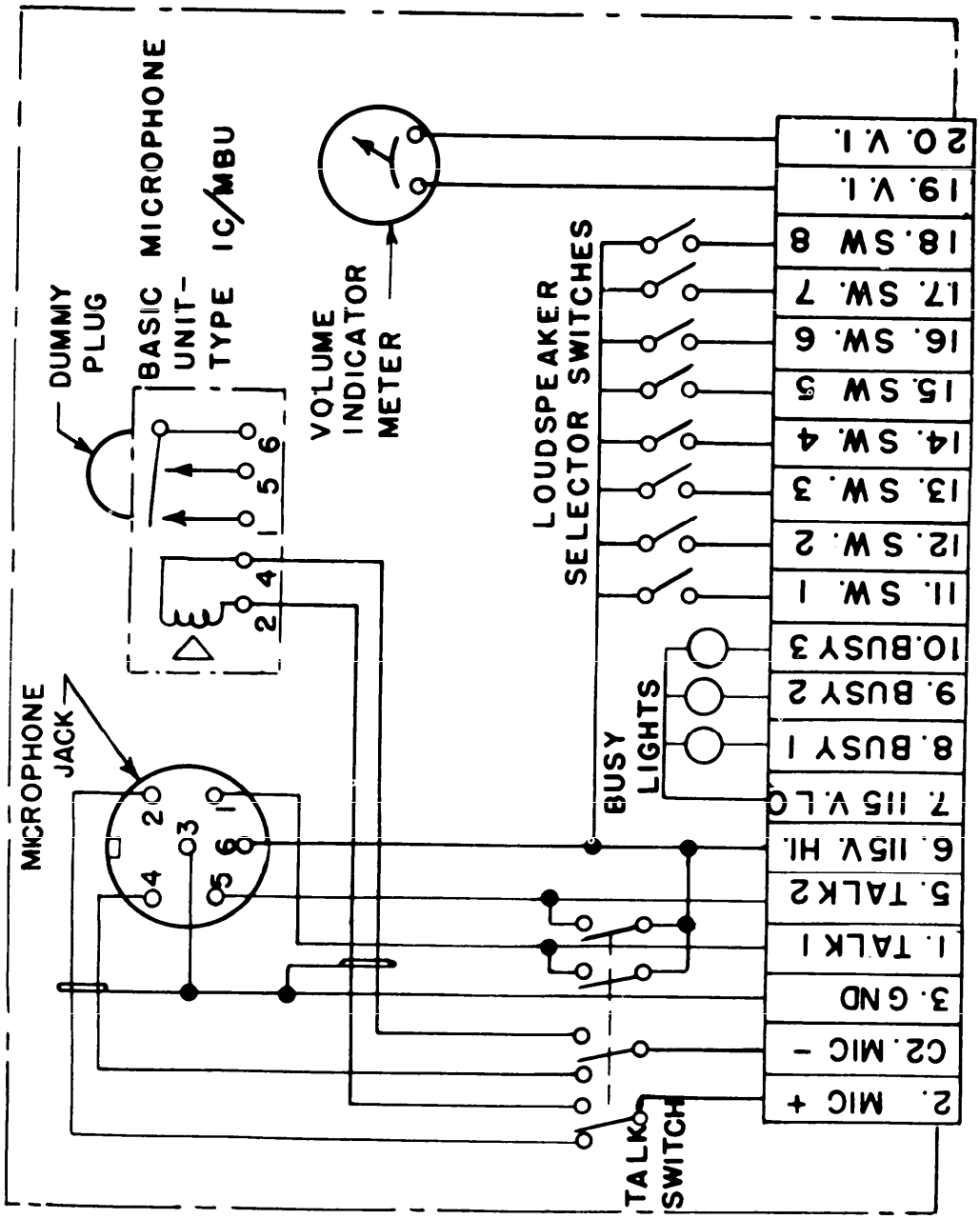


FIGURE 10

STANDARD CIRCUIT & TERMINAL MARKING
MICROPHONE CONTROL STATION, TYPE IC/MSA

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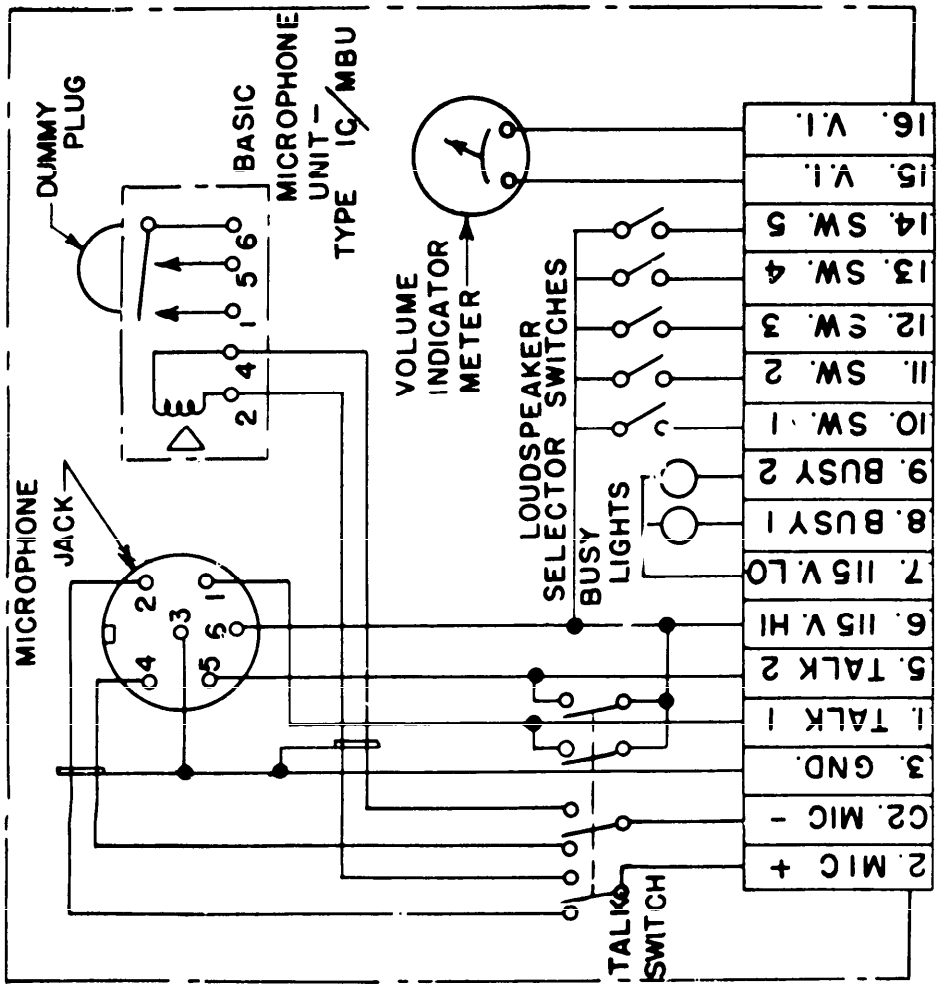
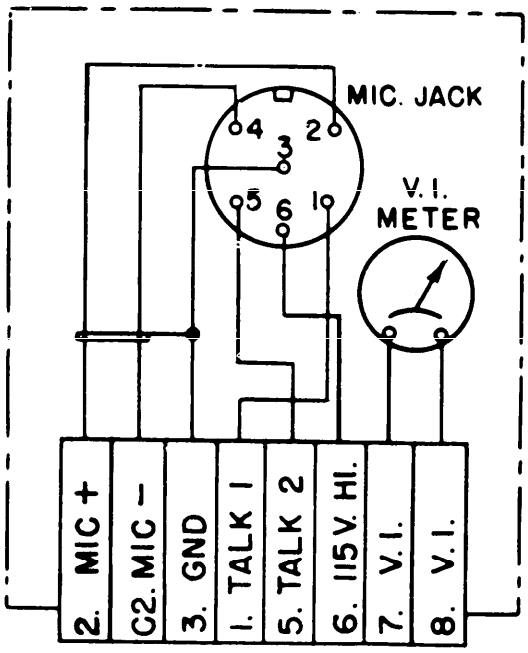
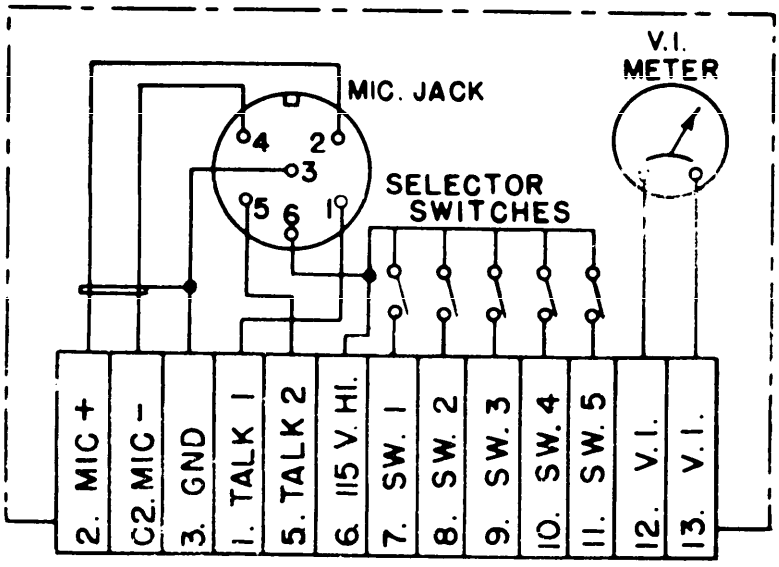


FIGURE II
STANDARD CIRCUIT & TERMINAL MARKING
MICROPHONE CONTROL STATION, TYPE IC/MSB

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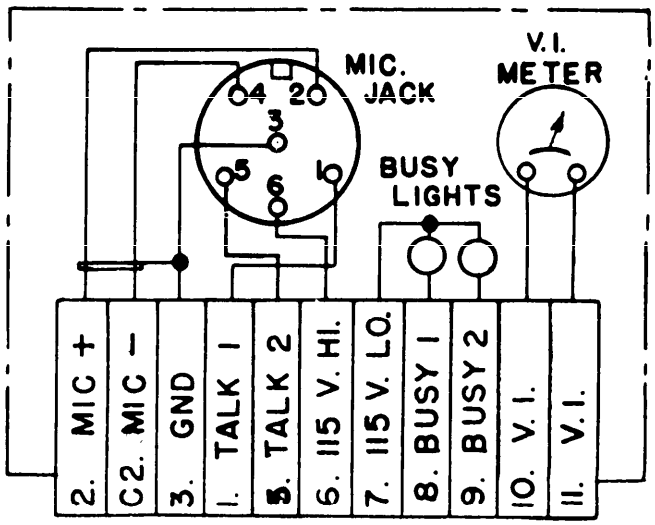
MICROPHONE
CONTROL BOX
TYPE 1C/MBA



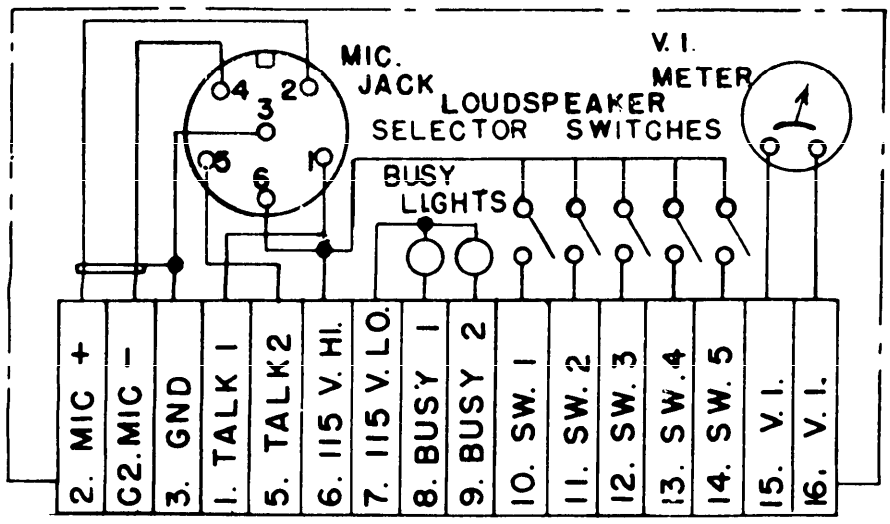
MICROPHONE
CONTROL BOX
TYPE 1C/MBB

FIGURE 12
STANDARD CIRCUITS & TERMINAL MARKINGS

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MICROPHONE CONTROL BOX TYPE IC/MBC



MICROPHONE CONTROL BOX TYPE IC/MBD

FIGURE 13
STANDARD CIRCUITS & TERMINAL MARKINGS

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STANDARD TERMINAL MARKINGS

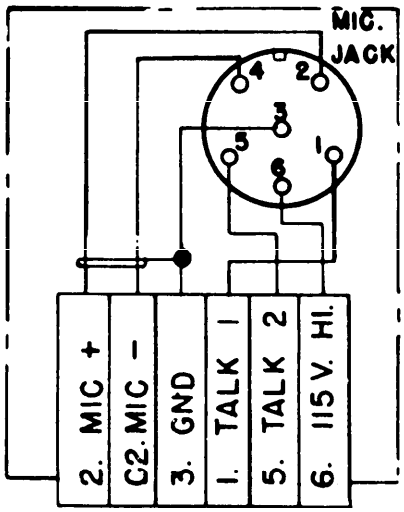
MICROPHONE CONTROL BOX, TYPE IC/MJT

TERMINAL MARKING	FUNCTION
MC + MCC -	115 Volt A. C. power supply from source.
MC 6 +	Power to switches at remote jack boxes (IC/MJT)
MC 7 -	Return from busy lights at remote jack boxes (IC/MJT) - fused
MC 1	Power from first talk switch contact at remote jack boxes (IC/MJT)
MC 2	Power from second talk switch contact at remote jack boxes (IC/MJT) - The use of this terminal is optional, depending upon circuit design employed.
MC 8	Power to busy lights at remote jack boxes (IC/MJT)
MC 9	Power to amplifier for "ready" condition
MC 22 MCC 22	Control circuit power for automatic transmission of IMC alarm signals
MC 11 MCC 11	Microphone Talk-Back Circuit from remote group No. 1
MC 12 MCC 12	Microphone Talk-Back Circuit from remote group No. 2
MC 13 MCC 13	Microphone Talk-Back Circuit from remote group No. 3
MC 14 MCC 14	Microphone Talk-Back Circuit from remote group No. 4
MC 15 MCC 15	Microphone Talk-Back Circuit from remote group No. 5
MC 16 MCC 16	Microphone Talk-Back Circuit from remote group No. 6
MC 17 MCC 17	Microphone Talk-Back Circuit from remote group No. 7
MC 18 MCC 18	Microphone Talk-Back Circuit from remote group No. 8
MC 19 MCC 19	Microphone Talk-Back Circuit from remote group No. 9
MC 110 MCC 110	Microphone Talk-Back Circuit from remote group No. 10
MC 10 MCC 10	Audio Input to Amplifier.
MC 30 MCC 30	Audio Output from Amplifier
MC 31 MCC 31	Audio Output to remote loudspeaker Group No. 1
MC 32 MCC 32	Audio Output to remote loudspeaker Group No. 2
MC 33 MCC 33	Audio Output to remote loudspeaker Group No. 3
MC 34 MCC 34	Audio Output to remote loudspeaker Group No. 4
MC 35 MCC 35	Audio Output to remote loudspeaker Group No. 5
MC 36 MCC 36	Audio Output to remote loudspeaker Group No. 6
MC 37 MCC 37	Audio Output to remote loudspeaker Group No. 7
MC 38 MCC 38	Audio Output to remote loudspeaker Group No. 8
MC 39 MCC 39	Audio Output to remote loudspeaker Group No. 9
MC 310 MCC 310	Audio Output to remote loudspeaker Group No. 10
MC 301 MCC 301	Audio Output to local loudspeaker
MC 3	Ground connection
MC 23 MCC 23	Audio Output from IMC amplifier for automatic transmission of IMC alarm signals

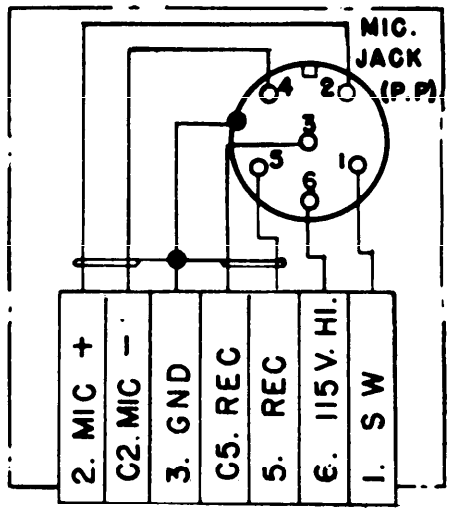
FIGURE 414

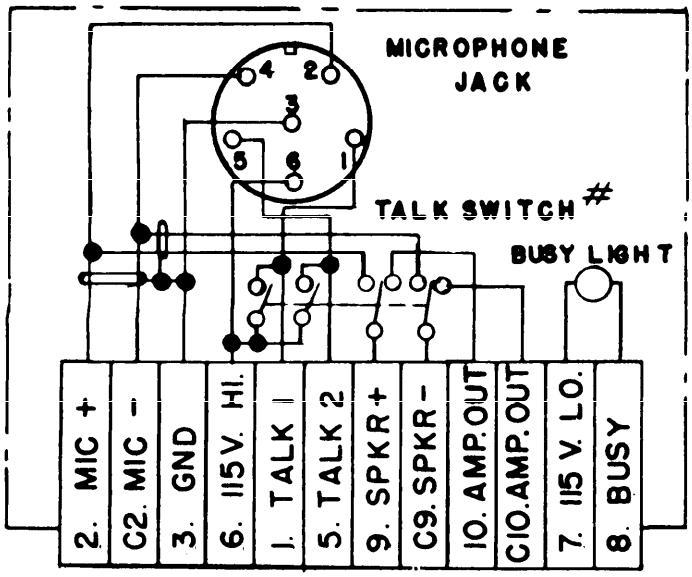
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MICROPHONE JACK BOX TYPE IC/MJB



MICROPHONE JACK BOX (P.P.) TYPE IC/MJP



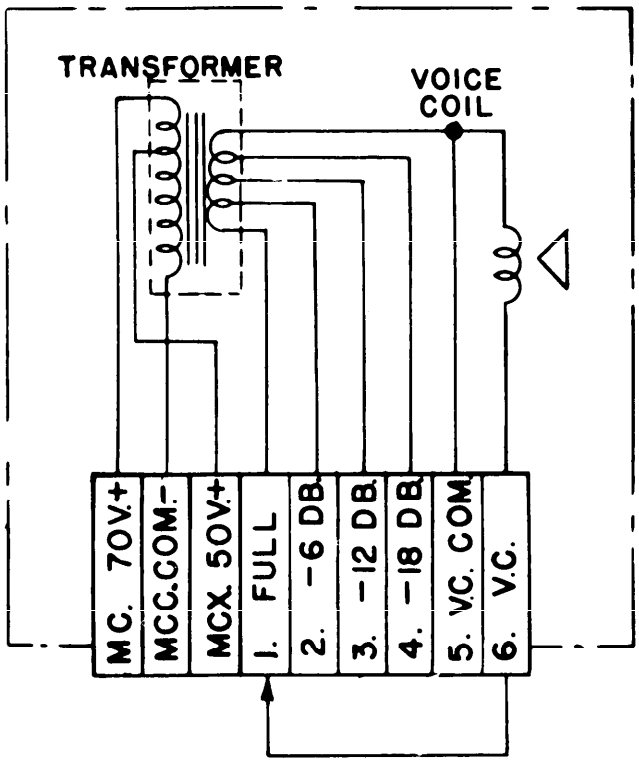


MICROPHONE JACK BOX TYPE IC/MJT

#NOTE: THE SWITCHING ARRANGEMENT INDICATED HEREIN IS ILLUSTRATIVE ONLY. DESIGN CONSIDERATIONS WILL DETERMINE ACTUAL ARRANGEMENT.

FIGURE 15
STANDARD CIRCUITS & TERMINAL MARKINGS

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NOTE:

THE CONVENTION ESTABLISHED FOR CIRCUIT AND MARKINGS SHALL BE FOLLOWED AS APPLICABLE FOR THE OTHER TYPES OF LOUDSPEAKERS.

FIGURE 16
STANDARD CIRCUIT & TERMINAL MARKINGS
LOUDSPEAKERS, TYPES IC/SAA & IC/SBA

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